The agro-hydrological & chemical and crop systems simulator (AHC) — Theoretical documentation

Version 2.1R

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Research report No. 2

May 2026

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Xu Xu, Chen Sun, Yutao Liu, Guanhua Huang (2026). The agro-hydrological & chemical and crop systems simulator (AHC) —Theoretical documentation (version 2.1R). China Agricultural University (Beijing), Research report for version 2.1R.

Abstract

This document introduces the Agro-Hydrological & chemical and Crop systems simulator (AHC) as an efficient tool for modeling and assessing agroecosystem processes. The AHC, as a one-dimensional physical model, numerically simulates the vertical transport of soil water and vapor, solute (salinity and/or nitrogen), and heat in variably saturated soils. It also includes modules to simulate surface runoff, evapotranspiration, soil nitrogen-carbon turnover, and plant growth and yield formation. The model is designed to simulate agro-eco-hydrological processes at the field scale level and during the whole agricultural year. The software consists of the AHC mathematical computer program and a menu-driven program as a user-friendly interface for efficient data manipulation. The model's ability to simulate the transport of soil water, salinity, nitrogen, gas emission, and crop growth has been tested and demonstrated in various soil-crop environments. The key/unique features of the AHC include: the simultaneous solution of heat, water, and solute fluxes with various boundary conditions; the optional function for simulating the fate of salt, nitrogen, or salt-nitrogen, also including soil carbon and nitrogen turnover; provision of two different levels of complexity crop modules (moderate and detailed); detailed simulation of soil freezing and thawing processes for seasonally frozen areas; the efficient global method for sensitivity analysis and parameter estimation; and applicability in a wide range of field conditions. Information from the model could be used to analyze the complex agroecosystem processes and to support the rational management of irrigation application, salinity control, nitrogen fertilizer use, and crop production. This manual is primarily used to describe the main structures and compositions of the AHC, the modeling concepts and algorithms (involving soil water flow, solute transport and transformation, heat flow, soil nitrogen/carbon turnover, evapotranspiration, crop growth, and surface water flow), and the management of field drainage, irrigation, and nitrogen fertilization.

Table of contents

Abstract ii

Table of contents iii

List of figures vii

List of tables viii

1. Introduction 9

    1.1 Background 9

    1.2 Overview of AHC and its components 12

2. Variably saturated water flow 18

    2.1 Soil water flow equation 18

    2.2 Soil hydraulic functions 20

    2.3 Vapor flux and its calculation 23

    2.4 Initial and boundary conditions 25

        2.4.1 Initial condition 25

        2.4.2 Top boundary 25

        2.4.3 Bottom boundary conditions and lateral drainage 28

    2.5 Sink term: root water uptake 34

    2.6 Treatment for soil frost and thaw conditions: the simple approach 39

    2.7 Mobile/immobile flow 40

3. Surface water flow and field drainage 42

    3.1 Surface runoff 42

    3.2 Interception of rainfall 47

    3.3 Field drainage 47

        3.3.1 Field scale drainage based on tabular relation 48

3.3.2 Drainage equations of Hooghoudt and Ernst: field-scale 49
3.3.3 General aspects of regional scale drainage 60
3.4 The snowpack 61
4. Solute transport 63
4.1 Governing solute transport equations 63
4.2 Initial and boundary conditions 67
4.3 Root solute uptake 69
4.4 Mobile/Immobile solute transport 69
5. Heat transport 71
5.1 Governing heat transport equation 71
5.2 Soil heat properties 73
5.2.1 Heat capacity and thermal conductivity 73
5.2.2 Freezing depression equation 75
5.3 Initial & boundary conditions and sink/source 76
6. Soil nitrogen 79
6.1 Mineralization and immobilization 79
6.2 Nitrification and NH_3 volatilization 83
6.3 Denitrification 85
6.4 N_2O emission 86
6.5 Urea hydrolysis 86
6.6 Plant N uptake 87
7. Evaporation and transpiration 89
7.1 Penman-Monteith equation 89
7.2 Radiation term 92
7.3 Aerodynamic term 95

7.4 Actual soil evaporation 98
7.5 Hargreaves ET_o equation 100
7.6 Canopy resistance correction: VPD and CO_2 100
8. Crop growth and yield (EPIC) 103
8.1 Phenological development 104
8.2 Crop growth and stress 104
8.3 Crop yield 108
8.4 Root respiration module 109
9. Numerical solution of water flow, solute and heat transport 112
9.1 Numerical solution of Richards equation 112
9.1.1 Space and time discretization 112
9.1.2 Treatment of top boundary conditions 116
9.1.3 Treatment of bottom boundary conditions 117
9.2 Numerical solution of advection-dispersion equation (ADE) for solute transport .. 119
9.2.1 Equation discretization: Non-linear adsorption 120
9.2.2 Equation discretization: Non-adsorption and linear adsorption 123
9.2.3 Immobile solute transport 123
9.3 Numerical solution of advection-dispersion equation (ADE) for heat transport 124
9.3.1 Equation discretization 124
9.3.2 Treatment of boundary conditions 126
9.3.3 Treatment of calculation stability 129
9.4 Numerical solution strategy: water-heat-solute calculation 131
9.4.1 Iterative scheme 131
9.4.2 Convergence and time Control 132
10. Model parameterization 134

10.1 Sensitivity-Analysis Process 134
10.2 Parameter-Estimation Process 136
11. Management aspects 140
11.1 Field Irrigation 140
11.1.1 Irrigation scheduling options 141
11.1.2 Timing criteria 141
11.1.3 Application depth criteria 143
11.2 Design of field drainage 144
11.3 Field fertilization 144
12. Graphical user interface and application description 146
12.1 Graphical user interface 146
12.2 Application description with field cases 148
Acknowledgments 157
References 158
Appendix 1: Notation（List of all variables) 159
Annex A Derivation of various partial derivatives 160
Annex B: Calculation of maximum liquid soil water content during freezing conditions 163
Annex C: Numerical solution of Richards equation using Picard iteration method 164
Annex D: Numerical solution of Richards equation using Newton-Raphson iterative technique (primary switching algorithm) 168
Annex E: Calculation of solute concentration (C_{im}) in immobile regions using the Newton iteration method 174
Annex F: Development process and history of the AHC model 175
Annex A: Derivation of various partial derivatives （程序中的单位, 备份用的） 177

List of figures

Figure 1.1 The calculation procedure of the Agroecosystem-Simulation (AESIM) Process in AHC. 17

Figure 2.1 Pseudo two-dimensional Cauchy lower boundary conditions, in the case of the local lateral drainage to ditches and the seepage from a deep aquifer through aquitard. 31

Figure 2.2 Reduction coefficient for root uptake, α_w, as a function of soil water pressure head h and potential transpiration rate T_p (left: Feddes et al. (1978); right: van Genuchten (1987)).......... 35

Figure 2.3 Reduction coefficient for root uptake, α_s, as a function of soil water electrical conductivity EC and osmotic head (left: Maas and Hoffman (1977); right: Dirksen and Augustijn (1988)). 36

Figure 3.1 Flow to vertically-walled drains reaching the impervious layer.......... 50

Figure 3.2 Drain pipe with gravel envelope in drain trench.......... 54

Figure 3.3 Geometry of two-dimensional flow toward drains, according to Ernst.......... 55

Figure 3.4 Five field (steady-state) drainage situations considered in AHC (adapted from Ritzema (2006)). The hydraulic head ϕ_{gwl} is defined as positive upward with $\phi = 0$ at the soil surface, and K_{top} and K_{bot} are the saturated hydraulic conductivity of the fine layer and coarse layer, respectively.......... 58

Figure 3.5 The water fluxes to and from the snow layer (adapted from Kroes and van Dam (2003)).......... 61

Figure 6.1 Conceptual diagram of the soil nitrogen transport and transformation in AHC. TON is the total organic nitrogen.......... 79

Figure 6.2 Schematic presentation of the soil organic matter turnover model included in AHC. 81
Figure 9.1 Spatial and temporal discretization with the time (*t*) and depth (*z*) for the solution of the Richards equation. 113
Figure 10.1 The flow chart of the GSA (Global-Sensitivity-Analysis) Process based on the LH-OAT method. 135
Figure 10.2 The flow chart of the GPE (Global-Parameter-Estimation) Process based on modified-MGA approach. 137
Figure 12.1 Graphical user interface for the AHC – representative windows for model input and output: (a) the window for control file; (b) the window for the AESIM Process; (c) and (d) are part of the display windows for the soil water and solute output and the crop modeling output, respectively. 147

List of tables

Table 2-1 Eight options for the lower boundary conditions 32
Table 3-1 Indicative CN values for various AMC II and their corresponding values for AMC I (dry) and III (wet) for various infiltration rates 44

1. Introduction

1.1 Background

Agroecosystem models can be defined as a quantitative scheme for predicting hydrological, chemical, and biological processes in soil-crop systems, with relevant human and environmental factors on various scales (Monteith, 1996; Sinclair and Seligman, 2000; Steduto et al., 2009). The functions and applications of the model mainly include evaluating field experiments, searching for the best management practices, and using as heuristic tools in teaching and research (Sinclair and Seligman, 1996; Cameira et al., 2007; Šimůnek et al., 2016). Current agroecosystem models have varying degrees of numerical sophistication, ranging from purely empirical to highly mechanistic (Monteith, 1996; Kucharik, 2003). In the aspect of agro-hydrological processes, the models can be classified as balanced and physically mechanistic models. The former includes the lumped balance models (e.g., SIMDualKc) and the cascading layer models (e.g., AquaCrop and DSSAT) (Rosa et al., 2012; Steduto et al., 2009; Jones et al., 2003). The latter is fully based on deterministic and physical laws, such as HYDRUS, SWAP, Daisy, and RZWQM (Šimůnek et al., 2005; van Dam et al., 1997; Hansen, 2002; Ahuja et al., 2000). For the aspect of crop growth and yield formation, the models might be divided into empirical models and process-based models. Empirical models consist of empirical functions that are chosen to fit measurements from the field. Process-based models often simulate the plant biomass or organ development based on the plant physiology or morphology, typically as EPIC, WOFOST, and DSSAT (Williams et al., 1984; van Diepen et al., 1989; Hoogenboom et al., 2015). So far, agroecosystem modeling is still facing some disputes and dilemmas in the generality of models and the balance of simplicity and complexity (Therond et al., 2011; Rosa et al., 2012; Sinclair and Seligman, 2000).

On the other hand, existing agroecosystem models generally have different functionalities for practical use. Models for irrigation management more focus on soil water, salinity, and crops (e.g., SIMDualKc, AquaCrop, SWAP, and SaltMod (Oosterbaan, 2001)), while the suite of specialized crop models always includes the simulation of the fate of nutrients and/or the soil carbon balance, such as EPIC, RZWQM, DSSAT, and APSIM (McCown et al., 1996). Moreover, some researchers conclude that the agroecosystem models can be classified into two types of approaches: scientific and engineering (Steduto et al., 2009). The first aims to improve the understanding of crop behavior based on physiology, and the second attempts to provide management advice to farmers and policymakers. Generally, a common consensus is that it is impossible to create universal models, as each season or new location brings new challenges not foreseen in the original model (Sinclair and Seligman, 2000). Therefore, considerable and continuous efforts of modification of model functions are necessary to make the model cope with discrepancies that derive from the changes in the peculiarities of the application environments (Sinclair and Seligman, 1996; Sinclair and Seligman, 2000; Coucheney et al., 2015; Xu et al., 2016).

Model calibration is another major challenge in agroecosystem modeling, involving sensitivity analysis (SA), parameter estimation (PE), and uncertainty analysis (Xu et al., 2016). The SA is a critical step in identifying the sensitive and important parameters (DeJonge et al., 2012). The most popular SA practice is that of the OAT (One-factor-At-a-Time) method, which has the limitation of local attributes of data sampling. In addition to local methods, global sensitivity analysis (GSA) methods are recently more preferred and applied to some agroecosystem models, typically as the extended FAST, Sobol' and LH-OAT methods (Zhao et al., 2014; Della Peruta et al., 2014; Xu et al., 2016). The automatic inverse optimization method is becoming a preferred alternative in parameter estimation. The local PE method has been widely used; however, it may prematurely terminate the search and thus is prone to find local rather than global minima (Šimůnek et al., 2012). The global PE methods become a better alternative when more parameters are to be estimated or a more complex model is adopted.

Some classic applications in agroecosystem modeling include the use of PEST, genetic algorithm, GLUE, Ensemble Kalman Filter, and shuffled complex methods (Doherty, 2004; Ines and Droogers, 2002; Abbaspour et al., 2001; Evensen, 2003; Duan et al., 1994). So far, some mature models have already released their own specific PE module for practical use, e.g., the inverse solution based on the gradient-type minimization method in HYDRUS-1D (Šimůnek et al., 2012) and the DSSAT GLUE program (Jones et al., 2011). An uncertainty analysis is also applied for model parameters (Xu et al., 2016), especially in watershed hydrologic modeling because of the high uncertainty at large scales (Yang et al., 2008).

Therefore, the agroecosystem models developed by different research institutions usually have relatively specific backgrounds, features, and adaptabilities. In particular, if a model is used in changed environments (different from its original development), it often requires certain or even significant degrees of modification of model functions. China has a vast land area and a variety of different farmland environments. We generally encounter various issues worldwide in agroecosystem management (e.g., water deficit, waterlogging, secondary salinity, and unreasonable nitrogen fertilization and carbon management), or even face several of them simultaneously. Hence, many different (commonly used) models have been selected and applied to solve specific problems depending on the research topics. In actuality, this also brings a heavy burden in model learning, application, and especially source code modification. Therefore, we introduce a powerful model/software (named AHC) for agroecosystem simulation to adapt to various agro-eco-hydrological environments, which is mainly based on the motivation of integrating our previous experimental and modeling work related to various agro-ecosystem conditions in the northern regions of China. The AHC is developed as a systematic tool to numerically simulate the soil water flow, the fate of solute (salt and/or nitrogen), heat transport, soil nitrogen and carbon, and crop growth and yield with the effects of ambient environmental changes. The efficient global methods for sensitivity analysis and inverse estimation of parameters are also involved. So far, the AHC has been tested and well applied in many cases for water-salt-nitrogen-crop simulation and strategy management. The conceptual framework, development process, and distinctive components and features of the AHC are briefly described in the following section of this chapter.

1.2 Overview of AHC and its components

The AHC (Agro-Hydrological & chemical and Crop systems simulator) is designed with two main parts, i.e., the mathematical model and the graphical user interface (GUI). The mathematical model as the core of the AHC could be seen as composed of three processes: Agroecosystem-Simulation (AESIM) Process, Global-Sensitivity-Analysis (GSA) Process, and Global-Parameter-Estimation (GPE) Process. Process could be defined as a kind of modularization entity, and each process is a part of code that performs certain functions by specific methods. These source codes are all written in FORTRAN, and the current version 2.1R is compiled with Intel® Fortran Compiler (Classic ifort or ifx) integrated in the Intel® oneAPI HPC Toolkit. The friendly GUI (version 2.1R) is developed for the pre-and post-processing of datasets using the Visual Basic.NET (VB.NET) programming language in Microsoft Visual Studio 2022. The mathematical model of FORTRAN codes is packaged into the GUI program. The friendly GUI is preliminarily developed by Neng (2016), and it is continuously being improved and updated with the development of the AESIM Process.

The framework of the AESIM Process is initially inherited from an early version of a well-known agro-hydrological model SWAP (version 2.0) (developed by van Dam et al. (1997), Wageningen Agricultural University). We also partly employ the experience gained with other models (i.e., EPIC, Daisy, SHAW, HYDRUS, RZWQM, and AquaCrop) and refer to the experimental results in many previous literature. So far, the AESIM Process has made comprehensive and profound improvements compared to the original SWAP v2.0, owing to various reasons of functionality, efficiency, or numerical stability. This primarily involves developing the more advanced/new soil-heat flux modules and a new solute transport module to replace the old ones; incorporating a new moderate crop model (EPIC_CGM), and the updated detailed crop model (WOFOST_CGM) from WOFOST v7.1; introducing new modules of soil nitrogen/carbon turnover and surface runoff; and adding new source codes for improving or describing more other processes (e.g., nitrogen fate, vapor transfer, CO_2 effects, evaporation & transpiration correction, crop dormancy, ground surface mulching, root water and solute uptake, etc.), according to problems and research needs.

For the current version, the AESIM Process consists of seven main modules related to surface runoff, evaporation and transpiration, soil water, solute, heat, soil nitrogen and carbon, and plant. The model is mainly based on the numerical solution of a series of partial differential equations to simulate the transport processes. The main calculation flow chart is presented in Fig. 1. An overview of the principles and modeling concepts is briefly described in this chapter, also corresponding to the contents in the subsequent chapters.

(1) The model employs the modified forms of the Richards equation for soil water movement in the variably saturated soil matrix, including the vapor transfer and presence of soil freezing (Chapter 2). The soil hydraulic functions are described by the analytical expressions of van Genuchten and Mualem or Brooks and Corey. AHC also considers the vapor flux q_v in soil, which is calculated from a vapor density gradient according to Fick's law. Atmospheric top boundary, eight types of bottom boundary, and lateral drainage are provided to flexibly treat the various field conditions. Root water extraction (as a sink term) is calculated from potential transpiration while considering water and salt stress; In addition, AHC also includes the function to consider the compensation effects of root water uptake.

(2) AHC uses a modification of the SCS curve number (SCS-CN) method to compute the surface runoff, which accounts for the effects of soil type, slope, land use, cover, and antecedent soil water conditions. In case the field practices (e.g., soil bounds) may limit or prevent the

surface runoff, the SCS-CN runoff subroutine is bypassed, and water that cannot infiltrate as a result of excessive rainfall or irrigation will be stored in a ponding reservoir. The part of water over the maximum ponding height is calculated as the surface runoff using an empirical formula. In either case, the groundwater level may rise above the soil surface and fill the ponding reservoir. Lateral field drainage fluxes to the drainage system are alternatively simulated using a simple linear or nonlinear (i.e., using the drainage equations of Hooghoudt and Ernst) relationship between groundwater level and drainage flux. The use of drainage equations allows the design or evaluation of drainage systems (Chapter 3).

(3) The model adopts the advection-dispersion equation (ADE) to simulate the solute transport and transformation processes in a variably saturated rigid porous medium (Chapter 4). It uses a generalized non-linear equation to describe the nonlinear adsorptions between the solid and liquid phases. The transport models also account for advection and dispersion in the liquid phase. The solute transport equations also consider the first-order degradation independent of other solutes, and the first-order decay reactions in the sequential decay chain of multiple solutes. In particular, AHC provides specific options for solute fate simulation, including only total salt transport, nitrogen transport, and salt-nitrogen migration.

(4) AHC uses the heat ADE for heat conduction and convection, considering phase changes of water and ice (Chapter 5). The heat transport equation considers movement by conduction as well as convection with flowing water. The vapor transfer and its effects on the soil system are included as well. In addition, the energy balance equation also considers the existence of soil freezing and thawing in soil layers, i.e., the heat by phase changes between liquid water and ice. The freezing point depression equation is used to define the relation between ice content and temperature. The thermal conductivity and soil heat capacity are calculated from the soil composition and the volume fractions of water, air, and ice.

(5) The model provides a detailed simulation for the transformation processes of soil carbon and nitrogen, including mineralization-immobilization, solid adsorption, nitrification, denitrification, volatilization, nitrous oxide (N_2O) emission, root nitrogen uptake (positive and active) involved in the chain equations. The urea hydrolysis to ammonia is described using the first-order kinetics equation. The mineralization and immobilization of nitrogen (N) are described by the quantitative transformation of different carbon and nitrogen pools (Chapter 6.1); meanwhile, the model calculates the emitted CO_2 flux from microbial respiration in soils.

(6) AHC uses a two-step approach to calculate evapotranspiration (Chapter 7). The first step involves the calculation of the potential evapotranspiration rate according to the Penman-Monteith equation, and partitions into potential soil evaporation and crop transpiration (based either on the leaf area index or the soil cover fraction). The Penman-Monteith equation involves two calculation ways: the direct method based on the given minimal canopy resistance, and the indirect method based on the reference FAO-ET_o and relevant conversion coefficient. Note that the canopy resistance term can be modified to reflect the impacts of the high vapor pressure deficit and the changes in CO_2 concentration on leaf conductance. In the second step, the actual soil evaporation and crop transpiration are determined based on water and/or salinity stress and the reduction due to limited soil evaporation capacity. Particularly, AHC introduces a soil moisture-based, mechanism approach to better calculate soil evaporation under shallow groundwater environments. Moreover, instead of data for Penman-Monteith, also a reference potential evapotranspiration rate (ET_o) could be optionally calculated by the Hargreaves equation (with only temperature and radiation data required), together with a relevant conversion coefficient.

(7) Two generic crop growth modules are provided in AHC, including the moderate EPIC crop module (called EPIC_CGM, Chapter 8) with less data demand, and the detailed WOFOST module (called WOFOST_CGM) with more data input. They are both driven by the cumulative

temperature and solar radiation, and applicable to be used for various plant species. EPIC_CGM module considers leaf area development, light interception, and the conversion of intercepted light into biomass and yield, together with the effects of temperature, water, and salt/nitrogen stress. Biomass is computed from the solar radiation intercepted by the crop leaf area, estimated with Beer's law. The actual crop yield is calculated using the harvest index concept based on dry above-ground biomass. The WOFOST_CGM module has more detailed biological processes compared to the EPIC_CGM module. It considers both photosynthesis and maintenance respiration in crop growth, as affected by various environmental factors. The dry matter produced is partitioned among roots, leaves, stems, and storage organs, using partitioning factors that are a function of the crop phenological development stage.

(8) For numerical simulation (Chapter 9), the two differential equations for soil water and heat flow are now solved by using the Newton-Raphson iterative technique for each time step, which can particularly make the source code more concise and efficient for soil frozen simulation. Hence, the model can simulate soil freezing and thawing processes using heat-water coupling iteration calculation (considering phase changes of ice and water). The numerical solution of solute transport is based on a fully implicit or Crank-Nicholson, central finite-difference scheme, which could ensure the model stability and calculation efficiency to meet the needs of GSA and GPE Processes.

Furthermore, the GSA process employs the Latin Hypercube One-factor-At-a-Time (LH-OAT) method to perform the parameter sensitivity analysis, which is a very efficient global method for identifying parameter sensitivity. After selecting the sensitive parameters, the GPE Process uses the modified Micro Genetic Algorithm (modified-MGA) to determine the values or ranges of the important model parameters (Chapter 10). Therefore, so far, the AHC (version 2.1R) has become a powerful software tool that, in brief, can be used to simulate water flow, solute transport and transformation, soil nitrogen/carbon, heat transport, and crop growth and yield in various/complex agricultural fields.

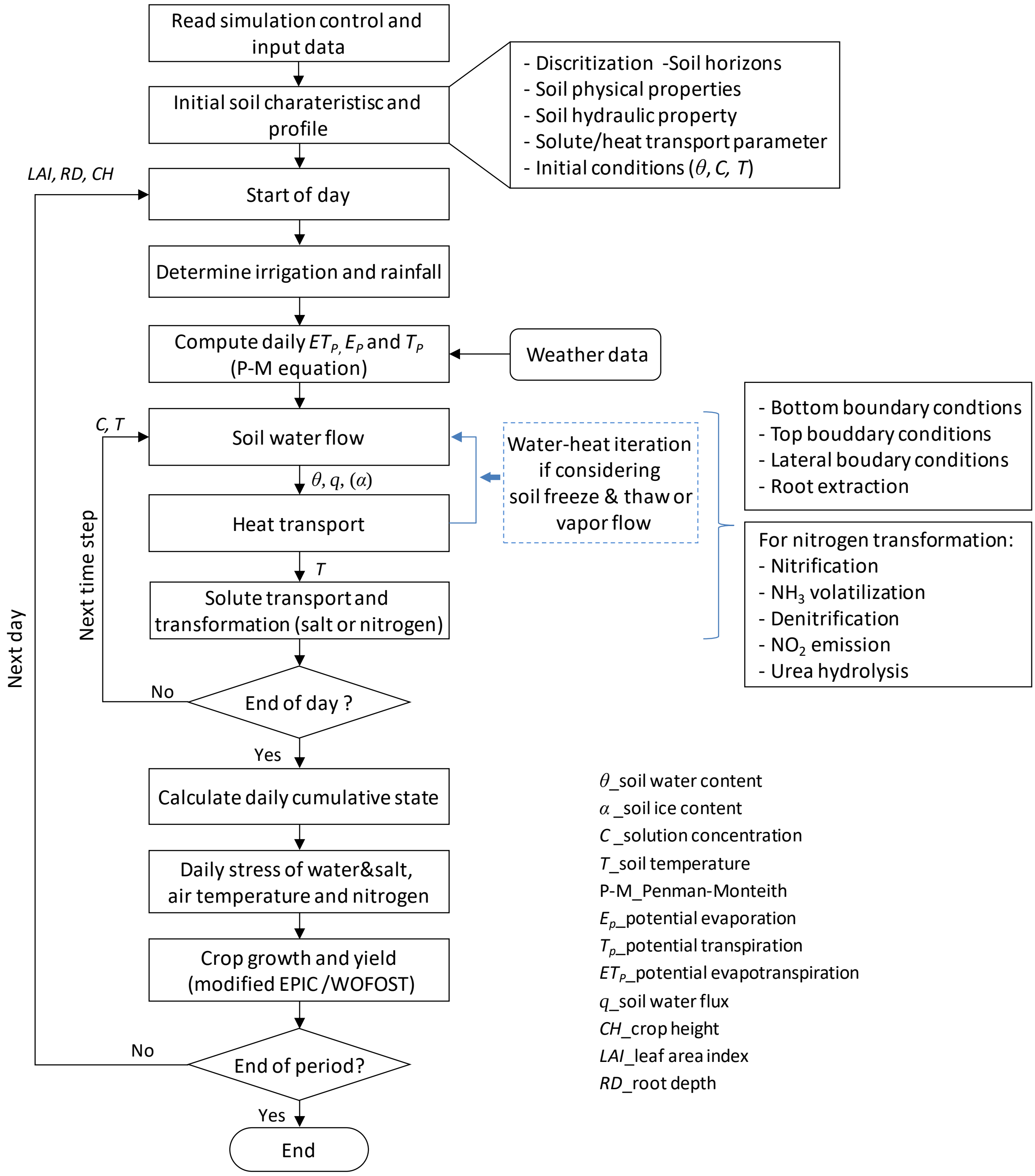


Figure 1.1 The calculation procedure of the Agroecosystem-Simulation (AESIM) Process in AHC.

2. Variably saturated water flow

2.1 Soil water flow equation

Darcy's law, though originally conceived for flow in saturated porous media, is extended and commonly used to describe the unsaturated water fluxes, with the provision that the conductivity is now a function of the matric suction head (Richards, 1931; Hillel, 1998):

$$q_l = -K(h)\nabla H \tag{2.1}$$

where q_l is liquid soil water flux density (positive upward) (cm d^{-1}), $K(h)$ is the unsaturated soil hydraulic conductivity (cm d^{-1}), h is the soil water pressure head (cm), H is the soil water potential (cm), and ∇H is the hydraulic head gradient (-) which may primarily include both suction and gravitational components in many cases.

Introducing the continuity principle, we can obtain the general flow equation and account for transient variably saturated flow processes:

$$\frac{\partial \theta}{\partial t} = -\nabla \cdot q_l \tag{2.2}$$

where θ is the volumetric soil water content (cm^3 cm^{-3}), and t is time (d). This equation, along with its various forms, is known as the Richards equation. For one-dimensional (1-D) vertical flow, Darcy's equation can be written as:

$$q_l = -K(h)\nabla\left(h+z\right) = -K(h)\frac{\partial\left(h+z\right)}{\partial z} \tag{2.3}$$

where z is the vertical coordinate (cm, positive upward).

For water balance consideration of an infinitely small soil volume, the soil water flow can be described with the one-dimensional (vertical) form of Richards equation:

$$\frac{\partial\theta}{\partial t} = -\frac{\partial q_l}{\partial z} - S_w(h) \tag{2.4}$$

with S_w the sink term of soil water flow (cm^3 cm^{-3} d^{-1}). It can be rewritten as the *h*-based formulation with the combination of Eq. (2.3), as follows:

$$\frac{\partial\theta}{\partial t} = \frac{\partial\theta}{\partial h}\frac{\partial h}{\partial t} = C_w(h)\frac{\partial h}{\partial t} = \frac{\partial}{\partial z}\left[K(h)\left(\frac{\partial h}{\partial z}+1\right)\right] - S_w(h) \tag{2.5}$$

with C_w the differential soil water capacity (cm^{-1}). The Richards equation considers only water flow in the liquid phase, which is applicable to many cases under non-frozen conditions. When considering the presence of soil freezing and thawing (with liquid water and solid ice), Eq. (2.5) can be further expressed as a modified form of Richards equation:

$$\frac{\partial\theta}{\partial t} + \frac{\rho_i}{\rho_l}\frac{\partial\alpha}{\partial t} = C_w(h)\frac{\partial h}{\partial t} + \frac{\rho_i}{\rho_l}\frac{\partial\alpha}{\partial t} = \frac{\partial}{\partial z}\left[K(h)\left(\frac{\partial h}{\partial z}+1\right)\right] - S_w(h) \tag{2.6}$$

where α is the volumetric ice content (cm^3 cm^{-3}), ρ_i is the bulk density of ice (g cm^{-3}), and ρ_l is the liquid water bulk density (g cm^{-3}).

Moreover, the above form of Richards equation (Eq. 2.5 or Eq. 2.6) ignores the water vapor transfer in soils. Note that while this assumption is reasonable in most applications, the effects of vapor flow could not be negligible in many problems. Vapor movement is often an important part of total water flux when the liquid water content is relatively low, such as in relatively dry soil conditions. Thus, when considering the soil water flux by vapor transfer, Eq. (2.6) could be further written into:

$$\frac{\partial\theta}{\partial t} + \frac{\rho_i}{\rho_l}\frac{\partial\alpha}{\partial t} = \frac{\partial}{\partial z}\left[K(h)\left(\frac{\partial h}{\partial z}+1\right)\right] - \frac{1}{\rho_l}\frac{\partial q_v}{\partial z} - S_w(h) \tag{2.7}$$

where q_v is the vapor flux in the soil (g cm^{-2} d^{-1}) and its calculation is described in detail in section 2.3. This equation is the basis for AHC to simulate the vertically 1-D variable-saturated water flow, including possible transient and perched groundwater levels. AHC solves Eq. (2.7) numerically using the finite-difference method, subject to specified initial and boundary conditions (BCs). In addition, Eq. (2.7) is highly non-linear, mainly due to dependencies of the water content and hydraulic conductivity on the pressure head (i.e., $\theta(h)$ and $K(h)$), and due to freezing-thawing effects that relate the ice content with the temperature (i.e., $\alpha(T)$). The solution of Eq. (2.7) requires knowledge of the soil hydraulic properties, i.e., with known relations among θ, h, K, and soil temperature (T) (discussed in section 2.2). Note that, in AHC, it is allowed to optionally use different forms of Richards equations (Eqs. 2.5, 2.6, or 2.7) to describe the soil water flow in the soil profile, according to the needs of actual environmental conditions.

2.2 Soil hydraulic functions

The soil water retention and hydraulic conductivity functions (i.e., $\theta(h)$ and $K(h)$ in Eq. 2.5) are necessary to solve the Richards equation, which are in general highly nonlinear functions of the pressure head (h). Researchers propose many different analytical expressions for soil hydraulic functions. The analytical $\theta(h)$ and $K(h)$ functions proposed by Brooks and Corey (1964) (i.e., B-C functions) have been used for a number of years. Then van Genuchten (1980) proposes a more flexible $\theta(h)$ function than Brooks and Corey (1964), and uses the statistical pore-size distribution model of Mualem (1976) to derive a close-form function for unsaturated hydraulic conductivity in terms of soil water retention parameters (i.e., VG-M functions). The van Genuchten functions are subsequently used in numerous studies, and adopted in AHC as the primary choice for describing the soil-hydraulic functions. The expressions of $\theta(h)$ functions according to van Genuchten (1980) are given by:

$$\theta(h) = \begin{cases} \theta_r + \dfrac{\theta_s - \theta_r}{\left(1 + |\alpha_L h|^n\right)^m} & h < 0 \\ \theta_s & h \geq 0 \end{cases} \tag{2.8}$$

Without losing much flexibility, m can be taken equal to:

$$m = 1 - \frac{1}{n} \qquad n > 1 \tag{2.9}$$

The corresponding $K(\theta)$ function according to Mualem (1976) is given as:

$$K(\theta) = K_s S_e^{\lambda_p} \left[1 - \left(1 - S_e^{n/(n-1)}\right)^{(n-1)/n}\right]^2 = K_s S_e^{\lambda_p} \left[1 - \left(1 - S_e^{1/m}\right)^m\right]^2 \tag{2.10}$$

with S_e (-) the effective saturation:

$$S_e(h) = \frac{\theta(h) - \theta_r}{\theta_s - \theta_r} = \frac{1}{\left(1 + |\alpha_L h|^n\right)^m} \tag{2.11}$$

where θ_r and θ_s denote the residual and saturated water contents (cm^3 cm^{-3}), respectively, K_s is the saturated hydraulic conductivity (cm d^{-1}), α_L (cm^{-1}), and n (-) are empirical shape parameters, and λ_p is a pore connectivity/tortuosity parameter (-). However, when the soil is frozen, the presence of ice in some pores may significantly increase the resistance of the porous medium to water flow and lead to an apparent blocking effect. The hydraulic conductivity is reduced to account for this blockage. Two approaches can be optionally used in AHC, as follows: (1) hydraulic conductivity is reduced linearly with ice content, assuming zero conductivity at an available porosity of 0.13 (Bloomsburg and Wang, 1969); and (2) hydraulic conductivity is reduced by introducing an impedance factor (Ω, -), as follows:

$$K(h) = 10^{-\Omega Q} K(h) \tag{2.12}$$

where Q is the ratio of ice content to total (minus the residual) water content (-), which accounts for the fact that the blocking becomes more effective as ice content part of the total water content increases. This equation shows that even a small value of Ω can have a significant effect on the conducitivity of the liquid phase as the ice portion increases.

Furthermore, the specific soil moisture capacity, $C_w(h)$, which is the slope of the soil water retention curve at any particular value of soil water pressure head, can be obtained by differentiating θ with respect to h, as follows:

$$C_w(h) = \frac{\partial \theta}{\partial h} = \frac{m \cdot \alpha_L \cdot n \cdot |\alpha_L h|^{n-1} (\theta_s - \theta_r)}{\left(1 + |\alpha_L h|^n\right)^{m+1}} \tag{2.13}$$

AHC also implements the B-C functions to describe soil water retention since they also have been adopted in many studies (especially in earlier modeling studies). The expressions of soil hydraulic functions according to Brooks and Corey (1964) are given as:

$$S_e(h) = \frac{\theta(h) - \theta_r}{\theta_s - \theta_r} = \begin{cases} |\alpha_L h|^{-n} & h < -1/\alpha_L \\ 1 & h \geq -1/\alpha_L \end{cases} \tag{2.14}$$

$$K(\theta) = K_s S_e^{2/n+l+2} \tag{2.15}$$

The corresponding specific soil moisture capacity is:

$$C_w(h) = \begin{cases} -(\theta_s - \theta_r) \cdot (-\alpha_L n) \cdot |\alpha_L h|^{-n-1} & h < -1/\alpha_L \\ 0 & h \geq -1/\alpha_L \end{cases} \tag{2.16}$$

The above equations contain six independent parameters: θ_r, θ_s, α_l, n, K_s, and l. Among them, the parameter l can be usually set to 0.5 which is suggested by Mualem (1976) and van Genuchten et al. (1991) to be the average value for many soils. The referenced values of the remaining five hydraulic parameters can be readily available from existing soil databases or generated using some general pedo-transfer functions (PTFs) according to the numerous literature, e.g., the ROSETTA PTFs (Schaap et al., 2001), the HYPRES PTFs (Wösten et al., 1999), and the Saxton & Rawls PTFs (Saxton and Rawls, 2006).

2.3 Vapor flux and its calculation

The vapor flux q_v in the soil is calculated from a vapor density gradient according to Fick's law:

$$q_v = -D_v \frac{\partial \rho_v}{\partial z} \tag{2.17}$$

where ρ_v is the vapor density (g cm^{-3}), and D_v is the vapor diffusivity in the soil (cm^2 d^{-1}). D_v is related to vapor diffusivity in air by:

$$D_v = D_v^{'} b_v \theta_a^{c_v} \tag{2.18}$$

where D_v' is the vapor diffusivity in the air (cm^2 d^{-1}), b_v and c_v are the coefficients (-) accounting for the standard tortuosity of the air voids with values of 0.66 and 1.0, respectively, and θ_a is air porosity (cm^3 cm^{-3}). The dependence of vapor diffusion on temperature and pressure may be expressed as:

$$D_v^{'} = D_v^0 \left(\frac{T_k}{273.16} \right)^2 \left(\frac{101.3}{P_{air}} \right) \tag{2.19}$$

where D_v^0 (cm^2 d^{-1}) is the vapor diffusion coefficient in air at standard temperature and pressure conditions of 0°C and 101.3 kPa, and P_{air} is the actual atmospheric pressure (kPa).

The thermodynamic relationship is introduced as:

$$\rho_v = \rho_{vs} h_r = \rho_{vs} \exp\left(\frac{Mg \cdot 0.01h}{RT_k} \right) \tag{2.20}$$

where ρ_{vs} is the density of saturated water vapor (g cm^{-3}), h_r is the relative humidity with in the soil (-), g is the gravitational acceleration (=9.81 m s^{-2}), R is the gas constant (=8.314 J mol^{-1} K^{-1}), M_w is the molecular weight of water (=0.018015 kg mol^{-1}), and T_k is the temperature of the soil in Kelvin (K).

Hence, we have:

$$\frac{\partial \rho_v}{\partial z} = \frac{\partial(\rho_{vs} h_r)}{\partial z} = h_r \frac{\partial \rho_{vs}}{\partial z} + \rho_{vs} \frac{\partial h_r}{\partial z} = h_r \frac{\partial \rho_{vs}}{\partial T} \frac{\partial T}{\partial z} + \rho_{vs} \frac{\partial h_r}{\partial z} \tag{2.21}$$

Putting Eq. (2.21) into Eq. (2.17), we can get an equation of the form:

$$q_v = -D_v \left(h_r \frac{\partial \rho_{vs}}{\partial T} \frac{\partial T}{\partial z} + \rho_{vs} \frac{\partial h_r}{\partial z} \right) \tag{2.22}$$

Eq. (2.22) means that the q_v in the soil can be separated into two components, that due to the temperature gradient and that due to the moisture gradient. When noting the slope of the saturated vapor density $s_v = \frac{\partial \rho_{vs}}{\partial T}$, it can be further written as:

$$q_v = q_{vm} + q_{vT} = -D_v \left(h_r s_v \frac{\partial T}{\partial z} + \zeta \rho_{vs} \frac{\partial h_r}{\partial z} \right) \tag{2.23}$$

where ζ is an enhancement factor (-). Observed vapor transfer in response to a temperature gradient exceeds that predicted by Eq. (2.22). The enhancement factor is included to give the correct vapor flux. The enhancement factor (ζ) is calculated by(Cass et al., 1984):

$$\zeta = E_1 + E_2 \left(\frac{\theta}{\theta_s} \right) - (E_1 - E_4) \exp\left(- \left(\frac{E_3 \theta}{\theta_s} \right)^{E_5} \right) \tag{2.24}$$

where E_1, E_2, E_4, and E_5 have assigned values of 9.5, 3.0, 1.0, and 4.0, respectively. E_3 is calculated from clay content.

$$E_3 = \theta_s \left(1 + \frac{2.6}{\sqrt{C_c}} \right) \tag{2.25}$$

where C_c is the clay content at the soil compartment (g g^{-1}).

The saturated vapor density (ρ_{vs}) is estimated from a seventh-order polynomial based on temperature:

$$\rho_{vs} = 10^{-3} \frac{100}{R \cdot T_k / M_w} \begin{pmatrix} 6.1104546 + 0.4442351T + 0.014302099T^2 + 2.6454708 \times 10^{-4} T^3 \\ +3.0357098 \times 10^{-6} T^4 + 2.0972268 \times 10^{-8} T^5 + 6.0487594 \times 10^{-11} T^6 \\ -1.469687 \times 10^{-13} T^7 \end{pmatrix} \tag{2.26}$$

The slope of the saturated vapor density (s_v, g cm^{-3} K^{-1}) curve is expressed very accurately for typical temperature ranges using the empirical equation given by:

$$s_v = 10^{-3}\left(0.0000165 + 4944.43\frac{\rho_{vs}}{T_k^2}\right) \qquad (2.27)$$

Above 45°C, however, this function diverges somewhat from the derivative of Eq. (2.27), in which case the derivative of Eq. (2.26) is used for the slope of the saturated vapor density curve.

2.4 Initial and boundary conditions

2.4.1 Initial condition

The solution of Eq. (2.6) requires knowledge of the initial distribution of soil pressure head within the flow domain:

$$h(z,t) = h_{\text{ini}}(z) \qquad t = t_0 \qquad (2.28)$$

$$\alpha(z,t) = \alpha_i(z) \qquad t = t_0 \qquad (2.29)$$

where h_{ini} (cm) is the initial condition for soil water pressure head which is the prescribed function of z at t_0, and t_0 is the time when the simulation starts. h_{ini} can also be converted from the initial soil water content, θ_{ini} (z) (e.g., using θ-h relationship in Eq. 2.8); or h_{ini} can be derived from the initial groundwater level (z_{gwi}, cm, negative below soil surface) when assuming a steady-state pressure head profile corresponding to a user-specified flux q_{ss}, (cm d^{-1}, positive upward). q_{ss}=0 means the hydrostatic equilibrium in the soil profile, and thus we have $h_{\text{ini}}(z) = z_{\text{gwi}} - z$. In AHC, initial conditions of soil water flow can be implemented with one of three input options: h_{ini}, θ_{ini}, or z_{gwi} & q_{ss}.

2.4.2 Top boundary

Appropriate criteria for the treatment of top boundary conditions are important for accurate simulation of rapidly changing soil water fluxes near the soil surface. The soil-air interface is exposed to atmospheric conditions, and thus the actual flux depends also on the soil moisture conditions near the surface. The soil surface (top) boundary condition depends on the rate of actual evaporation, and the fluxes of irrigation and precipitation. Note that evaporation occurs from the ponded water layer or the surface of the soil. In AHC, if there is surface ponded water, evaporation is first sustained by the ponded water and then by the surface soil water. *The top boundary condition may also change from a prescribed flux type (Neumann, in moderate weather and soil witness conditions) to a head-controlled type (Cauchy, in either very wet or very dry conditions), and vice versa.*

(1) **At moderate weather and soil wetness conditions**, the soil top boundary conditions will be flux-controlled:

$$-K\left(\frac{\partial h}{\partial z}+1\right)\bigg|_{z=0} = q_0(t) \qquad (2.30)$$

where q_0 is the soil water flux at the soil surface boundary (cm d^{-1}, positive upward). q_0 is calculated as the difference between evaporation (q_{eva}, cm d^{-1}) and incoming water from precipitation (q_{prec}, cm d^{-1}) and irrigation (q_{irri}, cm d^{-1}), following from:

$$q_0 = q_{\text{eva}} - q_{\text{prec}} - q_{\text{irri}} \qquad (2.31)$$

(2) **In case of very dry conditions**, the top flux q_0 should be limited to the maximum flux according to Darcy's law (i.e., $E_{\max}$, cm). If the flux q_0 is greater than $E_{\max}$, the top boundary will change into the head-controlled type condition:

$$-K\left(\frac{\partial h}{\partial z}+1\right) = f_0\left(h_{\text{atm}}, h\right) \qquad at\ z=0 \qquad (2.32)$$

where f_0 is the given function of h_{atm}, and h_{atm} (cm) is the soil water pressure head in equilibrium with the prevailing air relative humidity. h_{atm} can be calculated from the air humidity (H_r, -), as follows:

$$h_{\text{atm}} = \frac{100RT_K}{M_w g}\ln\left(H_r\right) \tag{2.33}$$

where M_w is the molecular weight of water (=0.018015 kg mol^{-1}), g is the gravitational acceleration (=9.81 m s^{-2}), and R is the gas constant (=8.314 J mol^{-1} K^{-1}). In addition, h_{atm} may also be approximated to -2.75×10^5 cm. In the numerical solution, the top boundary condition follows from:

$$-K\left(\frac{\partial h}{\partial z}+1\right)\bigg|_{z=0} \approx -K_{1/2}\left(\frac{h_{\text{atm}}-h_{1\text{n}}-z_1}{0.5\cdot\Delta z_1}\right) \tag{2.34}$$

where $K_{1/2}$ is the average hydraulic conductivity (cm d^{-1}) between the soil surface and the first node, h_{atm} is the soil water pressure head (cm) in equilibrium with the air relative humidity, h_{1n} is the soil water pressure head (cm) of the first node, z_1 (cm) and $\triangle z_1$ (cm) is the depth and thickness of the first node in the soil profile, respectively.

(3) **In the case of infiltration/runoff**, a head-controlled condition applies if the potential top flux exceeds the maximum infiltration rate. A "surface reservoir" boundary condition is applied if the surface ponding is expected to develop.

$$-K\left(\frac{\partial h}{\partial z}+1\right)\bigg|_{z=0} = \frac{dh_{\text{pond}}}{dt} - q_{\text{nin}}(t) \tag{2.35}$$

where q_{nin} is the difference between incoming water (precipitation and irrigation) and evaporation (cm d^{-1}) (i.e., q_{nin}=q_{prec}+q_{irri}-q_{eva}) for the surface water pond, and h_{pond} is the ponding depth of the surface water layer (cm). This equation shows that h_{pond} increases due to precipitation/irrigation, and reduces because of infiltration and evaporation. In AHC, water that does not infiltrate the soil in time could form ponding water on the soil surface; and, this is also applied to the case when the soil occasionally gets flooded in areas with shallow groundwater tables. In addition, the formation and change of surface ponding water are related to rainfall, runoff, and field practices, which are described in detail in section 3.1. The detailed processing of top boundary conditions is further described in the numerical implementation (section 9.1).

2.4.3 Bottom boundary conditions and lateral drainage

The bottom boundary of the one-dimensional model is either in the unsaturated zone or in the upper part of the saturated zone, where the transition takes place to three-dimensional groundwater flow (Kroes and van Dam, 2003). One of the following boundary conditions must be specified at the bottom of the soil profile (z=L):

$$\begin{aligned}
&h(z,t)\big|_L = h_L(t) \\
&-K\left(\frac{\partial h}{\partial z}+1\right)\bigg|_L = q_L(t) \\
&-K\left(\frac{\partial h}{\partial z}+1\right)\bigg|_L = f_L\left(h_{gwl},h\right) \\
&\frac{\partial h}{\partial z}\bigg|_L = 0
\end{aligned} \tag{2.36}$$

The first three equations correspond to three types of boundary conditions: the *Dirichlet* type (the pressure head h is specified), the *Neumann* type (the bottom flux q is specified), and the *Cauchy* type (the bottom flux depends on the groundwater level). The fourth one is the free drainage used in deep groundwater level conditions (belonging to Neumann type).

AHC further provides eight options to describe the above three types of bottom boundary conditions in response to various real-world situations (van Dam et al., 1997; Kroes and van Dam, 2003). Each of options has its typical scale of application. The relevant description and equations are presented in Table 2-1 for those eight options.

In AHC, the *Dirichlet* (first-type) condition is a predescribed pressure head, often determined from a recorded phreatic surface of a present groundwater table (options 1) or the recorded pressure head at/near the bottom (option 5) (both at a field level). The soil water pressure head is assigned to the lowest compartment. The main adavantage of the *Dirichlet* condition is the easy recording of the phreatic surface or pressure head in case of shallow groundwater tables. A drawn back is that (at shallow groundwater tables) the simulated phreactic surface fluctuations are sensitive to the soil hydraulic functions. This condition may result in significant fluctuations of the water fluxes across the lower boundary. Thus, it is important to obtain reasonable soil hydraulic properties (especially for the lower soil profile) and sufficient observation data.

The *Neumann* (second-type) condition is used with prescribing bottom flux (options 2, 6, 7, and 8). A bottom flux (q_{bot}, cm d^{-1}) of zero may be applied when an impervious layer exists at the bottom of the profile (option 6). A deep groundwater table usually results in free drainage of a soil profile (option 7); in this case, the unit gradient is assumed at the bottom boundary, and the bottom flux depends directly on the hydraulic conductivity of the lowest compartment. Options 6 and 7 are the two often-used Neuman-type boundary conditions. In addition, option 2 will be used when the bottom flux is known, such as for describing the interaction with a regional groundwater system.

The option 8 is a system-dependent type of boundary condition. This condition represents a seepage face (i.e., free outflow at soil-air interface) at the bottom of the soil profile through which water can leave the saturated part of the flow domain. This type of boundary condition assumes that a zero-flux boundary condition applies as long as the local pressure head at the bottom of the soil profile is negative. However, h_n will be set equal to zero as soon as the bottom of the profile becomes saturated. This means that drainage will only occur if the pressure head in the bottom compartment h_n increases until above zero, and q_{bot} is calculated by solving the Richards' equation. This type of boundary condition often applies to finite lysimeters that are allowed to drain under gravity. Besides, this condition is appropriate at fields when soil profile is drained by a coarse gravel layer.

The *Cauchy* (third-type) condition is used when the bottom flux depends on the groundwater level (options 3 and 4). The case of option 3 is used to calculate q_{bot} from an aquifer below an aquitard (Eq. T2-3). The case of option 4 is used to describe the *deep Drainage from the Soil Profile*. Vertical drainage across the lower boundary of the soil profile is sometimes approximated by a flux that depends on the position of the groundwater level (Hopmans and Stricker, 1989). In case of option 4, AHC calculates bottom flux (q_{bot}) as a function of groundwater level, using an exponential flux-average groundwater level relationship (Eq. T2-4). This relationship is found by Ernst and Feddes (1979) for the sandy soils in the eastern part of the Netherlands. This is especially valid for deep sandy areas. For additional data of q_{bot} -ϕ_{avg} relationships, see Massop and de Wit (1994).

In addition, AHC makes a distinction between the local (field) drainage flux to ditches and drains q_{dr} (cm d^{-1}) (as calculated by Eq. 3.13 according to section 3.3), and the bottom liquid flux q_{bot}. AHC assumes that the drainage flux is extracted laterally in the saturated zone of the soil profile. So in the case of options 1, 2, 3, 5, and 6, the bottom flux q_{bot} (defined by the user or calculated by the program) does not include the horizontal drainage flux q_{dr}; and this means that the lateral drainage flux q_{dr} can also be included in above five options as needed, in addition to the q_{bot}. However, in the case of option 4, Eq. T2-4 has already considered the lateral drainage to local ditches or drains, so q_{dr} calculation should not be included in this case. In the case of options 7 and 8, the simulated soil profile is unsaturated, so lateral drainage will not occur (van Dam et al., 1997).

Taking the case of option 3 as an example, Fig. 2.1 shows a soil profile that is drained by ditches and which received a seepage flux from a semi-confined aquifer. The drainage flux (q_{dr}) is calculated as described in section 3.3. The bottom flux (q_{bot}) is the seepage flux due to regional groundwater flow, which is calculated by Eq. T2-3. The Cauchy condition applies to the bottom boundary.

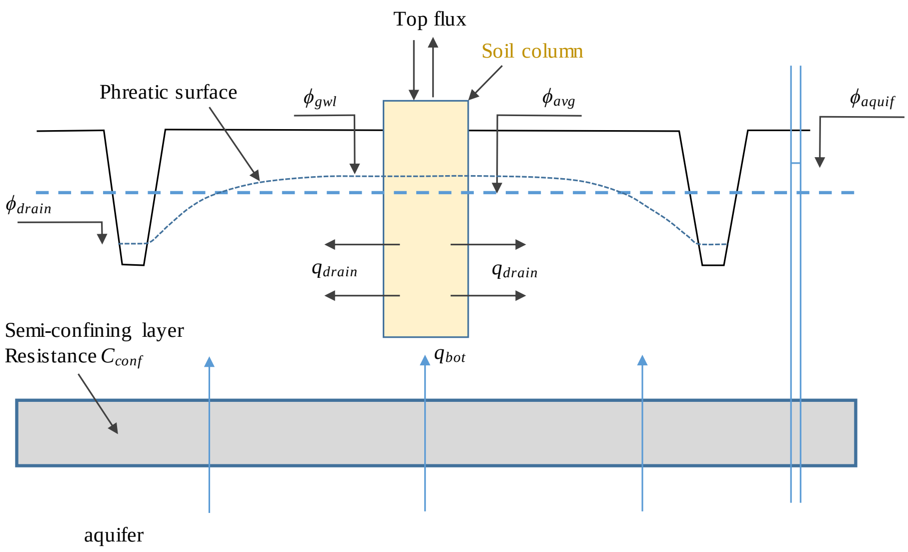


Figure 2.1 Pseudo two-dimensional Cauchy lower boundary conditions, in the case of the local lateral drainage to ditches and the seepage from a deep aquifer through aquitard.

Table 2-1 Eight options for the lower boundary conditions

Option	Description	Type condition	Application scale	Equation		Data input into model and other descriptions
1	Prescribe groundwater level (GWL)	Dirichlet	Field	$h_n = GWL - z_n - h_{resis}$	Eq. T2-1	GWLs are given as function of time; AHC will interpolate linearly for times between observations.
2	Prescribe bottom flux (q_{bot})	Neumann	Region/field	$q_{bot} = q_{ng}$	Eq. T2-2	q_{ng} is given as function of time, or calculated as function of time: $q_{ng} = q_{avg} + q_{\mathrm{Amp}} \cdot \cos\left[0.0172\left(1 - DN_{\max}\right)\right]$
3	Calculate the bottom flux from a deep aquifer below an aquitard (Fig. 2.1)	Cauchy	Region	$q_{bot} = (\phi_{aquif} - \phi_{avg}) / c_{conf}$ $\phi_{avg} = \phi_{drain} + \beta_{gwl}(\phi_{gwl} - \phi_{drain})$ $\phi_{aquif} = \phi_{aq,m} + \phi_{aq,a} \cos\left[\frac{2\pi}{\phi_{aq,p}}(t - t_{\max})\right]$	Eq. T2-3a Eq. T2-3b Eq. T2-3c	c_{conf}, ϕ_{drain}, β_{gwl}, $\phi_{aq,m}$, $\phi_{aq,a}$, $\phi_{aq,p}$, and t_{max} are required; A sinusoidal wave is assumed for hydraulic head ϕ_{aquif} in the semi-confiined aquifer.
4	Deep drainage from the soil profile as a function of GWL	Cauchy	Region	$q_{bot} = a_{qb} e^{b_{qb} \lvert \phi_{avg} \rvert}$	Eq. T2-4	a_{qb}, and b_{qb} are required. q_{bot} is calculated as function of GWL (Eq. T2-4). This type of exponential relationship is valid for deep sandy areas. Lateral drainage q_{dr} is not allowed in this option.
5	Prescribe soil water pressure head of bottom compartment (h_n)	Dirichlet	Field	$h_n = h_{ng}$	Eq. T2-5	h_{ng} are given as function of time; AHC will interpolate linearly for times between observations.
6	Bottom flux equals zero	Neumann	Field	$q_{bot} = 0$	Eq. T2-6	No data input required
7	Free drainage of the soil profile	Neumann	Field	$\frac{\partial(h+z)}{\partial z} = 1$ thus: $q_{bot} = -K_n(h)$	Eq. T2-7	No data input required

8	Free outflow at soil-air interface (seepage face)	Neumann	Field	when $h_L \approx h_n + 0.5 \cdot \Delta z_n \le 0$, $q_{\mathrm{bot}} = 0$; else when h_L increases to above zero, h_L is set to zero and drainage occurs	No data input required. This option is commonly applied for lysimieters, or specific field conditon with soil profile drained by a coarse gravel layer.

Note: where h_{ng} (cm) and q_{ng} (cm d^{-1}) are the given pressure head and flux at the bottom of the soil profile, respectively. z_n is the position of bottom nodal point (negative, cm), and h_{resis} is the head difference between the groundwater level and hydraulic head of the bottom nodal point in the previous time step (cm).

q_{avg} and q_{Amp} are the mean, amplitude of the flux sinus wave (cm d^{-1}), and DN_{max} is a time (d) at which q_{ng} reaches its maximum.

ϕ_{aquif} is the hydraulic heads in the semi-confined aquifer (cm), ϕ_{avg} is the average groundwater level in the region (cm), and c_{conf} is the semi-confining layer resistance (d). t_{max} is a time (d) at which ϕ_{aquif} reaches its maximum; $\phi_{aq,m}$, $\phi_{aq,a}$, and $\phi_{aq,p}$ are the mean (cm), amplitude (cm) and period (d) of the hydraulic head sinus wave in the semi-confined aquifer. ϕ_{drain} the hydraulic head of the drain (cm) and β_{gwl} the groundwater shape factor (-). Possible values for β_{gwl} are 0.66 (parabolic), 0.64 (sinusoidal), 0.79 (elliptic) and 1.00 (no drains).

h is the soil water pressure head (cm), z is the vertical coordinate (cm, positive upward), $K_n(h)$ is the hydraulic conductivity of the bottom nodal point (cm d^{-1}), and Δz_n is the thickness of the bottom compartment (cm).

2.5 Sink term: root water uptake

The actual soil water extraction rate by plant roots (S_a) is the main sink term. Its potential rate at a certain depth $S_p(z)$ is calculated by distributing T_p into each root zone compartment, as a function of the specified root length density. The actual root water uptake rate $S_a(z)$ is determined by considering the water and salt stress that is assumed to be multiplicative (Eq. 2.37). The integration of $S_a(z)$ over the rooting depth is equal to the T_a rate. In addition, the function of the compensated root water uptake could be considered following the conception of Jarvis (1989). It represents a threshold value, above which the root water uptake reduced in stressed parts of the root zone is fully compensated by uptake from other, less-stressed parts (Šimůnek and Hopmans, 2009). The detailed algorithm is provided in the following part.

The actual transpiration (T_a, cm d^{-1}) is governed by the root water uptake at a certain depth ($S_a(z)$, d^{-1}), which is calculated from the potential transpiration (T_p), rooting depth and distribution, and a possible reduction due to water and salt stress, as follows:

$$S_a(z) = \alpha_{ws} S_p(z) = \alpha_w \alpha_s S_p(z) \tag{2.37}$$

where α_w (-) and α_s (-) are the reduction coefficients relative to water stress and salt stress, respectively, α_{ws} (-) is the reduction coefficient of integrated water-salt stress, and $S_P(z)$ is the potential root water uptake rate at a certain depth (d^{-1}). A homogeneous root length density distribution is adopted over the rooting depth as often assumed in the modeling practice, because the distribution is usually not available (Kroes and van Dam, 2003).

Stress due to dry or wet conditions may cause stress on crop transpiration. AHC employs two types of functions to calculate water stress coefficient α_w: (1) the first one is a piecewise linear function (Eq. 2.38) proposed by Feddes et al. (1978), and (2) the second one is an S-shaped function (Eq. 2.39) proposed by van Genuchten (1987). Fig. 2.2 gives a schematic of the water stress response function for the above two equations.

$$\alpha_w(h)=\begin{cases}0 & h>h_1\\ (h_1-h)/(h_1-h_2) & h_2<h\le h_1\\ 1 & h_3\le h\le h_2\\ (h_4-h)/(h_4-h_3) & h_4\le h<h_3\\ 0 & h<h_4\end{cases} \tag{2.38}$$

$$\alpha_w(h)=\frac{1}{1+\left(h/h_{50}\right)^{p_1}} \tag{2.39}$$

where h_1 corresponds to h with no water extraction at higher pressure head (cm), h_2 is h below which optimum water uptake starts (cm), h_3 is h below which water uptake reduction starts (cm), h_4 is h corresponding to wilting point (on water uptake at lower pressure head) (cm), p_1 is an experimental constant, and h_{50} is the pressure head at which the water extraction rate is reduced by 50% during conditions of negligible osmotic stress.

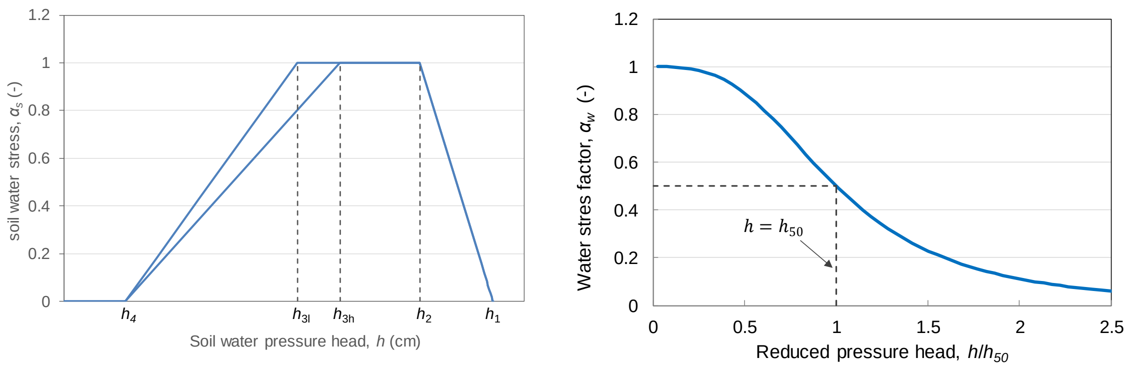


Figure 2.2 Reduction coefficient for root uptake, α_w, as a function of soil water pressure head h and potential transpiration rate T_p (left: Feddes et al. (1978); right: van Genuchten (1987)).

The stress coefficient α_s is described by the reduction function: (1) the empirical function of the electrical conductivity of a saturated soil extract, EC_e (Eq. 2.40), proposed by Maas and Hoffman (1977); and (2) the approximate S-shaped functions (see Eq. 2.41) involving a salinity threshold value, suggested by Dirksen and Augustijn (1988). Fig. 2.3 gives a schematic of the salt stress response function for the above two equations.

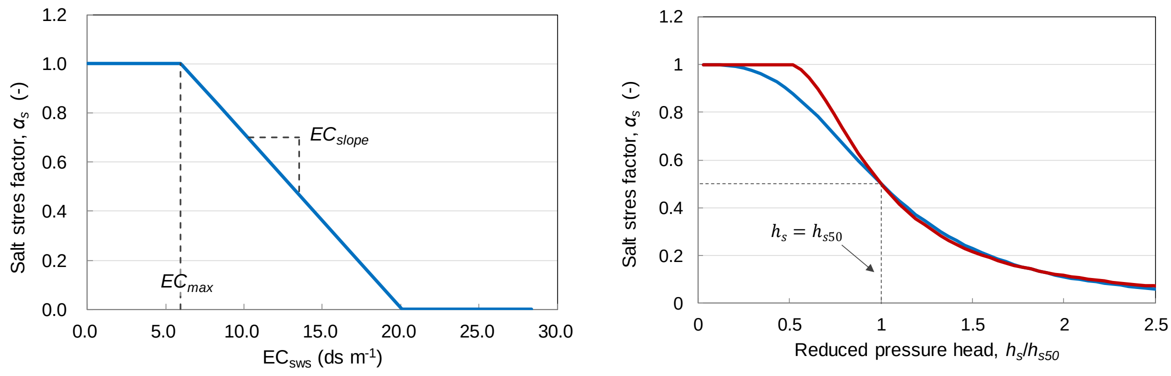


Figure 2.3 Reduction coefficient for root uptake, α_s, as a function of soil water electrical conductivity EC and osmotic head (left: Maas and Hoffman (1977); right: Dirksen and Augustijn (1988)).

$$\alpha_s(EC_{\text{sat}}) = 1 - \frac{EC_{\text{slope}}}{100}\left(EC_{\text{sat}} - EC_{\text{max}}\right) = 1 - \frac{EC_{\text{slope}}}{100}\left(\frac{c\theta}{f_{ECc}\theta_s} - EC_{\text{max}}\right) \tag{2.40}$$

$$\alpha_s(h_s) = \frac{1}{1 + ((h_o^* - h_s)/(h_o^* - h_{s50}))^{p_2}} \tag{2.41}$$

where EC_{sat} is the electrical conductivity of the solution in a saturated soil (dS m^{-1}) which could be assumed to be the value of EC_{e}, EC_{max} is the EC_{sat} level at which salt stress starts (dS m^{-1}), EC_{slope} is the decline of root water uptake above EC_{max} (%/dS m^{-1}), f_{ECc} is a global conversion factor (≈0.64-0.67) to convert the electrical conductivity of soil solution EC_{ss} (dS m^{-1}) to the soil solution concetration c (g cm^{-3}) (i.e., $c = f_{\text{ECc}} \cdot EC_{\text{ss}}$) (****), h_s is the soil solution osmotic head (cm), h_o^* is the threshold value of osmotic head (cm) above which there is no salt stress, h_{s50} is the soil solution osmotic head of soil salinity at which $\alpha_s(h_s)$ is reduced by 0.50 (cm), and p_2 is an empirical, presumably crop, soil and climate-specific dimensionless parameter described as (Homaee et al., 2002):

$$p_2 = \frac{h_{s50}}{h_{s50} - h_o^*} \tag{2.42}$$

In addition, the solution osmotic head (h_s) can be assumed to be a linear function of the concentration or electrical conductivity of soil solution (Shainberg and Oster, 1978; Homaee et al., 2002):

$$h_s \approx f_{\mathrm{conc}} c \approx f_{\mathrm{EC}} \cdot EC_{\mathrm{ss}} \tag{2.43}$$

where f_{conc} and f_{EC} are the conversion factors that are primarily affected by the chemical components of the solution. $f_{\mathrm{EC}} =$ -360 and f_{conc} = -537 are recommended values if there is no experimental data available (Shainberg and Oster, 1978).

Note that Eq. (2.41) is based on the soil solution osmotic head instead of the electrical conductivity of saturation extracts ($\mathrm{EC_e}$). If h_o^* is set to 0, it becomes the S-shaped function the same as that proposed by van Genuchten (1987) for calculating the salinity stress, as follows:

$$\alpha_s(h_s) = \frac{1}{1 + (h_s / h_{s50})^{p_3}} \tag{2.44}$$

where p_3 is an empirical parameter similar to p_2 and is specified by users (with recommended value = 3). The above two non-linear models could provide sufficient flexibility without introducing unwarranted complexity, compared to Eq. (2.40). Indeed, having a salinity threshold parameter h_o^*, Eq. (2.41) is in principle more reliable than Eq. (2.44) (Homaee et al., 2002). In addition, the h_s based equation is more reliable than EC_{sat} based equation. Moreover, it is essentially more realistic to use the h_s as the variable than to use the EC_{sat} in the salinity stress function. Because root water uptake is directly affected by salt concentration in solution (or h_s). But EC_{sat} is more about the soil salt contents not the salinity concentration in the solution.

$S_p(z)$ may be determined by introducing a non-uniform distribution of the potential water uptake rate over a root zone of arbitrary shape:

$$S_P(z) = \frac{b_r(z)dz}{\int_{-z_r}^{0} b_r(z)dz} T_p \Big/ dz = \frac{b_r(z)}{\int_{-z_r}^{0} b_r(z)dz} T_p = b_r^*(z) T_p \tag{2.45}$$

where Z_r is the root depth (cm), $b_r(z)$ is the arbitrary root water uptake distribution (user-specified with piecewise linear input) which is often related to the distribution of root length density, and $b_r^*(z)$ is the normalized water uptake distribution (-). This equation gives the spatial variation of the potential extraction term, S_p, over the root zone. When the potential water uptake rate is equally distributed over the root zone, $S_p(z)$ becomes:

$$S_P(z) = \frac{T_p}{Z_r} \tag{2.46}$$

For the whole root zone, it follows that $S_p(z)$ is related to T_p by the expression:

$$\int_{-z_r}^{0} S_P(z)dz = T_p \tag{2.47}$$

Integrating S_a(z) over the root layer yields the actual transpiration rate T_a (cm d^{-1}).

The AHC also includes the function of the compensated root water uptake. A critical value of the water stress index ω_c, i.e., a root adaptability factor, is further introduced to describe the compensated water uptake, referring to Jarvis (1989). ω_c represents a threshold value, above which the root water uptake reduced in stressed parts of the root zone is fully compensated for by uptake from other less-stressed parts (Eq. 2.48). The stress parts are just partly compensated if α_{ws} is below this critical value (Eq. 2.49).

$$\begin{aligned} T_a^* &= \int_{-z_r}^{0} S_a^* dz = \int_{-z_r}^{0} \frac{S_a}{\omega_{ws}} dz = \int_{-z_r}^{0} \frac{b_r(z) T_p \alpha_{ws}(z)}{\omega_{ws}} dz \\ &= \frac{T_p}{\omega_{ws}} \int_{-z_r}^{0} b_r(z)\alpha_{ws}(z)dz = \frac{T_p}{\omega_{ws}} \omega_{ws} = T_p, \quad \omega_{ws} \geq \omega_c \end{aligned} \tag{2.48}$$

$$\begin{aligned} T_a^* &= \int_{-z_r}^{0} S_a^* dz = \int_{-z_r}^{0} \frac{S_a}{\omega_c} dz = \int_{-z_r}^{0} \frac{b_r(z) T_p \alpha_{ws}(z)}{\omega_c} dz \\ &= \frac{T_p}{\omega_c} \int_{-z_r}^{0} b_r(z)\alpha_{ws}(z)dz = \frac{T_p}{\omega_c} \alpha_{ws}, \quad \omega_{ws} < \omega_c \end{aligned} \tag{2.49}$$

where T_a^* (cm d^{-1}) and S_a^* (d^{-1}) are the actural crop transpiration rate and the potential water uptake rate at a certain depth over the root zone, respectively, when considering the root water uptake compensation (cm d^{-1}), and $b_r(z)$ is the normalized water uptake distribution in the root zone (cm^{-1}).

2.6 Treatment for soil frost and thaw conditions: the simple approach

(1) Adjustment for soil hydraulic conductivity under frost conditions

As soil water freezes below a soil temperature of 0°C, the soil hydraulic conductivity is further adjusted under frost conditions, as follows (Kroes and van Dam, 2003):

$$K^{*}(z) = f_{ST}(z)\big(K(z) - K_{\min}\big) + K_{\min} \tag{2.50}$$

$$f_{ST}(z) = \begin{cases} 1.0 & T \geq T_1 \\ \dfrac{T(z) - T_2}{T_1 - T_2} & T_2 < T(z) < T_1 \\ 0 & T \leq T_2 \end{cases} \tag{2.51}$$

where $K^{*}(z)$ is the adjusted hydraulic conductivity at depth z (cm d^{-1}), $K_{\min}$ is a very small hydraulic conductivity (cm d^{-1}) with a default value of 10^{-10} cm d^{-1}, $f_{ST}(z)$ is a correction factor (-) for soil temperature at depth z, $T(z)$ is the soil temperature at depth z (°C), and T_1 and T_2 are the soil temperatures where reduction of hydraulic conductivity just begins and ends, respectively. The default values for T_1 and T_2 are taken as 0°C and -1°C.

(2) Soil thawing process: a simple approach

In the AHC, a variable active-node method is proposed to describe the simulation zone (i.e. active zone) and simulate the fate of soil water and solute during the soil thawing period in a simple way (Xu et al., 2013). This is particularly designed for the initial season when the upper soil layer fully thaws while the underlying soil layer is in thawing processes (e.g., for spring wheat growth in Northwest China). The active zone corresponded to the upper fully thawed soil layer, which is bounded by the soil surface at the top and the upper boundary of the frozen soil layer at the bottom. The rest nodes below are set as inactive nodes. The active zone depth (D_{act}, cm) is increased with soil thawing and estimated as follows:

$$\begin{aligned} D_{\mathrm{act}} &= a \cdot J^{b} + c \qquad & if\ D_{\mathrm{act}} < D_{\max} \\ D_{\mathrm{act}} &= D_{\mathrm{sp}} \qquad & if\ D_{\mathrm{act}} \geq D_{\max} \end{aligned} \tag{2.52}$$

where $D_{\max}$ is the maximum active zone depth (cm) and specified according to in-situ observation or experience, D_{sp} is the depth of the whole simulation domain (cm), J is the day number after soil starts to thaw, c is the value of D_{act} in initial simulation, and a and b are empirical coefficients (-). Given that the thawing rate tended to increase with time, b should be larger than one. Meanwhile, a non-flux bottom boundary condition is defined for the active zone during the thawing period, assuming that there is little water and solute exchange at the frozen interface. After the soil profile fully thaws, all nodes are activated and the bottom boundary condition is redefined as user-specified previously.

2.7 Mobile/immobile flow

In field soils, soil water may bypass large parts of the unsaturated soil domain, which is generally called preferential flow. This phenomenon may have large effects on the solute leaching. Preferential flow can be caused by macropores in structured soils or by unstable wetting fronts in unstructured soils resulting from soil stratification, air entrapment, and water repellency. This section primarily focuses on the water repellency, caused by the organic coatings of soil particles, to organic matter, and to specific microflora. The mobile-immobile concept is used to describe both the water flow and solute transport in transient conditions. This concept of preferential flow is easy to conceive and implement, which uses a limited number of physically based and easily measurable parameters (e.g., the soil volume fraction in which water is mobile) compared to the dual-porosity model (van Dam et al., 1997).

In the laboratory, soil samples are typically brought to saturation first, and relatively long equilibrium times are allowed during the experiment. These conditions suppress the effects of water repellency. Thus, the soil hydraulic functions measured in the laboratory will be denoted as $\theta_{\mathrm{lab}}(h)$ and $K_{\mathrm{lab}}(h)$.

In the field, the bulk field water retention function can be calculated as:

$$\theta_{bulk}(h) = f_m\theta_{\mathrm{lab}} + \left(1 - f_m\right)\theta_{im} \tag{2.53}$$

Meanwhile, the effective field conductivity curve $K(h)$ is related to $K_{\mathrm{lab}}(h)$ as:

$$K(h) = f_m K_{\mathrm{lab}}(h) \tag{2.54}$$

where θ_{im} is the soil water content in the immobile region (cm^3 cm^{-3}). The factor f_m can roughly be estimated by visual observation of dry and wet spots in the field shortly after precipitation, or by calibration processes. It is assumed that the θ_{im} remains constant in time. Therefore, when the immobile regions contain water, variation of f_{m} with h induces exchange of water between the mobile and immobile soil volumes. This exchange is included as an extra loss term $G_w=\theta_{im}\cdot\partial \mathrm{f}_m/\partial t$ in the Richards' equation:

$$\frac{\partial\theta}{\partial t} = \frac{\partial}{\partial z}\left[K(h)\left(\frac{\partial h}{\partial z}+1\right)\right] - S_w(h) + \frac{\partial f_m}{\partial t}\theta_{im} \tag{2.55}$$

where G_w is the amount of water (d^{-1}) transferred from the mobile to the immobile region.

3. Surface water flow and field drainage

3.1 Surface runoff

Surface runoff is regarded as the overland flow that runs off the landscape, and it results in the interaction between soil and surface water system. Surface runoff is commonly related to the part of excessive precipitation that does not infiltrate into the soil and flows over the land surface; besides, when the groundwater level rises above the soil surface, the soil surface also starts to function as a 'drainage medium' generating surface runoff (van Dam et al., 1997). In AHC, the amount of precipitation lost by surface runoff is computed using a modification of the SCS curve number (SCS-CN) method (USDA Soil Conservation Service, 1972) (section 3.1.1). In addition, in case the field is surrounded by soil bunds or ridges, another method based on “surface reservoir boundary condition” is optionally selected to describe the ponding water formation and the related surface runoff, while the SCS-CN subroutine is bypassed (section 3.1.2).

3.1.1 SCS-CN method for surface runoff

The SCS-CN method is an empirical model that came into common use in the mid-1950s. It was developed to provide a consistent basis for estimating the amounts of runoff under varying land use and soil types (Rallison and Miller, 1981). Note that the SCS-CN method is suitable for estimating the surface runoff that occurs along a sloping surface. In addition, the maximum amount that can infiltrate the soil (either as rainfall or irrigation), is however, limited by the saturated hydraulic conductivity of the topsoil layer. Excess water is then considered as lost by surface runoff.

The SCS curve number equation is (USDA Soil Conservation Service, 1972):

$$Q_{\mathrm{surf}} = \frac{\left(R_{\mathrm{day}} - I_{\mathrm{a}}\right)^2}{\left(R_{\mathrm{day}} - I_{\mathrm{a}} + S\right)} \tag{3.1}$$

$$S = 25.4\left(\frac{1000}{CN} - 10\right) \tag{3.2}$$

where Q_{surf} is the accumulated runoff or rainfall excess (mm), R_{day} is the (gross) rainfall depth for the day (mm), I_{a} is the initial abstraction (mm) which includes surface storage, interception (section 3.2), and infiltration prior to runoff, S is the retention parameter (mm) (i.e., potential maximum storage), and CN is the curve number for the day. The initial abstraction, I_a, is commonly approximated as 0.2S, and Eq. (3.1) becomes:

$$Q_{\text{surf}} = \frac{\left(R_{\text{day}} - 0.2S\right)^2}{\left(R_{\text{day}} + S - 0.2S\right)} \tag{3.3}$$

The above equations represent that rain that falls on unsaturated soil infiltrates, increasing the soil water content until the topsoil becomes saturated (P=0.2S), after which additional rainfall becomes surface runoff. A soil with a high Curve Number (CN) will have small potential storage (S) and may lose a large amount of rainfall as runoff (Raes et al., 2012). The curve number is a function of the soil type (mainly soil's permeability), slope, land use, cover, and antecedent soil water conditions (mainly the relative wetness of the topsoil). Especially, the curve number varies non-linearly with the moisture content of the soil. The curve number drops as the soil approaches the wilting point and increases to near 100 as the soil approaches saturation. Water that does not infiltrate becomes surface runoff (Neitsch et al., 2005).

The calculation of the relative wetness of the topsoil extends to a depth of 0.3 meters. In the calculation, the soil water content at the soil surface has a larger weight than the soil water content at 0.3 meters:

$$w_{\text{relat}} = 1 - \frac{\exp^{fd/dx} - 1}{\exp^{f} - 1} \tag{3.4}$$

where w_{relat} is the relative weight factor, f is the shape factor (fixed at -4), d is the soil depth (cm, d=-z), and dx is the maximum depth considered as relevant for adjustment of CN (default = 30 cm).

Antecedent Soil Moisture Condition

SCS defines three antecedent moisture conditions (AMC): I—dry (wilting point), II—average moisture, III—wet (field capacity). The moisture condition I curve number (CN_1) is the lowest value of the daily curve number assumed in dry conditions. The curve number for moisture conditions I and III (CN_2 and CN_3) are calculated with the equations (Neitsch et al., 2005):

$$CN_1 = CN_2 - \frac{20(100 - CN_2)}{\left(100 - CN_2 + \exp\left[2.533 - 0.0636(100 - CN_2)\right]\right)} \tag{3.5}$$

$$CN_3 = CN_2 \cdot \exp\left[0.0636(100 - CN_2)\right] \tag{3.6}$$

In AHC, the specified CN value is the value that belongs to the antecedent moisture class AMC-II (i.e. CN_2). Typical curve numbers for moisture conditions II are listed in Table 3-1 for various land covers and soil types (Rycroft and Smedema, 1983; Cronshey, 1986). These values are appropriate for a 5% slope.

Table 3-1 Indicative CN values for various AMC II and their corresponding values for AMC I (dry) and III (wet) for various infiltration rates

AMC	Soil water content	Infiltration rate (mm/day)			
		>250	250-50	50-10	<10
I	$\theta = \theta_{wp}$	45	56	63	70
II	Average moisture*	65	75	80	85
III	$\theta = \theta_{FC}$	84	88	91	93

Note: * average moisture means that $\theta = \theta_{wp} + \theta_{FC}$

The storage capacity of soil is indeed somewhat larger (i.e. smaller CN value) if it is dry than when it is wet. The retention parameter varies with the soil profile water content, calculated as follows:

$$S = S_{\max}\left(1 - \frac{SW'}{\left[SW' + \exp\left(w_1 - w_2 \cdot SW'\right)\right]}\right) \tag{3.7}$$

where $S_{\max}$ is the maximum value the retention parameter can achieve on any given day (mm), SW' is the soil water content of the entire profile excluding the amount of water held in the profile at wilting point (i.e. *WP*) (mm H_2O), and w_1 and w_2 are the first and second shape coefficients, respectively. The S_{max} value is calculated by solving Eq. (3.2) using CN_1.

The shape coefficients are determined by solving Eq. (3.7), assuming that: (1) the retention parameter for moisture condition I curve number corresponds to wilting point soil profile water content, (2) the retention parameter for moisture condition III curve number corresponds to field capacity soil profile water content, and (3) the soil has a curve number of 99 (S=2.54) when completely saturated.

$$w_1 = \ln\left[\frac{FC'}{1 - S_3/S_{max}} - FC'\right] + w_2 FC' \tag{3.8}$$

$$w_2 = \frac{\ln\left[\frac{FC'}{1 - S_3/S_{max}} - FC'\right] - \ln\left[\frac{SAT'}{1 - 2.54/S_{max}} - SAT'\right]}{SAT' - FC'} \tag{3.9}$$

where SAT' and FC' is the amount of water in the soil profile at saturation and field capacity (excluding *WP*), respectively (mm H_2O), S_3 is the retention parameter for the moisture condition III curve number, and 2.54 is the retention parameter value for a CN of 99.

Slope Adjustments

The default CN_2 value corresponds to be appropriate for 5% slopes. Williams (1995) developed an equation to adjust the curve number to a different slope:

$$CN_{2s} = \frac{(CN_3 - CN_2)}{3} \cdot \left[1 - 2 \cdot \exp(-13.86 \cdot slp)\right] + CN_2 \tag{3.10}$$

where CN_{2s} is the moisture condition II curve number adjusted for slope, *slp* is the average fraction slope of the ground surface, and CN_2 and CN_3 are the I-dry and II-wet curve number for the default 5% slope. AHC does not adjust curve numbers for slope. If the user wishes to adjust the curve numbers for slope effects, the adjustment must be done prior to entering the curve numbers in the management input file (Neitsch et al., 2005).

The SCS-CN method is also applied to calculate surface runoff due to excessive irrigation under sprinkler irrigation. After the subtraction of surface runoff, the remaining part of the rainfall and irrigation water (i.e., infiltration water) is allowed to infiltrate into the soil profile (Steduto et al., 2012). Further, the maximum amount that can infiltrate the soil is, however, still limited by the saturated hydraulic conductivity of the topsoil layer. This means that the part of the above infiltration water that does not infiltrate into the soil is also considered lost as another part of surface runoff (calculated by Eq. 3.11, with a defaulted value z_{sill}=0).

3.1.2 Ponding water and the related surface runoff

Another situation is that field practices (plowing practices, ridges, or soil bunds) might limit or prevent soil surface runoff. *In case the field is surrounded by soil bunds, the abovementioned SCS-CN runoff subroutine is bypassed* (i.e., Q_{surf}=0). Water that cannot infiltrate as a result of excessive rainfall or irrigation will be stored between the bunds, which results in a ponding reservoir. This ponding reservoir may be regarded as a thin layer of water on top of the soil surface, which can store water to a certain maximum (e.g., the storage capacity or ponding depth is limited by the height of the bund). In either case, the groundwater level may rise above the soil surface and fill the ponding reservoir until a certain threshold ponding level. Ponding water that overtops the maximum ponding height will be assumed to be lost by surface runoff (Steduto et al., 2012). Thus, in case the ponding depth is greater than a threshold value, the surface runoff is supposed to happen and estimated as follows:

$$q_{\mathrm{runoff}} = \frac{1}{\gamma_{\mathrm{sill}}}\left(h_{\mathrm{pond}} - z_{\mathrm{sill}}\right)^{\beta_{\mathrm{sill}}} \tag{3.11}$$

where q_{runoff} is the surface runoff from surface ponding water layer (cm d^{-1}), z_{sill} is the maximum ponding height below which no surface runoff occurrs (cm), γ_{sill} is the drainage resistance of surface runoff (d), and β_{sill} is an exponent (-). z_{sill} is determined by the irregularities of the soil surface. As surface runoff is a rapid process, the γ_{sill} will typically have values of less than 1 d (e.g. with a rough estimation of 0.1 d). z_{sill}=0 cm means that surface runoff will immediately occur when water overtops the ground surface. When the dynamics of surface runoff are relevant, the values of γ_{sill} and β_{sill} might be derived from experimental data or a hydraulic model of soil surface flow (Kroes and van Dam, 2003).

3.2 Interception of rainfall

Not all the precipitation will reach the surface, and a fraction of rainfall will be intercepted by the canopy (leaves, stems, etc). Canopy interception reduces the erosive energy of droplets and traps a portion of rainfall within the canopy, which is affected by the density of plant cover and the morphology of the plant species (see Eq. 3.12). An empirically general formula, proposed by von Hoyningen-Hüne (1983) and Braden (1985) (based on measured interception of precipitation for various crops), is applied to calculate canopy interception, as follows:

$$R_{\text{inter}} = a \cdot LAI \left(1 - \frac{1}{1 + \dfrac{b \cdot R_{\text{gross}}}{a \cdot LAI}} \right) \tag{3.12}$$

where R_{inter} is intercepted rainfall (cm), R_{gross} is gross rainfall (cm, $=R_{day}/10$), a is an empirical constant (cm), and b is the soil cover fraction (b=LAI/3.0) (-). This means that for increasing rainfall amounts, the R_{inter} asymptotically reaches the maximum (or saturation) amount $a{\cdot}LAI$. The parameter a is specified by the user, which must be determined experimentally in principle; and we may assume a = 0.25 in the case of ordinary crops (if no data is available). Note that the method of Von Hoyningen-Hüne and Braden is based on daily precipitation values. In addition, when irrigation water is applied through sprinklers, the interception by canopy should be also considered and calculated similarly to that for rainfall by Eq. 3.12. whilst, the solute concentration for the two water sources (rainfall and irrigation water) is generally different (van Dam et al., 1997).

3.3 Field drainage

In section 3.3.3 the bottom boundary has been described. In addition to the bottom flux (q_{bot}, cm d^{-1}), which accounts for regional drainage or seepage fluxes, it is possible to define lateral field drainage fluxes (q_{dr}, cm d^{-1}) to the local drainage system. The drainage flux is incorporated in the numerical solution to the Richards Equation by specifying it as another system-dependent boundary condition. The general formulation of the drainage equation is:

$$q_{dr} = \frac{h_{dr}}{\gamma_{drain}} = \frac{\phi_{gwl} - \phi_{drain}}{\gamma_{drain}} \quad \text{with } h_{dr} = \phi_{gwl} - \phi_{drain} \tag{3.13}$$

where the drainage base ϕ_{drain} is equal to the surface water level in the drainage ditch (cm below the soil surface), ϕ_{gwl} is the groundwater level midpoint between the drains or ditches (cm below the soil surface), γ_{drain} is the drainage resistance (d), and h_{dr} is the water-table height above the drain at the midpoint between the drains (cm), i.e., the hydraulic head difference needed for calculating subsurface flow into the drains.

q_{dr} can be calculated by the following three methods: (l) linear or tabulas $q_{dr}(\phi_{gwl})$ relation (section 3.3.1); (2) drainage equations of Hooghoudt and Ernst (section 3.3.2); and (3) drainage/infiltration to/from surface water systems (section 3.3.3).

In AHC, the drainage flux can be applied to the soil column with two options: (1) being partitioned over the saturated layers, or (2) leaving the model at the bottom compartment. The vertical distribution of drainage becomes important in the case of solute transport and deep, non-uniform soils, and the first option (i.e., vertical partition) should be considered. The second option is realistic if only water flow is simulated or a relatively small section of a deeply saturated profile is simulated (Kroes et al., 1999).

3.3.1 Field scale drainage based on tabular relation

In this option, lateral field drainage flux to the drainage system is calculated using a tabulated relationship between q_{dr} and ϕ_{gwl}. Tabular values of q_{dr} as a function of ϕ_{gwl} are required to be specified. One should specify a number of (ϕ_{gwl}, q_{dr}) data pairs. A linear relation is applied if only providing two data-pair suffices. Specifying more data pairs can reduce the inaccuracy which results from this type of linearization.

3.3.2 Drainage equations of Hooghoudt and Ernst: field-scale

The drainage equation of Hooghoudt and Ernst allows the evaluation of drainage design. The two equations are both steady-state drainage equations. They are derived based on the assumption that the drain discharge equals the recharge to the groundwater, and consequently that the water table remains in the same position. In irrigated areas or areas with highly variable rainfall, these assumptions are not met, and unsteady-state equations should be considered.

In this section, we discuss the flow of groundwater to parallel field drains under steady-state conditions. This is the typical situation in areas with a humid climate and prolonged periods of fairly uniform, medium-intensity rainfall, or areas with uniform, certain frequency irrigations. Two types of steady-state equations are applied in the model, i.e., the Hooghoudt equation and the Ernst equation. The theory behind these equations is clearly described in Ritzema (2006), and here we provide a brief introduction to their derivation.

(1) ***The Hooghoudt Equation***

Consider a steady-state flow to vertically-walled open drains reaching an impervious layer (Fig. 3.1). According to the Dupuit-Forchheimer theory, the Darcy's Equation can be applied to describe the flow (q_x) of groundwater through a vertical plane at a distance (x) from the ditch:

$$q_x = K_p \cdot y \cdot \frac{dy}{dx} \tag{3.14}$$

where q_x is the unit flow rate in the x-direction (cm^2 d^{-1}), K_{hp} is the horizontal saturated hydraulic conductivity of the soil above the drainage basis (cm d^{-1}), y is the height of the water table at x (cm), and dy/dx is the hydraulic gradient at x (-).

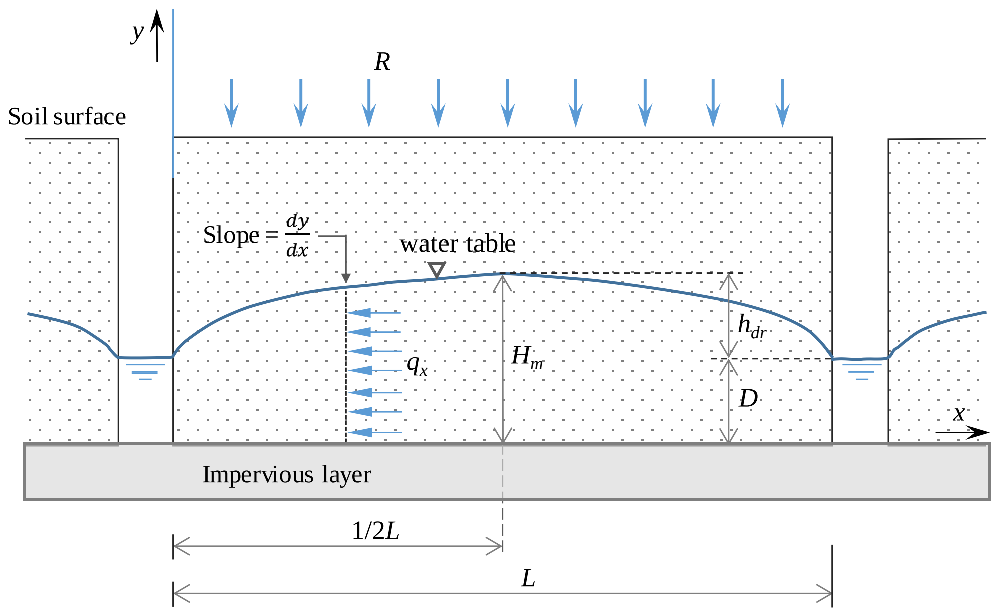


Figure 3.1 Flow to vertically-walled drains reaching the impervious layer

Under steady-state conditions, the continuity principle requires that all the water entering the soil in the surface area midway between the drains and the vertical plane (y) at distance (x) must pass through this plane on its way to the drain. If R is the rate of recharge per unit area, then the flow per unit time through the plane (y) is:

$$q_x = R\left(\frac{1}{2}L_{dr} - x\right) \tag{3.15}$$

where R is the rate of recharge per unit surface area (cm d^{-1}), and L_{dr} is drain spacing (cm). Since the flow in the two cases must be equal, we can equate the right side of Eqs. (3.14) and (3.15) :

$$K_{hp} \cdot y \cdot \frac{dy}{dx} = R\left(\frac{1}{2}L_{dr} - x\right) \tag{3.16}$$

which can also be written as:

$$K_{hp} \cdot y \cdot dy = R\left(\frac{1}{2}L_{dr} - x\right)dx \tag{3.17}$$

The limits of integration of this differential equation are: for $x = 0 \rightarrow y = D$ and for $x = 1/2L \rightarrow y = H_m$. D is the elevation of the water level in the drain (cm), and H_m is the elevation of the water table midway between the drains (cm). Integrating the differential equation and substituting the limit yields:

$$q_{dr} = R = \frac{4K_{hp}\left(H_m{}^2 - D^2\right)}{L_{dr}{}^2} \tag{3.18}$$

where q_{dr} is the drain discharge flux (cm d^{-1}). Since it follows that H_m-D=h_{dr}, this equation could be rewritten as:

$$q_{dr} = \frac{8K_{hp}Dh_{dr} + 4K_{hp}h_{dr}{}^2}{L_{dr}{}^2} \tag{3.19}$$

where h_{dr} is the height of the water table (at the midpoint between the drains) above the water level in the drain (cm). This equation is the most widely known form of **Hooghoudt's equation**. If the water level in the drain is very low, or the drain line level just coincides with the top of the impervious layer (i.e., D=0 m), the equation changes into:

$$q_{dr} = \frac{4K_{hp}h_{dr}{}^2}{L_{dr}{}^2} \tag{3.20}$$

If the impervious layer is far below drain level (D>>h), the second term in the enumerator of Eq. (3.19) can be neglected, giving:

$$q_{dr} = \frac{8K_{hp}Dh_{dr}}{L_{dr}{}^2} \tag{3.21}$$

This equation describes the flow below the drain level, neglecting the flow above the drain level.

If the soil profile consists of two layers with different hydraulic conductivities, and if the drain level is at the interface between the soil layers (e.g., case 3 in Fig. 3.4), Eq. (3.19) may be written as:

$$q_{dr} = \frac{8K_{hbot}Dh_{dr} + 4K_{htop}h_{dr}{}^2}{L_{dr}{}^2} \tag{3.22}$$

where K_{htop} is the horizontal hydraulic conductivity of the layer above the drain level (cm d^{-1}), and K_{hbot} is the horizontal hydraulic conductivity of the layer below the drain level (cm d^{-1}). This situation is quite common with the soil above drain level often being more permeable than the below drain level because the soil structure above drain level has been improved by many factors (e.g., formation of cracks and presence of roots and microbial).

If the pipe or open drains do not reach the imperious layer, the flow lines will converge towards the drain and will thus no longer be horizontal. This will result in longer, converging flow lines and extra head loss, and thus a higher water table. Hooghoudt (1940) introduced an equivalent depth (D_e, cm) to incorporate this extra head loss near the drains, assuming an imaginary impervious layer above the real one. Hence, Eq. (3.21) becomes:

$$q_{dr} = \frac{8K_{hp}D_e h_{dr} + 4K_{hp}{h_{dr}}^2}{{L_{dr}}^2} \tag{3.23}$$

and the equivalent depth equals:

$$D_e = \frac{\pi L_{dr}}{8\left(\ln \frac{L}{\pi r_0} + F(x)\right)} \tag{3.24}$$

where a typical length variable x is used:

$$x = \frac{2\pi D}{L_{dr}} = \frac{2\pi\left(\phi_{drain} - z_{imp}\right)}{L_{dr}} \tag{3.25}$$

and the function $F(x)$:

$$F(x) = 2\sum_{j=1}^{\infty} \ln\coth\left(jx\right) \tag{3.26}$$

If $x<10^{-6}$, then D_e is equal to D. If $10^{-6}<x<0.5$, then a comparison with Dagan’s formula results in an approximation that is highly accurate:

$$F(x) = \frac{\pi^2}{4x} + \ln\frac{x}{2\pi} \tag{3.27}$$

If $x>0.5$, then $F(x)$, which represents an infinite series of logarithms, can be modified to:

$$F(x)=\sum_{j=1,3,5}^{\infty}\frac{4e^{-2jx}}{j\left(1-e^{-2jx}\right)} \tag{3.28}$$

Therefore, the Hooghoudt Equation can be applied for (1) a homogenous soil profile above the impervious layer, or (2) a two-layered soil profile above the impervious layer, provided that the interface between the two layers coincides with the drain line level (Ritzema, 2006).

In addition, there are two other assumptions for Hooghoudt's equation: (1) the drains are running half-full, and (2) the drains have no entrance resistance. These assumptions imply that the entrance area, u, equals the wet perimeter of a semi-circle (the πr_0 in Eq. 3.24), so that:

$$r_0=\frac{u}{\pi} \tag{3.29}$$

where r_0 is the radius of the drain, and u is the wet perimeter (cm). For open drains, the equivalent radius (r_0) can be calculated by substituting the wet perimeter of the open drain for u in Eq. (3.29). For pipe drains laid in trenches, the wet perimeter is taken as:

$$u=b+2r_0 \tag{3.30}$$

where b is the width of the trench (cm). If an envelope material is used around the pipe drain (Fig. 3.2), Eq. (3.30) changes to

$$u=b+2\left(2r_0+m_{ev}\right) \tag{3.31}$$

where m_{ev} is the height of the envelope above the drain (cm).

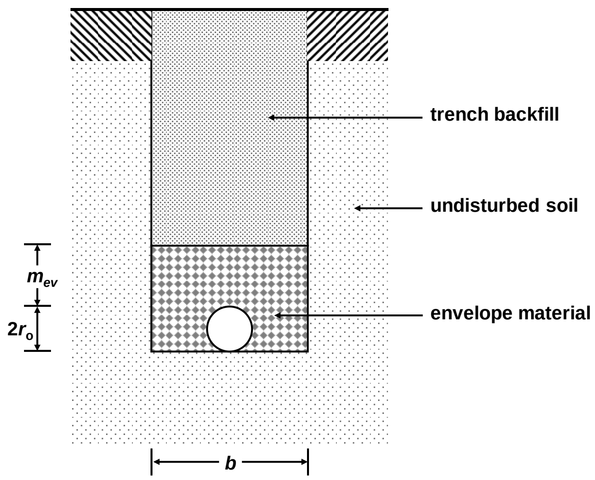


Figure 3.2 Drain pipe with gravel envelope in drain trench

We know that the entrance head loss (h_{entr}, cm) will be caused when water flows from the soil outside the pipe/ditch into the pipes or ditches (e.g. when passing through the envelope material). The second assumption implies that we are assuming an ideal drain (i.e., with no entrance resistance). In actual, the entrance resistance would be very small and could be negligible, if the hydraulic conductivity of the drain trench is at least 10 times higher than that of the undisturbed soil outside the trench (Rycroft and Smedema, 1983). If the hydraulic conductivity is less, an envelope material can be used to decrease the entrance resistance, so that a greater part of the total head will be available for the flow through the soil (Ritzema, 1994). However, if it is not possible to use an envelope material, the entrance head loss should be considered:

$$h_{entr} = q_{dr}\gamma_{entr} \tag{3.32}$$

The equation of Hooghoudt can be adjusted to take into account the entrance resistance to the flow of groundwater around the drain. The limits of integration of Eq. (3.17) are changed into: for $x = 0 \rightarrow y = D + h_e$ and $x = 1/2L \rightarrow y = H_m$. Then referring to the previous derivation method for Eq. (3.23), we can obtain the equation:

$$q_{dr} = R = \frac{8K_{hp}\left(D_e + h_e\right)h_h + 4K_{hp}h_h^2}{L_{dr}^{\ 2}} \tag{3.33}$$

where h_e is the entrance head (cm) and equal to the height of the water table just above the drain level, measured from the center of the drain, and h_h is the value of h_{dr} reduced with the entrance head (cm). This equation could be rewritten as:

$$h_h = \frac{q_{dr}L_{dr}^{\ 2}}{8K_{hp}\left(D_e + h_e\right) + 4K_{hp}h_h} = \frac{q_{dr}L_{dr}^{\ 2}}{8K_{hp}\left(D_e + h_e\right) + 4K_{hp}\left(h_{dr} - h_e\right)} \tag{3.34}$$

When considering the entrance head loss, the total available head h_{dr} can be divided into a head loss caused by the horizontal flow (h_h) and the entrance head loss (h_r), i.e., $h_{dr} = h_h + h_{entr}$.

Substituting Eqs. (3.32) and (3.34) into this equation results in:

$$h_{dr} = \frac{q_{dr} L_{dr}^{\ 2}}{8K_{hp}\left(D_e + h_e\right) + 4K_{hp}\left(h_{dr} - h_e\right)} + q_{dr}\gamma_{entr} = q_{dr}\left(\frac{L_{dr}^{\ 2}}{8K_{hp}\left(D_e + h_e\right) + 4K_{hp}\left(h_{dr} - h_e\right)} + \gamma_{entr}\right) \quad (3.35)$$

Note that in drainage evaluation, the h_e can be calculated by $h_e - r_0 = \frac{q_{dr} L_{dr}}{2\pi K} \ln\left(\frac{H_e}{r_0}\right)$.

Hence, when using the Hooghoudt Equation, the calculation formula for drainage resistance γ_{drain} in Eq. (3.13) can be expressed as:

$$\gamma_{drain} = \frac{L_{dr}^{\ 2}}{8K_{hp}\left(D_e + h_e\right) + 4K_{hp}\left(h_{dr} - h_e\right)} + \gamma_{entr} \quad (3.36)$$

If we assume that h_e is very small, this equation may be written as:

$$\gamma_{drain} = \frac{L_{dr}^{\ 2}}{8K_{hp} D_e + 4K_{hp} h_{dr}} + \gamma_{entr} \quad (3.37)$$

(2) *The Ernst Equation*

Ernst divided the flow to the drains into a vertical, a horizontal, and a radial component (Fig. 3.3). Consequently, the total available head h_{dr} can be divided into a head loss caused by the vertical flow (h_v), the horizontal flow (h_h), and the radial flow (h_r).

$$h_{dr} = h_v + h_h + h_r \quad (3.38)$$

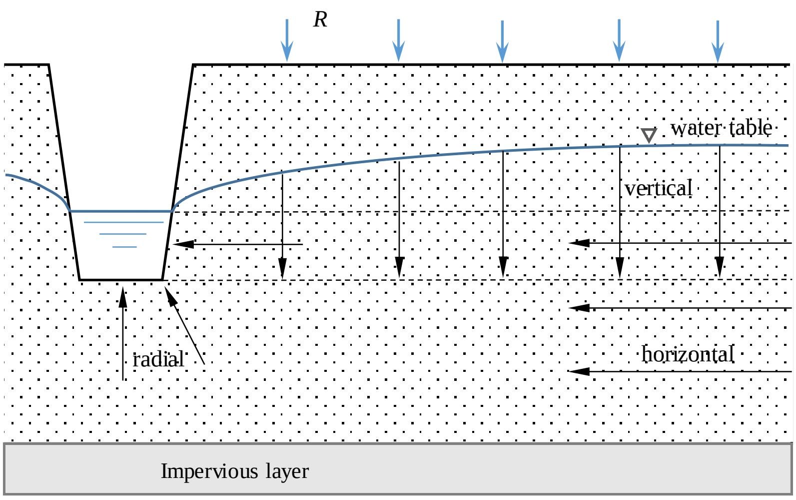


Figure 3.3 Geometry of two-dimensional flow toward drains, according to Ernst

Vertical flow:

Vertical flow is assumed to take place in the layer between the water table and drain level (Fig. 3.4). We can obtain the head loss caused by this vertical flow by applying Darcy's Law:

$$q_{dr} = K_v \frac{h_v}{D_v} = \frac{h_v}{\frac{D_1}{K_1} + \frac{D_2}{K_2}} = \frac{h_v}{\frac{\phi_{gwl} - z_{\text{int}}}{K_{vtop}} + \frac{z_{\text{int}} - \phi_{drain}}{K_{vbot}}} \quad \text{or} \quad h_v = q_{dr} \frac{D_v}{K_v} = q_{dr}\gamma_v \tag{3.39}$$

$$\frac{D_v}{K_v} = \gamma_v = \frac{\phi_{gwl} - z_{\text{int}}}{K_1} + \frac{z_{\text{int}} - \phi_{drain}}{K_2} \tag{3.40}$$

where D_v is the thickness of the layer in which vertical flow is considered (cm), and K_v is the vertical hydraulic conductivity (cm d^{-1}).

Horizontal flow:

The horizontal flow is assumed to take place below the drain level (Fig. 3.3). Based on the simplified version of Hooghoudt equation (Eq. 3.21), the horizontal loss h_h can be described by:

$$h_h = q_{dr} \frac{L_{dr}^2}{8\sum(KD)_h} = q_{dr} \frac{L_{dr}^2}{8K_{htop}D_{top} + 8K_{hbot}D_{bot}} \tag{3.41}$$

where $\sum(KD)_h$ is the transmissivity of the soil layers through which the water flows horizontally (cm^2 d^{-1}).

Radial flow:

The radial flow is also assumed to take place below the drain level (Fig. 3.3). The head loss caused by the radial flow can be expressed as:

$$h_r = q_{dr} \frac{L_{dr}}{\pi K_{rad}} \ln \frac{g_{dr} D_r}{u} \tag{3.42}$$

where K_{rad} is the radial hydraulic conductivity of the drain (cm d^{-1}), g_{dr} is the geometry factor of the radial resistance (-), and D_r is the thickness of the layer in which the radial flow is considered (cm) (i.e. the contributing layer below the drain level). This equation has the same restriction for the depth of the impervious layer as the equation for horizontal flow (i.e. D_r< ¼

L_{dr}). This means that D_r is calculated as the minimum of (ϕ_{drain} - z_{imp}) and ¼ L_{dr}, where z_{imp} (cm) is the depth of the impervious layer

Therefore, substituting the Eqs. (3.39), (3.41), and (3.42) into Eq. (3.38), the expression for h_{dr} can be rewritten as:

$$\begin{aligned} h_{dr} &= q_{dr}\frac{D_v}{K_v} + q_{dr}\frac{L_{dr}^2}{8\sum(KD)_h} + q_{dr}\frac{L_{dr}}{\pi K_r}\ln\frac{g_{dr}D_r}{u} \\ &= q_{dr}\left(\frac{D_v}{K_v} + \frac{L_{dr}^2}{8\sum(KD)_h} + \frac{L_{dr}}{\pi K_r}\ln\frac{g_{dr}D_r}{u}\right) \end{aligned} \tag{3.43}$$

This equation is generally known as the **Ernst Equation**. The Ernst equation is derived for a heterogeneous (i.e., any type of two-layered) soil profile above the impervious layer. In addition, if the entrance head loss (i.e., $h_{entr} = q\gamma_{entr}$) is further considered as a part of h_{dr}:

$$h_{dr} = h_v + h_h + h_r + h_{entr} \tag{3.44}$$

Then referring to the derivation of Eq. (3.43), we can obtain the equation including entrance head loss:

$$h_{dr} = q_{dr}\left(\frac{D_v}{K_v} + \frac{L_{dr}^2}{8\sum(KD)_h} + \frac{L_{dr}}{\pi K_r}\ln\frac{g_{dr}D_r}{u} + \gamma_{entr}\right) \tag{3.45}$$

Hence, when applying the Ernst Equation, the calculation formula for drainage resistance γ_{drain} (in Eq. 3.13) can be expressed as:

$$\gamma_{drain} = \frac{D_v}{K_v} + \frac{L_{dr}^2}{8\sum(KD)_h} + \frac{L_{dr}}{\pi K_r}\ln\frac{g_{dr}D_r}{u} + \gamma_{entr} \tag{3.46}$$

(3) ***Five typical drainage situations in the model***

Five typical drainage situations are distinguished (see Fig. 3.4). For each drainage situation the drainage resistance γ_{drain} in Eq. (3.13) can be defined according to the Hooghoudt Equation or the Ernst Equation). The Hooghoudt Equation is applied for a homogenous soil profile above the impervious layer (cases 1 and 2), and a two-layered soil profile above the impervious layer when the interface between the two layers coincides with the drain line level (case 3); whilst, the Ernst equation is used for a heterogeneous (two-layered) soil profile above the impervious layer (cases 4 and 5).

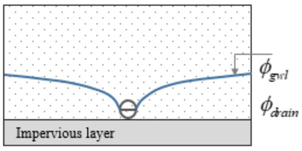
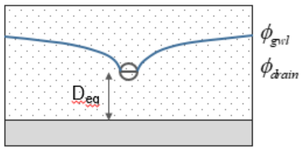
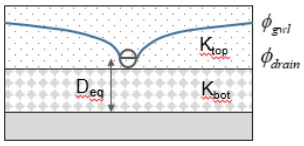
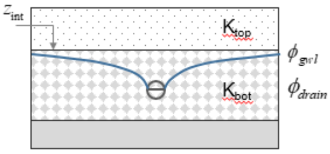
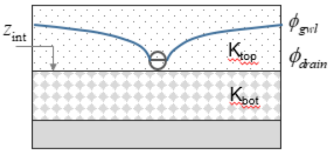
Case	Schematization	Soil profile	Drain position	Theory
Case 1		Homogeneous	Drain on top of impervious layer	Hooghoudt, Donnan
Case 2		Homogeneous profile	Drain above impervious layer	Hooghoudt with equivalent depth
Case 3		Two layers	Drain at interface between both soil layers	Hooghoudt
Case 4		Two layers (K_{top}<K_{bot})	Drain in bottom layer	Ernst
Case 5		Two layers (K_{top}<K_{bot})	Drain in top layer	Ernst

Figure 3.4 Five field (steady-state) drainage situations considered in AHC (adapted from Ritzema (2006)). The hydraulic head ϕ_{gwl} is defined as positive upward with ϕ = 0 at the soil surface, and K_{top} and K_{bot} are the saturated hydraulic conductivity of the fine layer and coarse layer, respectively.

Case 1: Homogeneous profile, drain on top of an impervious layer

The drain discharge q_{dr} can be calculated with Eq. (3.20) (with D equaling zero). Thus, based on Eq. (3.37), the drainage resistance is calculated as:

$$\gamma_{drain} = \frac{L_{dr}^{\ 2}}{4K_{hp}h_{dr}} + \gamma_{entr} \qquad (3.47)$$

with $h_{dr} = \phi_{gwl} - \phi_{drain}$.

Case 2: Homogeneous profile, drain above impervious layer

This drainage situation corresponds to that described by the general form of the Hooghoudt Equation, where q_{dr} can be calculated with Eq. (3.23) or (3.35). Referring to Eq. (3.37), the drainage resistance follows from:

$$\gamma_{drain} = \frac{L_{dr}^{\ 2}}{8K_{hp}D_e + 4K_{hp}h_{dr}} + \gamma_{entr} \tag{3.48}$$

Case 3: Heterogeneous soil profile, drain at the interface between both soil layers

Referring to Eq. (3.37), the drainage resistance follows from:

$$\gamma_{drain} = \frac{L_{dr}^{\ 2}}{8K_{hbot}D_e + 4K_{htop}h_{dr}} + \gamma_{entr} \tag{3.49}$$

where K_{hbot} and K_{htop} are the horizontal saturated hydraulic conductivity (cm d^{-1}) of the lower and upper soil layer, respectively.

Case 4: Heterogeneous soil profile, drain in bottom layer

Based on Darcy's equation applied for the vertical flow through the layered soils, the term K_v/D_v in Eq. (3.39) can be expressed as:

$$\frac{K_v}{D_v} = \frac{1}{\dfrac{D_1}{K_1} + \dfrac{D_2}{K_2}} = \frac{1}{\dfrac{\phi_{gwl} - z_{\mathrm{int}}}{K_{vtop}} + \dfrac{z_{\mathrm{int}} - \phi_{drain}}{K_{vbot}}} \tag{3.50}$$

In addition,

$$K_{rad} = \sqrt{K_{htop}K_{vtop}} \tag{3.51}$$

$$D_r = D_{bot} = \min\left(\phi_{drain} - z_{imp}, \tfrac{1}{4} L_{\mathrm{dr}}\right) \tag{3.52}$$

The horizontal drain flow contributed from the top fine layer is very small and can be neglected:

$$\sum (KD)_h = K_{htop}D_{top} + K_{hbot}D_{bot} \approx K_{hbot}D_{bot} \tag{3.53}$$

Substituting the above equations into Eq. (3.46), the drainage resistance can be estimated as follows:

$$\gamma_{drain} = \left(\frac{\phi_{gwl} - \zeta_{\text{int}}}{K_{htop}} + \frac{\zeta_{\text{int}} - \phi_{drain}}{K_{vtop}} \right) + \frac{L_{dr}^{\ 2}}{8K_{hbot}D_{bot}} + \frac{L_{dr}}{\pi\sqrt{K_{hbot}K_{vbot}}} \ln\left(\frac{D_{bot}}{u_{drain}} \right) + \gamma_{entr} \qquad (3.54)$$

Case 5: Heterogeneous soil profile, drain in the top layer

Again the approach of Ernst (1956) is applied (Eq. 3.46), in which:

$$D_r = D_{top} = \phi_{drain} - z_{\text{int}} \qquad (3.55)$$

with z_{int} the level of the transition (cm) between the upper and lower soil layer. Then integrating the Eqs. (3.50)-(3.53) into Eq. (3.46), the drainage resistance follows from:

$$\gamma_{drain} = \frac{\phi_{gwl} - \phi_{drain}}{K_{vtop}} + \frac{L_{dr}^{\ 2}}{8K_{htop}D_{top} + 8K_{hbot}D_{bot}} + \frac{L_{dr}}{\pi\sqrt{K_{hbot}K_{vbot}}} \ln\left(g_{dr} \frac{\phi_{drain} - z_{\text{int}}}{u_{drain}} \right) + \gamma_{entr} \qquad (3.56)$$

3.3.3 General aspects of regional scale drainage

A simple, basic interaction between groundwater and a maximum of 5 surface water systems may be simulated. The drainage/infiltration (q_{dr}, cm d^{-1}) to/from each surface water system *i* is calculated as:

$$q_{dr,i} = \frac{\phi_{gwl} - \phi_{drain,i}}{\gamma_{drain,i}} \qquad (3.57)$$

where the drainage base $\phi_{drain,i}$ is equal to the surface water level of system *i* (cm below the soil surface), ϕ_{gwl} is the groundwater level (cm below the soil surface), $\gamma_{drain,i}$ is the drainage or infiltration resistance from system *i* (d). $\gamma_{drain,i}$ can be simply specified as a constant value.

3.4 The snowpack

In case of snowfall, the water accumulates in a snowpack. The water will be released by snowmelt, during which a large volume of water becomes available for runoff or infiltration into the soil. There are various ways existing to consider the snow conditions, from a delay in precipitation to a complete water and energy balance of the snowpack. In AHC, a relatively simple way is employed and adapted from SWAP v3.03 (Kroes and van Dam, 2003), since it requires just the daily weather data. With the option to calculate snow accumulation and snowmelt for each day, the water balance of the snowpack on the soil surface will be calculated. This balance consists of several fluxes and a storage change in the snow layer (Fig. 3.5).

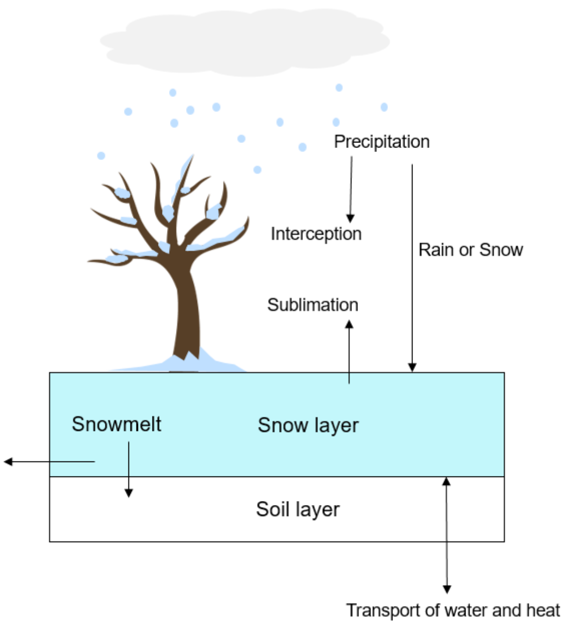


Figure 3.5 The water fluxes to and from the snow layer (adapted from Kroes and van Dam (2003))

The incoming fluxes are the rain and snowfall. The outgoing fluxes are the snowmelt and sublimation. The snowmelt q_{melt} (cm d^{-1}) is calculated when the air temperature rises above 0 °C (Kustas et al., 1994) with:

$$q_{melt} = C\left(T_{av} - T_s\right) \tag{3.58}$$

where C is a constant which can be specified by the user (default value = 0.2) (d °C cm^{-1}), T_{av} is the daily average air temperature (°C) and T_s is the temperature of the snow (°C). The assumption is made that the maximum snow temperature is 0 °C when the air temperature is above 0 °C.

In case of rainfall on the snowpack P_r (cm d^{-1}) additional melt will occur due to heat released by splashing raindrops. This amount of snowmelt $q_{melt,r}$, (cm d^{-1}) is calculated with (Fernández, 1998; Singh et al., 1997):

$$q_{melt,r} = \frac{P_r C_{ww} \left(T_{av} - T_s \right)}{L_m} \tag{3.59}$$

where C_{ww} is the specific heat of water (4180 J kg^{-1} K^{-1}) and L_m is the latent heat of melting (333580 J kg^{-1}). The melt fluxes leave the snowpack as runoff or infiltrate into the soil.

The snow can also evaporate directly into the air, a process called sublimation. The sublimation rate is taken equal to the potential evaporation rate (section 7.1 and section 7.5). When a snowpack exists, the evapotranspiration from the soil and vegetation is set to zero.

The snow storage (S_{snow}, cm) is calculated as the storage of the previous day plus the precipitation (P_r and P_s) minus the melt (q_{melt} and $q_{melt,r}$) and sublimation (E_s) amounts:

$$S_{snow}^{t+1} = S_{snow}^{t+1} + \left(P_r + P_s - q_{melt} - q_{melt,r} - E_s \right) \Delta t \tag{3.60}$$

In addition, in AHC, when there is ponding water on the soil surface, we assume that if the avarege daily air temperature is less than 0.5 °C and surface soil temperature is below 0 °C, all the ponding water will be frozen into ice. This amount of ice is added directly to the snow storage (S_{snow}) to simplify the calculation.

4. Solute transport

4.1 Governing solute transport equations

There are three principal mechanisms/modes of solute movement in soil water: convection, diffusion, and dispersion. The bulk transport of solutes occurs when solutes are carried along with the moving soil water. A convection flux of solute J_{con} (g cm^{-2} d^{-1}), carried by the mass flow of soil water (also called *Darcian flow*), is proportional to the concentration *c*:

$$J_c = qc = -c \cdot K\left(\frac{dH}{dz}\right) \tag{4.1}$$

where *c* is the solute concentration in the soil water (liquid phase) (g cm^{-3}), and *q* is the Darcy flux (cm d^{-1}) discussed in section 2.1. *q* is expressed as the volume of liquid flowing through a unit area (perpendicular to the flow direction) per unit time.

Diffusion is solute transport which is caused by the solute gradient, in consequence of the random thermal motion (often called *Brownian motion*). Solute molecules tend to diffuse from zones where their concentration is higher to where it is lower. In bulk water at rest, the rate of diffusion J_d is related by Fick's first law to the gradient of the concentration *c* (Hillel, 1998):

$$J_d = -D_w \frac{\partial c}{\partial z} \tag{4.2}$$

where D_w is the diffusion coefficient for a particular solute diffusing in bulk (free) water (cm^2 d^{-1}), and $\partial c/\partial z$ is that solute's effective concentration gradient. For diffusion in the soil, if the sole factors affecting the diffusion coefficient are fractional water volume and tortuosity, the solute flux J_{dif} can be generally described by the Fick's first law as:

$$J_{dif} = -\theta D_{dif} \frac{\partial c}{\partial z} \tag{4.3}$$

in which D_{dif} is the diffusion coefficient in the soil (cm^2 d^{-1}). For diffusion in the soil liquid phase, the effective diffusion coefficient is generally less than the diffusion coefficient in free water (D_w). There are two main factors: (1) the liquid phase occupies only a fraction of the soil volume, and (2) the soil's pore passages are tortuous so that the actual path length of diffusion is significantly greater than the apparent straight-line distance. Thus, D_{dif} is estimated by the approach proposed by Millington and Quirk (1961):

$$D_{dif} = D_w \frac{\theta^{7/3}}{\phi_{\mathrm{por}}^2} \frac{T_k}{273.16} \tag{4.4}$$

where D_w is the diffusion coefficient of a given solute in free water at 0°C (cm^2 d^{-1}), and ϕ_{por} is the soil porosity (cm^3 cm^{-3}).

Dispersion is the solute transport that results from the microscopic non-uniformity of flow velocity inside the soil's conducting pores. Because water moves faster through wide than through narrow pores and faster in the center of each pore than its walls. Some portions of a flowing solution move faster than others, causing an incoming solution to mix with or disperse within an antecedent solution. The degree of solution mixing is affected by the average flow velocity, pore size distribution, degree of saturation, and concentration gradient. The dispersion flux J_{dis} (g cm^{-2} d^{-1}) is proportional to the solute concentration gradient (Hillel, 1998; Bear, 1972):

$$J_{dis} = -\theta D_{dis} \frac{\partial c}{\partial z} \tag{4.5}$$

with D_{dis} the dispersion coefficient (cm^2 d^{-1}). Under laminar flow conditions, D_{dis} has been found more or less linearly on the pore water velocity (Bolt, 1979):

$$D_{dis} = L_{dis} v \tag{4.6}$$

where L_{dis} is the dispersion length (cm), and v (m s^{-1}) is the pore water velocity (i.e., average apparent velocity, taken as the straight-line length of path traversed within the soil in unit time, with $v = q/\theta$). v represents the average speed at which water moves through the interconnected pore spaces of a porous medium. Unless the water is flowing very slowly through repacked soil, the relative effect of hydrodynamic dispersion is usually much larger than that of molecular diffusion. When the soil solution is at rest, the hydrodynamic dispersion effect does not come into play at all (Hillel, 1998). L_{dis} has typical values of 0.5-2.0 cm in packed laboratory columns and 5-20 cm in the field, in spite of that they could be considerably larger in groundwater transport on a regional scale (Jury et al., 1991).

Therefore, to take into account the above three mechanisms of solute movement, the total solute flux J (g cm^{-2} d^{-1}) is described by the combination of Eqs. (4.1), (4.3) and (4.5):

$$J = J_c + J_{dif} + J_{dis} = qc - \theta\left(D_{dif} + D_{dis}\right)\frac{\partial c}{\partial z} \tag{4.7}$$

The rate of change of the solute mass in a volume element of soil should equal the difference among the incoming fluxes, outgoing fluxes, and sinks/sources. The sinks/sources refer to the solute that is generated within the soil (e.g. nitrate ion from nitrification of organic matter) or disappears from the soil volume (e.g. nitrate being taken up by plants). In addition, we may consider the existence of a dynamic storage for the solute adsorbed by the soil solids. Invoking the conservation principle, the continuity equation for transient-state solute transport can be expressed as:

$$\frac{\partial\left(\theta c + \rho_b Q\right)}{\partial t} = -\frac{\partial J}{\partial z} - S_s \tag{4.8}$$

where the term on the left side refers to the change rate of the total solute mass (X, g cm^{-3}) in the soil system, ρ_b is the dry soil bulk density (g cm^{-3}), Q is the amount absorbed in soil particles (g g^{-1}), and S_s is the solute sink/source term (g cm^{-3} d^{-1}) accounting for uptake by roots, mineralization, drainage, etc.

The solute sink term S_s can be written as:

$$S_s = \mu_{\mathrm{dw}}\theta c + \mu_{\mathrm{ds}}\rho_b Q + K_{\mathrm{r}} S_w c \tag{4.9}$$

where μ_{dw} and μ_{ds} is the first-order solute decomposition rate (d^{-1}) for liquid and solid phases, respectively, and K_{r} is the root uptake preference factor (-). In addition, S_s also includes some other solute sink/source terms, such as those from drainage/infiltration flux, mobile/immobile exchange, and crack/matrix exchange.

Combing Eqs. (4.7) and (4.8) yield the general formulation of the advection and dispersion equation (ADE) governing the solute transport in a vertical soil profile：

$$\frac{\partial\left(\theta c+\rho_b Q\right)}{\partial t}=\frac{\partial}{\partial z}\left[\theta\left(D_{dif}+D_{dis}\right)\frac{\partial c}{\partial z}\right]-\frac{\partial qc}{\partial z}-S_s \tag{4.10}$$

Taking into account a sequential first-order decay chain, we obtain the mass conservation and transport equations applied in AHC which is valid for dynamic, one-dimensional, non-equilibrium chemical transport of solutes during transient water flow in a variably saturated rigid porous medium:

$$\begin{aligned}\frac{\partial\left(\theta c_1+\rho_b Q_1\right)}{\partial t}=&-\frac{\partial qc_1}{\partial z}+\frac{\partial}{\partial z}\left[\theta(D_{dif,1}+D_{dis,1})\frac{\partial c_1}{\partial z}\right]-\left(\mu_{w,1}+\mu'_{w,1}\right)\theta c_1-\left(\mu_{s,1}+\mu'_{s,1}\right)\rho_b Q_1\\&-S_{s,1} \qquad\qquad (k=1)\end{aligned} \tag{4.11}$$

$$\begin{aligned}\frac{\partial\left(\theta c_k+\rho_b Q_k\right)}{\partial t}=&-\frac{\partial qc_k}{\partial z}+\frac{\partial}{\partial z}\left[\theta(D_{dif,k}+D_{dis,k})\frac{\partial c_k}{\partial z}\right]-\left(\mu_{w,k}+\mu'_{w,k}\right)\theta c_k-\left(\mu_{s,k}+\mu'_{s,k}\right)\rho_b Q_k\\&-S_{s,k}+\mu'_{w,k-1}\theta c_{k-1}+\mu'_{s,k-1}\rho_b Q_{k-1} \qquad\qquad (2\le k\le n_s)\end{aligned} \tag{4.12}$$

where the subscripts w and s refer to liquid and solid phases, respectively, and subscript k represents the k^{th} chain number, μ is the first-order rate constant of transformation (d^{-1}), μ' is the similar first-order rate constant of transformation providing connections between individual chain species. Based on this, AHC provides four specific options for solute fate simulation: (1) total salt transport, where k=1 corresponds to total salt; (2) nitrogen transport and transformation, where k=1 and k=2 refer to NH_4-N and NO_3-N; (3) multicomponent solute transport, where eight kinds of ions are considered with k equals from 1 to 8; and (4) salt-Nitrogen migration process, where k=1, k=2, and k=3 corresponds to NH_4-N, NO_3-N, and total salt, respectively.

The adsorption is assumed to be instantaneous (equilibrium approach). The adsorption isotherm relating Q_k (g g^{-1}) and c_k is described by a generalized non-linear equation:

$$Q_k=\frac{k_{s,k}c_k^{\beta_k}}{1+\eta_k c_k^{\beta_k}} \tag{4.13}$$

where $k_{s,k}$ (cm^3 g^{-1}), β_k (-) and η_k (cm^3 g^{-1}) are empirical coefficients. When β_k =1, this equation becomes the Langmuir equation; when η_k =0, it becomes the Freundlich equation; and when η_k =0 and β_k =1, it represents the linear adsorption isotherm. There is no solid adsorption when $k_{s,k}$=0. Note that there is almost no adsorption for NO_3^-, while the adsorption should be considered for NH_4^+ which tends to be absorbed by soil particles.

The fully implicit or Crank-Nicholson, central finite-difference scheme is applied to solve the Eqs. (4.11) and (4.12). Grid Peclet number and Courant number are used together to ensure numerical stability. The third-type boundary condition is applied for both top and bottom boundaries for solute transport. The drainage/infiltration to/from the ditches is also included in the model. The k should be equal to 1 for salinity simulation, without considering the reaction chain and solid adsorption. For nitrogen simulation, the transformation and transport processes, including mineralization-immobilization, solid adsorption, nitrification, denitrification, volatilization, nitrous oxide (N_2O) emission, root nitrogen uptake (positive and active), and leaching, are involved in the chain equations. The urea hydrolysis to ammonia is also considered and described using the first-order kinetics equation. The detailed calculation principles related to nitrogen transformation and plant root uptake can be found in Chapter 6. The concentration flux boundary conditions (third-type) are provided for both upper and bottom boundaries, consistent with the actual field conditions. In addition, since AHC simulates non-linear adsorption and first-order decomposition, it also permits the simulation of some kinds of pesticides.

4.2 Initial and boundary conditions

The solution of Eqs. (4.10)-(4.12) requires knowledge of the initial distribution of solute concentration within the flow domain:

$$\begin{aligned} c(z,t=0) &= c_i(z) \\ Q^k(z,t=0) &= Q_i^k(z) \end{aligned} \tag{4.14}$$

where c_i and Q_i^k are the prescribed functions of z. in AHC. The initial condition for Q_i^k could be calculated from c_i under equilibrium adsorption by Eq. (4.13), and thus not required to be specified in AHC. The subscript k is dropped in Eq. (4.14) and throughout the remainder of this report, thus assuming that the transport-related equation in the theoretical development and the numerical solution apply to each of the solutes in the decay chain.

The boundary conditions are defined according to the actual field conditions. For the top boundary condition, no solute enters the soil profile at the surface during evaporation; and during the infiltration process, the third-type (Cauchy type) condition is applied to the upper boundaries:

$$\left(-\theta D\frac{\partial c}{\partial z}+qc\right)\Bigg|_{z=0}=q_0(t)\cdot c_0(t) \tag{4.15}$$

with c_0 the concentration of soil water flux at the soil surface boundary (g cm^{-3}). The solute concentration of irrigation water and rainfall, c_{irr} and c_{pre} (g cm^{-3}), need to be specified to determine c_0.

The third-type (Cauchy type) condition is applied to the bottom boundary condition:

$$\left(-\theta D\frac{\partial c}{\partial z}+qc\right)\Bigg|_{z=L}=q_L(t)\cdot c_L(t) \tag{4.16}$$

with c_L the concentration of soil water flux at the bottom boundary of the soil domain (g cm^{-3}). In case of upward flow, c_L equals the average solute concentration in the groundwater, c_{gw} (g cm^{-3}); and in case of downward flow, c_L equals the solute concentration at the location of the bottom of the soil profile (i.e., the solute concentration in the lowest compartment, c_n (g cm^{-3}), in numerical solution).

The lateral drainage boundary condition is treated analogously to the source/sink term. The drainage solute flux J_{drain} follows from (1) during drainage,

$$J_{drain}=q_{drain}c \tag{4.17}$$

and (2) during infiltration,

$$J_{drain} = q_{drain} c_{gw} \tag{4.18}$$

In numerical solution, AHC assumes that the lateral drainage flux leaves the soil profile laterally at the lowest compartment, or distributed from the saturated compartments submerged in groundwater.

4.3 Root solute uptake

Root solute uptake is the main sink term for solute transport. Root takes the solute from the soil system with two mechanisms of passive uptake through mass flow and positive uptake by the root itself, which are taken into consideration in the model. Solute can be passively taken into the plant in proportional to the plant transpiration rate.

$$S_{s,r} = K_r \cdot S_a \cdot c \tag{4.19}$$

where K_r is the root uptake preference factor (-). For the nutrient solutes, the active uptake will occur if not enough nutrient is brought into the plant by passive uptake. A K_r value greater than 1.0 can be used to represent the occurrence of active root uptake. In addition, we provide another approach to estimate the active root uptake for nitrogen with the Michaelis-Menten submodel (see details in section 6.6), in which the K_r value is set to 1.0.

4.4 Mobile/Immobile solute transport

The water flow in soils with mobile/immobile flow has been described in section 2.7. In the mobile region, the transport of solutes is affected by convection, dispersion, adsorption, decomposition, root water uptake, and drainage/infiltration. These processes are included in the solute transport equation, but corrections are needed as only the soil volume fraction f_m is mobile. Thus, when considering the mobile/immobile solute transport, the Eq. (4.10) will be written as:

$$\frac{\partial\left(\theta c + \rho_b \dfrac{f_m k_s c^{\beta}}{1+\eta c^{\beta}}\right)}{\partial t} = \frac{\partial}{\partial z}\left[\theta\left(D_{dif} + D_{dis}\right)\frac{\partial c}{\partial z}\right] - \frac{\partial qc}{\partial z} - \mu_w \theta c - \mu_s \rho_b \frac{f_m k_s c^{\beta}}{1+\eta c^{\beta}} - K_r S c - G_c \tag{4.20}$$

where G_c is the transfer rate of solutes from the mobile to the immobile region (g cm^{-3} d^{-1}). In the immobile region, transport of solutes is assumed to occur only by diffusion since water flow is absent. In addition, we assume that few roots exist in the immobile zone, and thus root solute uptake in the immobile region is small and can be neglected. The change of solute amounts in the immobile region is therefore governed by solute transfer between mobile and immobile regions and by solute decomposition:

$$\frac{\partial(1-f_m)\left(\theta_{im}c_{im}+\rho_b\frac{f_m k_s c_{im}^{\beta}}{1+\eta c_{im}^{\beta}}\right)}{\partial t}=-(1-f_m)\left(\mu_w\theta_{im}c_{im}+\mu_s\rho_b\frac{f_m k_s c_{im}^{\beta}}{1+\eta c_{im}^{\beta}}\right)+G_c \tag{4.21}$$

where c_{im} is the solute concentration in the immobile region (g cm^{-3}).

$$G_c=K_{dif}\left(c-c_{im}\right) \tag{4.22}$$

$$G_c=\begin{cases}\min\left(G_c,\theta c-\theta\bar{c}\right) & \text{if } c\geq c_{im}\\ \max\left(G_c,\theta c-\theta\bar{c}\right) & \text{if } c<c_{im}\end{cases} \tag{4.23}$$

$$\bar{c}=\frac{\theta_{im}c_{im}\left(1-f_m\right)+\theta c}{\theta_{im}\left(1-f_m\right)+\theta} \tag{4.24}$$

where K_{dif} is the effective diffusion coefficient (d^{-1}) between the mobile and immobile region, and $\bar{c}$ is the concentration value when the concentrations of the mobile and immobile parts are equal after the solute exchange.

5. Heat transport

5.1 Governing heat transport equation

Radiation, convection, and conduction are the three principal modes of heat energy transfer. ***Convection*** involves the movement of a heat-carrying mass. An example relevant to soil physics is the infiltration of warm wastewater from a power plant into initially cold soil. ***Conduction*** is the propagation of heat within a body by internal molecular motion. Because temperature is an expression of the kinetic energy of a body's molecules, the existence of a temperature difference within a body will normally cause the transfer of kinetic energy (by the numerous collisions of rapidly moving molecules) from the warmer region of the body to their neighbors in the colder region. The process of heat conduction is thus analogous to diffusion, which tends to equilibrate a body's internal distribution of molecular kinetic energy - that is to say its temperature (Hillel, 1998). Commonly, heat flow can be modeled by the convection and dispersion/diffusion conduction equations.

For heat conduction in the porous media, the soil heat flux $q_{h,cd}$ can be generally described by the first law of heat conduction, known as Fourier's law. $q_{h,cd}$ is in the direction of, and proportional to, the temperature gradient:

$$q_{h,cd} = -\lambda \nabla T \tag{5.1}$$

where λ is the thermal conductivity of soil (J cm^{-1} °C^{-1} d^{-1}) (i.e. coefficient of the apparent thermal conductivity of the soil), and T is the soil temperature (°C). For the vertically one-dimensional soil heat flux, q_{heat} (J cm^{-2} d^{-1}), is described as:

$$q_{h,cd} = -\lambda_{heat} \frac{\partial T}{\partial z} \tag{5.2}$$

A convection flux of heat $q_{h,cv}$ (J cm^{-2} d^{-1}), carried by the mass flow of soil water (also called *Darcian flow*), is proportional to the soil temperature T:

$$q_{h,cv} = C_{wv} q_l T \tag{5.3}$$

with C_{wv} is the volumetric heat capacity (J cm^{-3} °C^{-1}) of the liquid water. According to De Vries (1975), the rate of heat transfer by water vapor diffusion is small and proportional to the temperature gradient. Therefore, such diffusion might be taken into account by slightly increasing the soil thermal diffusivity. This is a simplied approach that eliminates the need for water vapor transport calculations. Apparent thermal properties rather than real thermal properties are assumed to account for both conductive and non-conductive heat flow. Conservation of energy results in:

$$C_s \frac{\partial T}{\partial t} = \frac{\partial \left(q_{h,cd} + q_{h,cv} \right)}{\partial z} \tag{5.4}$$

with C_s the volumetric heat capacity of soil porous media (J cm^{-3} °C^{-1}). Therefore, the 1-D soil heat transfer which includes the term of heat convection flux by liquid can be written as:

$$\frac{\partial C_s(\theta) T}{\partial t} = \frac{\partial}{\partial z}\left[\lambda(\theta) \frac{\partial T}{\partial z} \right] - \rho_l C_l \frac{\partial q_l T}{\partial z} - S_h - \rho_l C_l S_w T_w \tag{5.5}$$

where C_l is mass heat capacity (i.e. specific heat) of liquid water (J g^{-1} °C^{-1}) (with C_{wm}=$\rho_l C_l$), S_h is the sink/source term of heat (J cm^{-3} d^{-1}), and T_w is the temperature of inflow or outflow water (°C). The form of this equation can be conveniently solved with given initial and boundary conditions and sinks/sources in the numerical solution.

One the basis of Eq. (5.5), the energy balance equation which can also consider the existence of soil freezing and thawing in soil layers is written as:

$$\frac{\partial C_s T}{\partial t} - L_f \rho_i \frac{\partial \alpha}{\partial t} = \frac{\partial}{\partial z}\left(\lambda \frac{\partial T}{\partial z} \right) - \rho_l C_l \frac{\partial q_l T}{\partial z} - S_h - \rho_l C_l S_w T_w \tag{5.6}$$

where L_f is the latent heat of ice fusion (335000 J kg^{-1}). If further considering the effect of the water vapor diffusion on heat transport, the Eq.(5.6) could be expressed as:

$$\frac{\partial C_s T}{\partial t} - L_f \rho_i \frac{\partial \alpha}{\partial t} + L_v \left(\frac{\partial \rho_v}{\partial t} + \frac{\partial q_v}{\partial z} \right) = \frac{\partial}{\partial z}\left(\lambda \frac{\partial T}{\partial z} \right) - \rho_l C_l \frac{\partial q_l T}{\partial z} - C_v \frac{\partial q_v T}{\partial z} - S_h - \rho_l C_l S_w T_w \tag{5.7}$$

where L_v is the latent heat of evaporation (J g^{-1}), and C_v is the mass heat capacity (i.e. specific heat) of water vapor (J g^{-1} °C^{-1}). This equation is the basis for AHC to simulate the heat transfer in the soil domain for both freezing and non-freezing soils with the effect of vapor transfer.

So far, the unknowns in Eqs. (2.7) and (5.7) for coupled soil water-heat transfer description are temperature, water content, ice content, and soil water pressure head. So, besides the soil water retention function (e.g., Eq. 2.8), an additional equation is still needed for the equation solution: the knowledge of the soil freezing properties, i.e., the known relations among θ, α and soil temperature (T). This is provided by the Clausius-Clapeyron equation that could express the relationship between unfrozen liquid water and ice (see Eq. (5.13) in section 5.2.2).

The numerical solution of Eq. (5.7) is given in section 9.3 in detail. A fully implicit finite difference scheme is applied to solve Eq. (5.7), with specified initial and boundary conditions. The soil heat capacity and thermal conductivity are calculated from the soil composition and the volume fractions of air, water, and ice for each compartment, referring to the approach of De Vries (1975). Their calculation is described in the following paragraphs. In addition, AHC provides calculation methods of different complexities: soil freeze-thaw or water vapor transfer can be considered selectively.

5.2 Soil heat properties

5.2.1 Heat capacity and thermal conductivity

The volumetric heat capacity, which depends on the soil composition (solid including ice, water, and air), is calculated as a weighted mean of the heat capacities of the individual components (De Vries, 1963):

$$C_s = f_{sand}C_{sand} + f_{silt}C_{silt} + f_{clay}C_{clay} + f_{organic}C_{organic} + \theta C_{water} + f_{air}C_{air2} + \alpha C_{ice} \tag{5.8}$$

where f and C on the right-hand side of this equation are the volume fraction (cm^3 cm^{-3}) and volumetric heat capacity (J cm^{-3} °C^{-1}) of each component, respectively. The reference values of

C_i for sand, silt, clay, organic matter, water, air, and ice are 2.218, 2.394, 2.385, 2.496, 4.180, 0.001212, and 1.932 J cm^{-3} °C^{-1}, respectively.

The thermal conductivities of the various soil components differ very markedly. The apparent thermal conductivity λ of soil depends on its mineral composition and organic matter contents, and volume fractions of water, ice, and air. The thermal conductivity of air is very much smaller than of water or solid matter, a high air content corresponds to a low thermal conductivity. The λ is calculated as the weighted average of conductivities for each component (De Vries, 1963; Tarnawski and Wagner, 1993):

$$\lambda = \frac{k_w \theta \lambda_w + k_a \sigma_a \lambda_a + \sum_{i=1}^{j} k_i \sigma_i \lambda_i}{k_w \theta + k_a \sigma_a + \sum_{i=1}^{j} k_i \sigma_i} \tag{5.9}$$

where σ is the volumetric fraction of the component (cm^3 cm^{-3}), k_{ic} is the ratio of the average temperature gradient in the kind of particular soil component to the average temperature gradient in a continuous medium (soil water or air). The subscript *ic* refers to *a*=air, *w*=unfrozen soil water, *i*=soil solid component (*i*=1…*j* - soil minerals and ice), and *j*=number of individual types of soil solid components. The reference values of λ_i for sand, silt, clay, organic matter, water, air, and ice are 7603, 2523, 2523, 216, 492, 22, and 1901 J cm^{-1} °C^{-1} d^{-1}, respectively.

The weighting factor k_{ic} is estimated upon the shape of the particles and thermal conductivities of soil components, according to the approach described in Tarnawski and Wagner (1992) and Tarnawski et al. (2005). It is assumed that the soil granules have the shape of ellipsoids of rotation and their axes (*a*, *b*, and *c*) comply with the relation (*a*= *b* = *nc*), and k_{ic} can be estimated as:

$$k_{ic} = \frac{2}{3}\left[1 + \left(\frac{\lambda_i}{\lambda_{cm}} - 1\right) g_i\right]^{-1} + \frac{1}{3}\left[1 + \left(\frac{\lambda_i}{\lambda_{cm}} - 1\right)(1 - 2g_i)\right]^{-1} \tag{5.10}$$

where g_i (-) is the shape factor of the ellipsoid along the a-axis for various soil components, and λ_{cm} is thermal conductivities for continuous medium (soil water or air) (J cm^{-1} °C^{-1} d^{-1}). g_i is calculated by:

$$g_i = \begin{cases} 0.5\left[\dfrac{1}{1-p^2} - \dfrac{p^2}{2(1-p^2)^{1.5}} \ln\left(\dfrac{1+(1-p^2)^{0.5}}{1-(1-p^2)^{0.5}}\right)\right] & p<1 \\ \dfrac{1}{3} & p=1 \\ \dfrac{p^2}{2(p^2-1)^{1.5}}\left[\dfrac{\pi}{2} - \arctan\dfrac{1}{(p^2-1)^{0.5}} - \dfrac{(p^2-1)^{0.5}}{p^2}\right] & p>1 \end{cases} \tag{5.11}$$

where p (-) refers to the shape value that equals the ratio of the length of the a-axis to that of the c-axis of rotated ellipsoids (i.e., $p=a/c$).

5.2.2 Freezing depression equation

Due to matric and osmotic potentials, soil water exists in equilibrium with ice at temperatures below the normal freezing point of bulk water and over the entire range of soil freezing temperatures normally encountered (Ahuja et al., 2000). Thermodynamic reasoning leads to a generalized form of the *Clapeyron equation* which relates liquid water pressure and ice pressure to equilibrium temperature in soil system (Loch, 1978; Miller, 1980):

$$\frac{P_w}{\rho_l} - \frac{P_i}{\rho_i} = \frac{L_f T}{T_k} \tag{5.12}$$

where P_w and P_i are the water pressure and ice pressure, respectively (10^5 Pa), and T is the actual soil temperature (°C). If the process is mainly related to the migration of moisture or heat in partially frozen soil rather than freezing-induced deformation, it could be assumed that the pressure in the ice phase (P_i) is atmospheric (zero gauge pressure) (Sheshukov and Nieber, 2011; Kurylyk and Watanabe, 2013). If further considering the osmotic effects, Eq. (5.12) can be written as (Cary and Mayland, 1972; Miller, 1980):

$$\varphi = h + h_{so} = 100\frac{L_f T}{g T_k} \tag{5.13}$$

where φ is the total potential of the soil water with ice present (cm) (which is controlled by the vapor pressure over ice), and h_{so} is the osmotic potential of the soil solution (cm). This equation is referred to as the freezing point depression equation (or freezing temperature equation). It can be used to define the relation between ice content and temperature. In dilute solutions, the osmotic pressure, h_s, is proportional to the concentration of the solution and to its temperature according to the following equation (Hillel, 1998):

$$h_{so} = -\frac{i_v R T_k}{10 \cdot \rho_l g} c_M = -\frac{i_v R T_k}{10 \cdot \rho_l g} \frac{c}{M} \tag{5.14}$$

where i_v is the van't hoff factor (-), c_M is the molar concentration of the solute in the solution (mol cm^{-3}), R is the universal gas constant (8.3143 J mol^{-1} K^{-1}), g is the gravitational acceleration (9.81 m s^{-2}), and M_s is the molecular weight of specific solute (g mol^{-1}). From Eqs. (2.8), (5.13), and (5.14), the matric potential and, therefore, (allowed) maximum liquid soil water content could be defined by temperature during freezing conditions (see details in Annex A); soil water content greater than the computed from these relations is assumed to be ice. Thus, if the total water content is known, ice content and the latent heat term can be determined.

5.3 Initial & boundary conditions and sink/source

The solution of Eq. (5.5), (5.6), or (5.7) requires knowledge of the initial distribution of soil temperature within the flow domain, i.e.,

$$T(z,t) = T_{ini}(z) \qquad t = t_0 \tag{5.15}$$

with T_{ini} a prescribed function of z (°C).

At the soil surface, the daily average air temperature and the temperature of irrigation water are used to define the boundary conditions. In general, third-type (or Cauchy type) boundary conditions can be specified at the top boundary for the field soil profile, i.e. the temperature-controlled type condition:

$$\left(-\lambda \frac{\partial T}{\partial z} + T C_w q\right)\bigg|_{z=0} = f_{h1}\left(T_{atm}, T\right) + T_0 C_w q \qquad at\ z = 0 \tag{5.16}$$

where f_{h1} is the given function of T_{atm}, T_{atm} (cm) is the atmospheric air temperature above the soil surface (°C), and T_0 is either the temperature of the incoming fluid or the soil temperature at the top boundary. In the numerical solution, the f_0 function follows from:

$$f_{h1}\left(T_{atm},T\right) \approx \lambda_{1/2}\left(\frac{T_{atm}-T_1}{z_1}\right) \tag{5.17}$$

where $\lambda_{1/2}$ is the average thermal conductivity between the soil surface and the first node (J cm^{-1} °C^{-1} d^{-1}), and T_1 is the soil temperature of the first node (°C). When irrigation or large rainfall occurs at the top surface soil, the third-type boundary condition may become:

$$\left(-\lambda\frac{\partial T}{\partial z}+TC_w q\right)\Bigg|_{z=0} = T_{0,in}C_w q_0 \tag{5.18}$$

where T_0 is just the temperature of the incoming fluid (i.e., irrigation or rainfall). The daily fluctuations in soil temperature can be represented by a sine function as follows (Kirkham and Powers, 1972):

$$T_{atm} = \bar{T} + A_{mp}\sin\left(\frac{2\pi t}{p_t}-\frac{7\pi}{12}\right) \tag{5.19}$$

where $\bar{T}$ is the daily average temperature near the soil surface (or atmospheric air) (°C), A_{mp} is the amplitude of the sine wave (°C), and p_t is a period of time (d) necessary to complete one cycle of the sine wave (taken to be 1 day).

The bottom boundary condition of the soil profile is specified as the third-type boundary condition, which is dependent on the prescribed temperature at the bottom:

$$\left(-\lambda\frac{\partial T}{\partial z}+TC_w q\right)\Bigg|_{z=L} = \begin{cases} f_{h2}\left(T_{bs},T\right)+C_w q T_L & q<0\ (downward\ flux) \\ f_{h2}\left(T_{bs},T\right)+C_w q T_{bs} & q\geq 0\ (upward\ flux) \end{cases} \quad at\ z=L \tag{5.20}$$

where T_{bs} is the soil temperature at the bottom of the soil column (°C), f_{h2} is the given function of T_{bs} and T at the soil bottom, and T_L (°C) is the temperature of incoming fluid (q_L>0, upward flux) or the soil temperature at the bottom boundary (q_L<0, downward flux). T_{bs} is usually specified as remaining at some constant (measured temperature) at sufficient soil depth. The function f_{h2} is defined based on the Fourier's law (see details in section 9.3 of the numerical solution). In numerical solution, it follows from:

$$f_{h2}\left(T_{bs},T\right) \approx \lambda_{N+1/2}\left(\frac{T_N - T_{bs}}{0.5\Delta z_N}\right) \tag{5.21}$$

where $\lambda_{N+1/2}$ is the average thermal conductivity at the bottom interface (J cm^{-1} °C^{-1} d^{-1}), T_N is the soil temperature of the bottom node (°C), and Δz_N is the thickness of the bottom node/compartment (cm).

6. Soil nitrogen

The nitrogen transformation and transport processes include mineralization-immobilization, solid adsorption, nitrification, denitrification, volatilization, nitrous oxide (N_2O) emission, root nitrogen uptake (positive and active), and leaching in AHC. The model can consider ammonia nitrogen, nitrate nitrogen, and nitrite nitrogen; however, in actual field applications, only the nitrate N and ammonia N are considered. The processes are involved in the chain equations of solute transport, which are calculated and coupled with the soil water flow equation in each time step. The urea hydrolysis to ammonia is considered and described with the first-order kinetics equation. The mineralization-immobilization is related to the soil organic matter turnover. The conceptual diagram of soil nitrogen transport and transformation is presented in Fig. 6.1, and the detailed calculation principles are described in the following sections of this chapter.

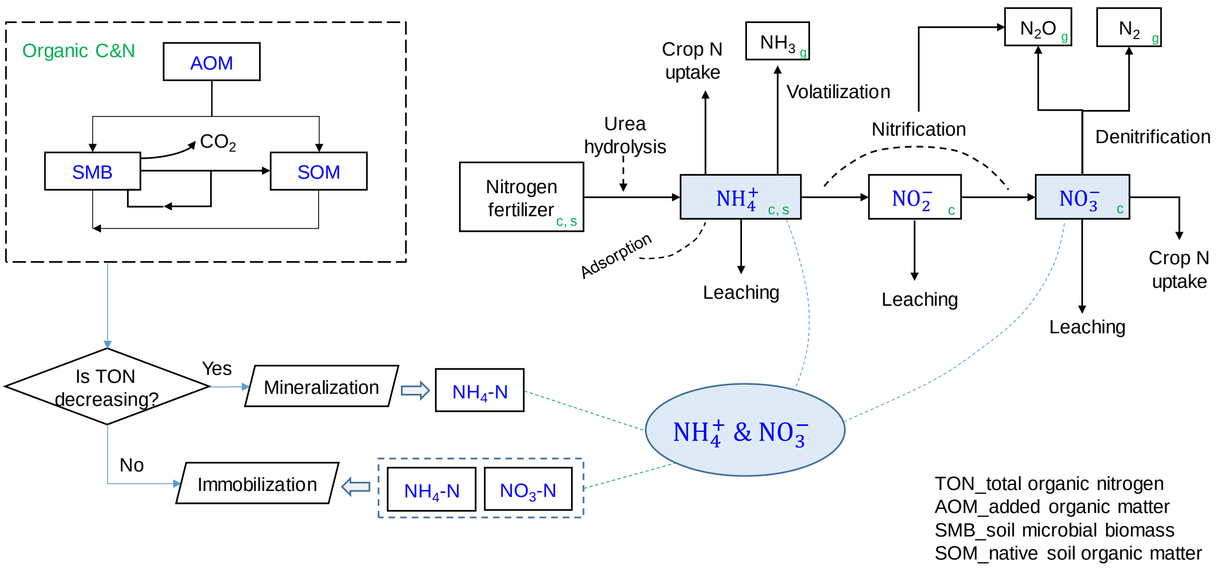


Figure 6.1 Conceptual diagram of the soil nitrogen transport and transformation in AHC. TON is the total organic nitrogen.

6.1 Mineralization and immobilization

The concepts of mineralization and immobilization of nitrogen (N) are mainly adapted from the Daisy (Hansen, 2002) while also referring to the APSIM (McCown et al., 1996). The organic matter is divided into three main pools as follows: (1) soil organic matter is divided into two main pools, viz., dead soil organic matter (SOM) and microbial biomass (SMB); and (2) the added fresh organic matter (AOM) is defined as the third main pool (Fig. 6.2). Each main pool is subdivided into two or three subpools with different decay rates. The SOM is similar to the humus and is further subdivided into three subpools SOM_0, SOM_1 and SOM_2. The SOM_0 consists of almost inert soil organic matter. The active subpools SOM_1 and SOM_2 are assumed to consist of chemically and physically stabilized organic matter, respectively. Their decomposition rates are both simulated by first-order reaction kinetics (Eqs. 6.5 and 6.6). The active SOM pool is assumed to change from about 60% to 20% of total soil organic matter with depth increase for shallow soils (e.g., 0-60 cm) and 0-5% for deeper arable soils(McCown et al., 1996). The microbial biomass in the soil which usually accounts for less than 4% of the active humus is subdivided into two subpools designated SMB_1 and SMB_2, respectively. SMB_1 is considered to be more stable while SMB_2 is a more dynamic part of the microbial biomass. The simulation of biomass turnover is based on growth efficiency, maintenance respiration, and death rate coefficients, described using the first-order reaction kinetics equations (Eqs. 6.3 and 6.4) (Hansen et al., 1991). The AOM is the organic fertilizer, such as plant residual, farmyard manure, and slurry, which is allocated to the subpools AOM_1, AOM_2, and SOM_2. Organic input is subdivided into structural, metabolic, and recalcitrant organic matter. The subpools AOM_1 and AOM_2 are assumed to consist of mainly cell wall material and mainly water-extractable cell material, respectively. The decomposition rate of each subpool is calculated by first-order reaction kinetics (Eqs. 6.1 and 6.2). The SOM_2, SMB_2, and AOM_2 have a faster turnover compared with the SOM_1, SMB_1, and AOM_1.

$$\frac{dC_1}{dt} = C_0 f_0 - \alpha_1 k_1 C_1 \tag{6.1}$$

$$\frac{dC_2}{dt} = C_0 f_{00} - \alpha_2 k_2 C_2 \tag{6.2}$$

$$\frac{dC_3}{dt} = \alpha_1 k_1 C_1 E_1 f_1 + \alpha_5 k_5 C_5 E_5 + \alpha_6 k_6 C_6 E_6 \left(1 - f_6\right) - \alpha_3 k_3 C_3 - \alpha_3 \beta_1 C_3 \tag{6.3}$$

$$\frac{dC_4}{dt} = \alpha_1 k_1 C_1 E_1 \left(1 - f_1\right) + \alpha_2 k_2 C_2 E_2 + \alpha_3 k_3 C_3 E_3 \left(1 - f_3\right) - \alpha_4 k_4 C_4 \left[1 - E_4 \left(1 - f_4\right)\right] - \alpha_4 \beta_2 C_4 \tag{6.4}$$

$$\frac{dC_5}{dt} = \alpha_6 k_6 C_6 E_6 f_6 - \alpha_5 k_5 C_5 \tag{6.5}$$

$$\frac{dC_6}{dt} = C_0 \left(1 - f_0 - f_{00}\right) + \alpha_3 k_3 C_3 E_3 f_3 + \alpha_4 k_4 C_4 E_4 f_4 - \alpha_6 \beta_6 C_6 \tag{6.6}$$

where C_i is the carbon content of the i^{th} sub-pool (kg C ha^{-1}), k_i is a first-order decomposition rate coefficient which is modified according to the considered abiotic factors (kg C ha^{-1} d^{-1}), α_i is the stress factor for carbon turnover (-), f_i is partitioning coefficient for carbon flow (-), and E_i is the substrate utilization efficiency (-).

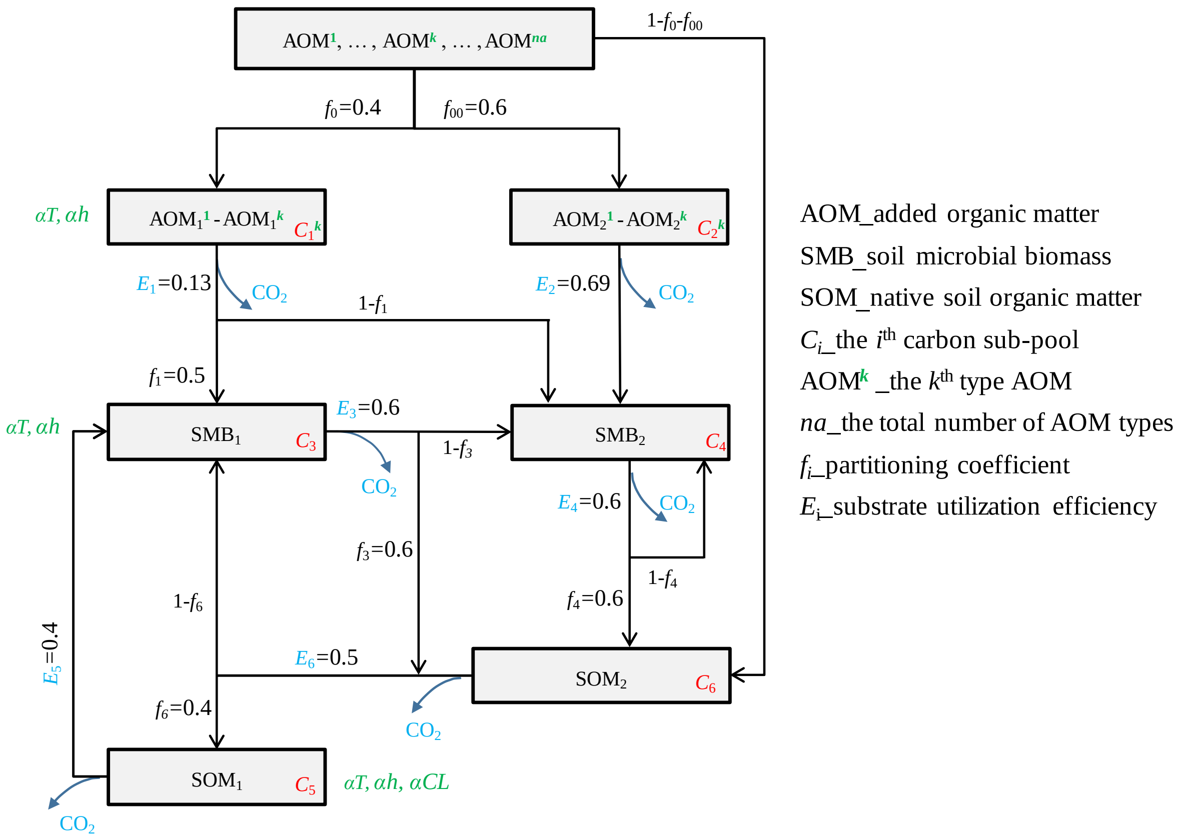


Figure 6.2 Schematic presentation of the soil organic matter turnover model included in AHC.

The soil temperature and moisture are two main abiotic factors influencing the above carbon turnover, while the clay content is a specific factor to SOM_1, SOM_2, and BOM_1. These factors are calculated for each soil compartment using Eqs. (6.7)-(6.10). It is assumed that there are no interaction effects between the different factors, and the combined effect is multiplicative (Hansen et al., 1991). The model allows flows of carbon among different subpools during the decomposition of the organic matter.

$$\alpha_1 = \alpha_2 = F_m(T)F_m(h) \tag{6.7}$$

$$\alpha_3 = F_m(T)F_m(h)F_m(C_c) \tag{6.8}$$

$$\alpha_4 = F_m(T)F_m(h) = \alpha_1 \tag{6.9}$$

$$\alpha_5 = \alpha_6 = F_m(T)F_m(h)F_m(C_c) = \alpha_3 \tag{6.10}$$

The stress function F_m (-) is calculated as follows:

$$F_m(T) = \begin{cases} 0 & T \le 0 \\ 0.1T & 0 < T \le 20 \\ \exp\left(0.47 - 0.027T + 0.00193T^2\right) & T > 20 \end{cases} \tag{6.11}$$

$$F_m(h) = \begin{cases} 0.6 & h \ge -1 \\ 0.6 + 0.4\log\left(-h\right)/1.5 & -10^{1.5} \le h < -1 \\ 1.0 & -10^{2.5} \le h < -10^{1.5} \\ 1.0 - \log\left(-10^{-2.5}h\right)/4 & -10^{6.5} \le h < -10^{2.5} \\ 0 & h < -10^{6.5} \end{cases} \tag{6.12}$$

$$F_m(C_c) = \begin{cases} 1.0 - aC_c & 0 < C_c \le C_c' \\ 1.0 - aC_c' & C_c > C_c' \end{cases} \tag{6.13}$$

where C_c' is set equal to 0.25 kg kg^{-1} and a is equal to 2.0.

The turnover of organic soil nitrogen is closely linked to the turnover of organic carbon. The flows of matter between different pools are calculated in terms of carbon, and thus the corresponding nitrogen flows are calculated based on the carbon/nitrogen ratio (*CNR*) of the receiving pool. The *CNR* value is set constant for each subpool. A carbon balance for each pool of organic matter can be established resulting in an expression for the rate of change of carbon in each pool. As each pool of organic matter is characterized by a particular carbon-to-nitrogen ratio (CNR_i), an overall organic nitrogen balance can be established resulting in an equation for the net mineralization of nitrogen as flows:

$$R_{\min} = -\sum_i CNR_i \frac{dC_i}{dt} \tag{6.14}$$

It is assumed that carbon in the form of CO_2 is lost from the soil system due to microbial respiration and decomposition of SOM (i.e., growth and maintainance respiration of SMB). Production of carbon dioxide results from all C-fluxes into the soil microbial biomass (SMB) pools (substrate utilization efficiencies being less than unity). Furthermore, soil microbial maintenance respiration produces carbon dioxide. Thus, the emitted CO_2 flux from above microbial respiration is obtained directly from the above soil organic matter turnover process. In addition, another part of the emitted CO_2 flux is from the respiration of the root system (i.e., the root-derived CO_2), which is described in detail in section 8.4. Moreover, we assume that the microbial respiration in this section may not only involve microbial maintenance respiration and decomposition of SOM, but also include the decomposition of rhizodeposits for the current AHC (Morell et al., 2012).

6.2 Nitrification and NH_3 volatilization

Nitrification is a microbial process related to the conversion of ammonia N to nitrate N, while NH_3 volatilization, the loss of ammonia to the atmosphere, often takes place simultaneously. They are estimated using a combination of the methods of Reddy et al. (1979) and Godwin et al. (1984) which are also adopted by the EPIC model (Williams, 1995). This approach is based on the first-order kinetic equations (Reddy et al., 1979), described in Eqs. (6.15)-(6.27). The combined rate is first calculated using Eq. (6.15) and then partitioned into the nitrification and volatilization rates using the subsequent equations. Nitrification, the conversion of ammonia N to nitrate N, is regulated by the soil water content, temperature, and pH. The volatilization, as the loss of ammonia to the atmosphere, is estimated simultaneously with nitrification. This loss from the soil surface is affected by the soil temperature and wind speed, while that from subsurface soils is influenced by the depth, cation exchange capacity (CEC, cmol kg^{-1}), and soil temperature (Williams, 1995).

$$\frac{dWNH_4}{dt} = R_{nit+vol} = WNH_4\left[1-\exp\left(-k_{nit}-k_{vol}\right)\right] \tag{6.15}$$

$$k_{\mathrm{nit}} = f_{\mathrm{T\_nit}}\, f_{\mathrm{sw\_nit}}\, f_{\mathrm{pH\_nit}} \tag{6.16}$$

$$f_{\mathrm{T\_nit}} = 0.41\frac{T-5}{10} \qquad \mathrm{T}_{\mathrm{soil}} > 5°C \tag{6.17}$$

$$f_{\mathrm{sw\_nit}} = \begin{cases} \dfrac{SW-WP}{SW25-WP} & SW < SW25 \\ 1.0 & SW25 \le SW \le FC \\ 1.0-\dfrac{SW-FC}{\phi_{por}-FC} & SW > FC \end{cases} \tag{6.18}$$

$$f_{\mathrm{pH\_nit}} = \begin{cases} 0.307\mathrm{pH}-1.269 & \mathrm{pH} < 7.0 \\ 1.0 & 7.0 \le \mathrm{pH} \le 7.4 \\ 5.367-0.599\mathrm{pH} & \mathrm{pH} > 7.4 \end{cases} \tag{6.19}$$

$$k_{\mathrm{vol}} = \begin{cases} f_{\mathrm{T\_nit}}\, f_{\mathrm{wind}} & \text{surface} \\ f_{\mathrm{T\_nit}}\, f_{\mathrm{CEC}}\, f_{\mathrm{depth}} & \text{subsurface} \end{cases} \tag{6.20}$$

$$f_{\mathrm{wind}} = 0.335 + 0.16\ln u_2 \tag{6.21}$$

$$f_{\mathrm{CEC}} = \max\left(0.0,\ 1.0-0.038\mathrm{CEC}\right) \tag{6.22}$$

$$f_{\mathrm{depth}} = 1-\frac{10\cdot(-z)}{10\cdot(-z)+\exp\left[4.706-0.305\cdot(-z)\right]} \tag{6.23}$$

$$R_{nit} = WNH_4\left[1-\exp\left(-k_{nit}\right)\right] \tag{6.24}$$

$$R_{vol} = WNH_4\left[1-\exp\left(-k_{vol}\right)\right] \tag{6.25}$$

$$R_{vol} = R_{vol}\frac{R_{nit+vol}}{R_{nit}+R_{vol}} \tag{6.26}$$

$$R_{nit} = R_{nit+vol} - R_{vol} \tag{6.27}$$

where WNH_4 and WNO_3 are respectively NH_4-N and NO_3-N contents in the soil layer (kg N ha^{-1}), k_{nit} and k_{vol} are respectively the nitrification regulator and volatilization regulator (d^{-1}), R_{nit}, R_{vol} and $R_{nit+vol}$ are respectively the nitrification rate, volatilization rate and combined rate of nitrification and volatilization (kg N ha^{-1} d^{-1}), f_{T_nit}, $f_{sw\_nit,}$ and f_{pH_nit} are respectively the factors (-) of soil temperature, soil water and pH for N nitrification, f_{wind}, f_{depth} and f_{CEC} are respectively the factors (-) of wind, soil depth and CEC for N volatilization (0-1), and u_2 here is the mean wind speed at 2 m height in a day (m s^{-1}). SW is the soil water content of the soil layer (cm^3 cm^{-3}), WP and FC are the soil water content at the plant wilting point and field capacity (cm^3 cm^{-3}), respectively, and $SW25$ is the soil water content at WP+ 0.25(FC−WP) (cm^3 cm^{-3}).

6.3 Denitrification

Nitrate N from nitrification or commercial fertilizer is subject to denitrification in anaerobic environments by microbes that can utilize the N in NO_3^- and NO_2^- as terminal electron acceptors. Denitrification is a complicated microbial process in which nitrate N is lost in the form of N_2 and N_2O gas from the soil to the atmosphere. Soil denitrification is assumed to only take place in the top 20 cm soil layer. The denitrification rate is calculated using the first-order equation as a function of soil temperature and organic carbon content, as follows:

$$R_{den} = \begin{cases} WNO_3\left[1-\exp(-1.4 f_{T\_den} OC)\right] & wfps \geq 0.80 \\ 0 & wfps < 0.80 \end{cases} \tag{6.28}$$

$$f_{T\_den} = 0.1 + 0.9\frac{T}{T + \exp(9.93 - 0.312T)} \tag{6.29}$$

where R_{den} is the total denitrification rate (kg N ha^{-1} d^{-1}), OC is the percent content of soil organic carbon (%), $wfps$ is the fraction of soil pore space filled with water (0-1) (with the value equal to θ/θ_s), and f_{T_den} is the soil temperature factor for denitrification (0-1).

6.4 N_2O emission

The nitrous oxide (N_2O) emitted from soils is an important part affecting stratospheric ozone levels and greenhouse effects. Both nitrification and denitrification contribute to the release of N_2O, which are both considered in the AHC. In nitrification, N_2O emission is estimated as a fraction of the nitrification rate, adjusted by the soil temperature and moisture (Eq. 6.30). N_2O emission during the denitrification process is calculated using Eq. (6.32) for saturated and unsaturated conditions (Xu et al., 1998).

$$N_2O_{nit} = \alpha_{nit} R_{nit} f_{\mathrm{T\_nit}} f_{\mathrm{sw\_nit}} \tag{6.30}$$

$$f_{sw\_den} = \exp\left(-23.77 + 23.77 wfps\right) \tag{6.31}$$

$$N_2O_{den} = \begin{cases} 0.05 R_{den} & wfps \geq 1.0 \\ \alpha_{den} R_{den} (1 - f_{sw\_den}) & 0.8 \leq wfps < 1.0 \\ 0 & wfps < 0.8 \end{cases} \tag{6.32}$$

where α_{nit} denotes the maximum fraction of nitrogen loss as N_2O through nitrification (-), f_{sw_den} is the soil moisture factor (-), and α_{den} is the maximum fraction of the total denitrification rate R_{den} occurring as N_2O (-), with a default value of 0.5. α_{nit} is set at a default value of 0.002 and can also be calibrated from the field measurement. The fraction of N_2O lost in nitrification varies from 0.001 to 0.05 as reported by Goodroad and Keeney (1984).

6.5 Urea hydrolysis

Urea is the major N fertilizer used for crop production in world agriculture. In the AHC, the rate for urea hydrolysis to ammonia ($HYDR_{urea}$, kg N ha^{-1} d^{-1}) is estimated using the first-order kinetics equation (Li et al., 2007):

$$HYDR_{urea} = WUREA\left[1.0 - \exp(-5.0 f_{T\_hy} wfps)\right] \tag{6.33}$$

where $WUREA$ is the urea content in the soil layer (kg N ha^{-1} d^{-1}), and f_{T_hy} is assumed to be equal to f_{T_den}. This implies that urea hydrolysis is generally faster (2-3 days) in hot weather and wet soil while slower under cool and dry conditions (Li et al., 2007). This treatment is different from some other models (e.g., NLEAP and EPIC) which assume that the process of urea hydrolysis is so fast that the urea can be directly treated as NH_4^+ as model input.

6.6 Plant N uptake

The nitrogen uptake is estimated using a demand and supply approach, referring to the EPIC model (Williams, 1995). The daily crop N demand is the difference between the crop N content and the ideal N content for that day and is estimated using Eq. (6.34). The optimal crop N concentration (C_{NB}) declines with increasing crop growth stage (Jones, 1983). It is calculated using Eq. (6.35) in which dn_1, dn_2 and dn_3 are crop parameters calculated from the crop-specific concentration of N in the plant at seedling, halfway through the season, and maturity, respectively.

$$UND_i = C_{NB,i} B_i - \sum_{k=1}^{i-1} UN_k \tag{6.34}$$

$$C_{NB,i} = dn_1 + dn_2 \exp(-dn_3 HUI_i) \tag{6.35}$$

where UND is the N demand rate of the crop (kg N ha^{-1} d^{-1}), C_{NB} is the optimal N concentration of the crop (kg kg^{-1}), B is the accumulated biomass of the crop (kg ha^{-1}), UN is the actual N uptake rate (kg N ha^{-1} d^{-1}), and the subscript i refers to the day number.

N is taken up as NH_4^+ and NO_3^- by plants, considering both the active process and passive processes in the AHC. The amount of N that passively enters the plant (N_t^{Pass}) is determined by the plant transpiration rate and the concentration of N in the soil water that enters the plant. If passive N uptake is not enough to meet the plant demand, then active N uptake occurs. Using the Michaelis-Menten kinetics could provide the active nitrogen uptake rates. A rectangular hyperbola equation is used to estimate the rate of active N uptake (N_t^{Act}, kg N plant^{-1} d^{-1}) (Ahuja et al., 2000; Jungk, 2002):

$$N_t^{Act} = \mu_{1p} \frac{N_t^{Soil}}{\mu_{2p} + N_t^{Soil}} \tag{6.36}$$

where N_t^{Soil} is the amount of N available in the soil layer (kg N ha^{-1}), μ_{1p} is the maximum proportion of N that can be removed from the soil for a plant (kg N plant^{-1} d^{-1}), and μ_{2p} the half maximum nitrogen-uptake amount for the species being simulated (kg N ha^{-1}). The total amount of additional nitrogen available to the plant through uptake is the sum of N_t^{Pass} and N_t^{Act}. In AHC, the N_t^{Act} is calculated as follows:

$$N_t^{Act} = \sum_{i=1}^{N_{root}} \left(\mu_1 \cdot 10^6\right) \cdot w_{r,i} \cdot \frac{c_i \theta_i \Delta z_i \cdot P_{area} \cdot 10^3}{\mu_2 \rho_l \theta_i \Delta z_i \cdot P_{area} + c_i \theta_i \Delta z_i \cdot P_{area} \cdot 10^3} \cdot P_{Den} \cdot 10^{-9} \quad (6.37)$$

where μ_1 is the maximum proportion of N that can be removed from the soil for a plant (g N plant^{-1} d^{-1}), and μ_2 the half maximum nitrogen-uptake amount for the species being simulated (ppm, i.e. μg N (g water)$^{-1}$), $w_{r,i}$ is partition weight of root water uptake for the i^{th} compartment, N_{root} is the total number of compartments in the root zone, P_{Den} is the plant density (plant ha^{-1}), and P_{area} is the area occupied by a plant (cm^2 plant^{-1}) ($P_{area} = 10^8/P_D$). μ_1 and μ_2 are the user-specified parameters in AHC.

7. Evaporation and transpiration

7.1 Penman-Monteith equation

The Penman-Monteith (P-M) equation could well describe the evapotranspiration rate from a dry, extensive, horizontally-uniform vegetated surface, which is optimally supplied with water. The original form P-M equation can be written as (Monteith, 1965; Monteith, 1980):

$$ET_p = \frac{\frac{\Delta_v}{\lambda_w}\left(R_n - G\right) + \frac{p_{c1}\rho_{air}C_{air}}{\lambda_w}\frac{\left(e_s - e_a\right)}{r_{air}}}{\Delta_v + \gamma_{air}\left(1 + \frac{r_c}{r_{air}}\right)} \tag{7.1}$$

where ET_p is the potential evapotranspiration rate for a canopy (mm d^{-1}), Δ_v is the slope of the saturation vapor pressure & temperature curve (kPa °C^{-1}), λ_w is the latent heat of vaporization (J kg^{-1}), R_n is the net radiation flux at canopy surface (J m^{-2} d^{-1}), G is the soil heat flux (J m^{-2} d^{-1}), p_{c1} accounts for unit conversion (=86400 s d^{-1}), ρ_{air} is the air density (kg m^{-3}), C_{air} is the heat capacity of moist air (J kg^{-1} °C^{-1}), e_s is the saturation vapor pressure (kPa), e_a is the actual vapor pressure (kPa), γ_{air} is the psychrometric constant (kPa °C^{-1}), r_c is the crop resistance (s m^{-1}), and r_{air} is the aerodynamic resistance (s m^{-1}). (e_s-e_a) represents the vapor pressure deficit of the air. The P-M equation requires four daily meteorological data in the calculation: solar radiation, air temperature, relative humidity, and wind speed. It is ranked as the best for all climatic conditions.

In AHC, the P-M equation is used to calculate the daily average of the potential evapotranspiration (ET_p, cm d^{-1}); and ET_p will be partitioned into potential crop transpiration rate (T_p, cm d^{-1}) and potential soil evaporation rate (E_p, cm d^{-1}) for further calculation. The model provides two approaches to determine the potential ET with the P-M equation: (1) direct method, giving the minimal r_c ($r_{c,min}$) and actual crop height (h_c, cm) of the specific crop. h_c can be used to determine the actual r_a; and (2) indirect method, based on the reference ET rate from a grass reference surface and relevant conversion coefficients. Three kinds of ET_p under actual soil-crop conditions are calculated using Eq.(7.1): (1) ET_{w0} (cm d^{-1}), potential evapotranspiration rate of a wet canopy, completely covering the soil and not short of water; (2) ET_{p0} (cm d^{-1}), potential evapotranspiration rate of a dry canopy, completely covering the soil and not short of water; and (3) E_{p0} (cm d^{-1}), potential evaporation rate of a wet, bare soil.

For direct method: (a) ET_{w0} is calculated with a crop resistance r_c (0 s m^{-1}) and actual h_c. In the case of a wet canopy, the r_c is set to zero; (b) ET_{p0} is calculated with a user-specified r_c (e.g. 30-150 s m^{-1}) and actual h_c. In the case of a dry crop with optimal water supply in the soil, $r_{c,min}$ varies between 30 s m^{-1} for the arable crop to 150 s m^{-1} for trees in a forest (Allen et al., 1989); and (c) E_{p0} is calculated with assuming r_c = 0 s m^{-1} and h_c = 0.1 cm.

For the indirect method: (a) ET_{w0} was calculated by multiplying ET_{o2} with a crop factor ($K_{c,max}$ = 1.0-1.3). ET_{o2} is the reference evapotranspiration rate of wet grass surface, with r_c = 0 s m^{-1} and h_c = 12 cm. (b) ET_{p0} is calculated by multiplying ET_o with K_{tm}. ET_o is the reference evapotranspiration rate from a grass reference surface, with r_c = 70 s m^{-1} and h_c = 12 cm (Allen et al., 1998). ET_o is an index for the evaporating power of the atmosphere. K_{tm} is the conversion factor that serves as an aggregation of the physical and physiological differences between crops (non-stress) and the reference grass surface (Allen et al., 1998; Steduto et al., 2009).

(1) Division of ET_p: potential evaporation E_p

The potential ET is partitioned into potential soil evaporation rate (E_p, cm d^{-1}) and potential crop transpiration rate (T_p, cm d^{-1}) based on the Beer's law. The aerodynamic term for soil under a standing crop will be small because the wind velocity near the soil surface is relatively small, which makes the aerodynamic resistance r_a very large (Ritchie, 1972). Thus, the only source for soil evaporation is net radiation that reaches the soil surface. Thus, E_p is calculated based on the fraction of solar energy reaching the soil surface:

$$E_p = K_{em} \cdot E_{s0} \cdot e^{-k_{gr} LAI} \quad (7.2)$$

where K_{em} is the conversion factor for bare soil (=0.5-1.5), k_{gr} (-) is a coefficient governing the extinction of global (solar) radiation by the canopy, and LAI (-) is the leaf area index. k_{gr} is the function of sun angle, the distribution of plants, and the arrangement of leaves (between 0.5-0.75). k_{gr} can also be estimated as a factor (generally equal to 0.75) times the extinction coefficient of diffuse radiation (k_{dif}), as k_{dif} can be measured directly under diffuse sky conditions. The extinction coefficient for global solar radiation (GSR) is less than that for photosynthetic active radiation (PAR), because the absorption of near-infrared radiation by the canopy is less efficient (Monteith, 1969; de Wit et al., 2020). The k_{dif} for GSR is about 2/3 of that for PAR. Crops with more erect leaves (like many cereals) have lower k_{gr} values, whereas crops with more prostrate leaves show higher values of k_{gr} (de Wit et al., 2020). In AHC, for the LAI descending period, the E_P is allowed to be further adjusted for the sheltering effect of the withered canopy when green leaf LAI declines in the late-season stage. Therefore, the E_p could be further reduced when LAI starts to decline in the late season, as follows:

$$E_p = \left(1 - f_{wc}\right) K_{em} \cdot E_{s0} \cdot e^{-k_{gr} LAI} \tag{7.3}$$

$$f_{wc} = \begin{cases} 0 & \text{when } LAI \le LAI_{top} \\ f_{cc} \cdot CC_{top} = f_{cc} \cdot \left(1 - e^{-k_{gr} LAI_{top}}\right) & \text{when } LAI > LAI_{top} \end{cases} \tag{7.4}$$

where f_{wc} is the reduction factor due to the effect of the withered canopy (-), CC_{top} is canopy cover prior to senescence (-), LAI_{top} is the LAI at the start of the leaf decline stage, and f_{cc} (-) is a coefficient expressing the sheltering effect of the dead/withered canopy cover (Eq. 7.5).

$$f_{cc} = \frac{f_{wcc} \cdot CC}{\left(fr_{red} - 1\right) \cdot CC_{top}} + \frac{f_{wcc}}{1 - fr_{red}} \tag{7.5}$$

where CC is the canopy cover (-), f_{wcc} (-) is a coefficient representing the maximum value that f_{cc} can reach in the late stage of crop growth, and fr_{red} is a canopy cover fraction (-) indicating that the f_{cc} reaches to its final value f_{wcc} when CC/CC_{top} equals to this fraction. The value of f_{cc} increases gradually from 0 (at the start of the late-season stage) to its final value (default value

= 0.6) when the canopy cover is half of the CC_{top} (i.e., with f_{wcc}=0.6 and fr_{red} =0.5) (Raes et al., 2012). In AHC, the value of f_{wcc} should be specified by users.

(2) Division of ET_p: potential transpiration T_p

T_p is calculated based on a negative exponential function of LAI (Ritchie, 1972):

$$T_p = ET_{p0} \cdot \left(1 - e^{-k_{gr} LAI}\right) \tag{7.6}$$

In addition, AHC assumes that the transpiration rate through the leaf stomata is negligible during evaporation of intercepted water. The evaporation rate of the water intercepted by the vegetation approximates ET_{p0}. Then the fraction of the day that the crop is wet (W_{frac}, -) follows from the ratio of daily interception (R_{inter}, cm d^{-1}) and ET_{w0}, i.e. W_{frac} = min (1.0, R_{inter} /ET_{w0}). Therefore, T_p is corrected as:

$$T_p = ET_{p0} \cdot \left(1 - e^{-k_{gr} LAI}\right)\left(1 - W_{frac}\right) \tag{7.7}$$

Moreover, if the daily values of four meteorological data are not available, the model allows the use of the reference potential ET rate (ET_{ref}, cm d^{-1}). The above-mentioned potential ET is calculated as follows: ET_{w0} = ET_o = $K_{cr} \cdot ET_{ref}$ and E_{po}= $K_{sl} \cdot ET_{ref}$. The ET_{ref} rate can be determined in several ways, such as pan evaporation, the P-M equation, the Priestly-Taylor equation, or the Hargreaves equation. The corresponding crop and soil factors are required to be specified by the user in different approaches.

7.2 Radiation term

The average solar radiation at the top of the atmosphere is estimated at 1370 W m^{-2}. A daily solar radiation constant can then be calculated as a cosine times the average solar radiation at the top of the atmosphere multiplied by a correction factor to correct for the elliptical orbit of the earth around the sun. This correction factor is estimated to be 0.033.

$$S_{c,d} = S_c \left[1 + 0.033 \cos\left(2\pi \frac{t_d}{365}\right)\right] \tag{7.8}$$

where $S_{c,d}$ is the solar constant at the top of the atmosphere for a certain day (J m^{-2} d^{-1}), S_c is the average solar radiation at the top of the atmosphere (J m^{-2} d^{-1}) with S_c=1370 J m^{-2} s^{-1}, and t_d is the number of day since 1 January (-). The astronomical day length can be calculated as:

$$N_s = 12 + \frac{24}{\pi}\arcsin\left(\frac{\sin LD}{\cos LD}\right) \tag{7.9}$$

where N_s is day length (i.e. the maximum possible duration of sunshine hours) (h), sinLD (-) is the seasonal offset of sine of solar height (=sinδ sinλ_g), and cosLD (-) is the amplitude of sine of solar height (= cosδ cosλ_g). δ is the solar declination (Rad), and λ_g is the geographic latitude (Rad).

The solar declination during the year can be approached by a cosine function. The distance of the sun to the Earth is considered to be infinite; therefore, declination can be considered equal for all places on Earth.

$$\delta = -\arcsin\left[\sin\left(23.45\frac{\pi}{180}\right)\cdot\cos\left(2\pi\frac{t_d+10}{365}\right)\right] \tag{7.10}$$

The integral of the solar height over the day can be obtained as twice the integral from sunrise (β = 0 in degree) to 12 o'clock solar time (β = 90° + (δ –λ_g)·180/π in degree):

$$\int \sin\beta dt_h = 3600\left(L_d \sin LD + \frac{24}{\pi}\cos LD\sqrt{1-\left(\frac{\sin LD}{\cos LD}\right)^2}\right) \tag{7.11}$$

where ∫sinβ is the integral solar height, and β is the solar elevation (Rad). L_d is the day length (h).

Multiplication of Eq. (7.8) with Eq. (7.11) yields the extraterrestrial radiation ($R_{a,d}$, J m^{-2} d^{-1}), which is also known as the Angot radiation (received at the top of the earth’s atmosphere on a horizontal surface), as follows:

$$R_{a,d} = S_{c,d}\int \sin\beta dt_h \tag{7.12}$$

The net radiation R_n is the difference between incoming and outgoing radiation of both short and long wavelengths. It is the balance between the energy absorbed, reflected, and emitted by the earth's surface:

$$R_n = (1-\alpha)R_s - R_{nl} \tag{7.13}$$

where α is the albedo or canopy reflection coefficient (-), which is 0.23 for the hypothetical grass reference crop, and R_s is the incoming solar radiation (J m^{-2} d^{-1}). If the global solar radiation, R_s, is not measured, it can also be estimated from the extraterrestrial radiation and relative sunshine duration (Allen et al., 1998):

$$R_s = \left(a_s + b_s \frac{n_s}{N_s}\right) R_{a,d} \tag{7.14}$$

where n_s is the actual duration of sunshine or daylight hours (h), and a_s and b_s are empirical coefficients. a_s is the regression constant, expressing the fraction of extraterrestrial radiation $R_{a,d}$ reaching the earth on overcast days. a_s+b_s is the fraction of extraterrestrial radiation reaching the earth on clear days. The Angstrom values a_s and b_s will vary depending on atmospheric conditions (humidity, dust) and solar declination (latitude and month). Where no actual solar radiation data are available and no calibration has been carried out for improved a_s and b_s parameters, the values a_s = 0.25 and b_s = 0.50 are recommended (Allen et al., 1998).

The earth which has a much lower temperature than the sun emits longwave radiation. This emitted longwave radiation is absorbed by the atmosphere or is lost into space. Meanwhile, the atmosphere radiates energy of its own, and parts of radiation find their way back to the earth’s surface. Consequently, the earth’s surface both emits and receives longwave radiation. The net longwave radiation (R_{nl}, J m^{-2} d^{-1}) is the difference between outgoing and incoming longwave radiation and represents an energy loss from the soil surface. R_{nl} is proportional to the absolute temperature of the surface raised to the fourth power, which is expressed quantitatively by the Stefan-Boltzmann law. In addition, the net energy flux (R_{nl}) leaving the earth's surface is, however, less than that emitted and given by the Stefan-Boltzmann law due to the absorption and downward radiation from the sky. Water vapour, clouds, carbon dioxide, and dust are absorbers and emitters of longwave radiation. Their concentrations should be known when assessing the net outgoing flux. Therefore, the Stefan-Boltzmann law is corrected by considering the two main factors (i.e. humidity and cloudiness) to calculate the R_{nl}, as follows (Penman, 1956; Allen et al., 1998):

$$R_{nl} = \sigma_{SB}\left(\frac{T_{\max,K}^4 + T_{\min,K}^4}{2}\right)\left(0.34 - 0.14\sqrt{e_a}\right)\left(0.1 + 0.9N_{rel}\right) \quad (7.15)$$

where σ_{SB} is the Stefan-Boltzmann constant (4.903 10^{-3} J K^{-4} m^{-2} d^{-1}), $T_{max,K}$ is the maximum absolute temperature of the air during the 24-hour period [K], $T_{min,K}$ is the minimum absolute temperature of the air during the 24-hour period [K], and N_{rel} is the relative sunshine duration (-). N_{rel} is derived from the measured global solar radiation R_s and the extraterrestrial radiation $R_{a,d}$, as follows:

$$N_{rel} = \frac{n_s}{N_s} \quad \text{or} \quad N_{rel} = \frac{R_s}{b_s R_{a,d}} - \frac{a_s}{b_s} \quad (7.16)$$

7.3 Aerodynamic term

Latent heat of vaporization, λ_w (J kg^{-1}), depends on the air temperature T_{air} (°C) (Harrison, 1963):

$$\lambda_w = \left(2501 - 2.361 \cdot T_{air}\right) \cdot 10^3 \quad (7.17)$$

Saturation vapor pressure is also related to air temperature, and can be calculated as (Tetens, 1930):

$$e^0\left(T_{air}\right) = 0.6108 \exp\left(\frac{17.27 T_{air}}{T_{air} + 237.3}\right) \quad (7.18)$$

with $e^o(T_{air})$ the saturation vapor pressure (kPa) at air temperature T_{air}. The mean saturation vapor pressure for a day is estimated as the saturation vapor pressure at maximum and minimum temperature, as follows:

$$e_s = \frac{e^0(T_{\max}) + e^0(T_{\min})}{2} \tag{7.19}$$

The slope of the saturated vapor pressure curve, Δ_v, is calculated as (Murray, 1967):

$$\Delta_v = \frac{4098\left[0.6108\exp\left(\frac{17.27T_{\text{air}}}{T_{\text{air}} + 237.3}\right)\right]}{\left(T_{\text{air}} + 237.3\right)^2} \tag{7.20}$$

In addition, if the actual vapor pressure, e_{a}, is not measured, it can also be estimated from the measured relative humidity of air (H_{r}, -):

$$e_{\text{a}} = e_{\text{s}} \cdot H_{\text{r}} \tag{7.21}$$

The psychrometric constant, γ_{air}, (kPa °C^{-1}), of the instrument, follows from (Brunt, 2011):

$$\gamma_{\text{air}} = \frac{c_p p_{\text{air}}}{\varepsilon\lambda_w} = 0.00163\frac{p_{\text{air}}}{\lambda_w 10^{-6}} \tag{7.22}$$

where c_p is the specific heat at constant pressure (=1.013×10^{-3} kJ g^{-1} °C^{-1}), ε (-) is the ratio molecular weight of water vapor/day air (=0.622), and P_{air} is the atmospheric pressure (kPa) at elevation z_0 (m). P_{air} is calculated from (Burman et al., 1987):

$$P_{\text{air}} = 101.3\left(\frac{T_{\text{air},K} - 0.0065z_0}{T_{\text{air},K}}\right)^{5.256} \tag{7.23}$$

Employing the ideal gas law, the atmospheric/air density, ρ_{air} (kg m^{-3}), can be shown to depend on P_{air} and the virtual temperature T_{vir} (K):

$$\rho_{\text{air}} = 3.486\frac{P_{\text{air}}}{T_{\text{vir}}} \tag{7.24}$$

where the virtual temperature is derived from:

$$T_{\mathrm{vir}} = \frac{T_{\mathrm{air},K}}{1 - 0.378\dfrac{e_{\mathrm{a}}}{P_{\mathrm{air}}}} \quad (7.25)$$

The heat capacity of moisture air, C_{air} (J kg^{-1} °C^{-1}), follows from:

$$C_{\mathrm{air}} = 622\frac{\gamma_{\mathrm{air}}\lambda_w 10^{-3}}{P_{\mathrm{air}}} \quad (7.26)$$

The aerodynamic resistance, r_{a} (s m^{-1}), determines the transfer of heat and water vapor from the evaporating into the air above the canopy. It depends on the wind speed profile and the roughness of the canopy. Under neutral stability conditions, r_{a} can be calculated as (Allen et al., 1998):

$$r_{\mathrm{a}} = \frac{\ln\left(\dfrac{z_m - d}{z_{om}}\right)\ln\left(\dfrac{z_h - d}{z_{oh}}\right)}{\kappa^2 u_z} \quad (7.27)$$

where z_m is height of wind speed measurements (m), z_h is height of temperature and humidity measurements (m), d is zero plane displacement height (m), z_{om} is roughness length governing momentum transfer (m), z_{oh} is roughness length governing transfer of heat and vapor (m), κ (-) is von Karman's constant (= 0.41), and u_z is wind speed measured at height z_m (m s^{-1}). The parameters d, z_{om} and z_{oh} are defined as:

$$d = \frac{2}{3}h_c \quad (7.28)$$

$$z_{om} = 0.123h_c \quad (7.29)$$

$$z_{oh} = 0.1z_{om} \quad (7.30)$$

Note that for ET_{o} calculation, z_m and z_h are default to 2 m, and h_c is 0.12 m for reference grass. In addition, the wind speed at a height of 2 m (u_2, m s^{-1}) is used in Eq. (7.27) for ET_{o} calculation.

Wind speed is slowest at the soil surface and increases with height, due to surface friction tends to slow down wind passing over it. Anemometers are often placed at a standard altitude of 10 m in meteorological stations and 2 or 3 m in agro-meteorological stations. When assuming neutral stability conditions, the wind speed measured at any altitude can be transformed to u_2 by the following formula (Monin and Obukhov, 1954):

$$u_2 = u_z \ln\left(\frac{2-d}{z_{om}}\right) \Big/ \ln\left(\frac{z_m - d}{z_{om}}\right) \approx u_z \frac{4.868}{\ln\left(67.75 z_m - 5.42\right)} \tag{7.31}$$

In addition, the wind speed at the altitude of actual underlying surface height (e.g. crop height) is required for the calculation of $ET_{\mathrm{po}}/ET_{\mathrm{wo}}$ and E_{po}, when using Eq. (7.27). The wind speed is corrected with the following assumptions: (a) a uniform wind pattern at an altitude of 100 meters; (b) wind speed measurements are carried out above grassland; (c) a logarithmic wind profile is assumed; (d) below 2 m wind speed is assumed to be unchanged with respect to a value at altitude of 2 m; applying a logarithmic wind profile at low altitudes is not carried out due to the high variation below 2 m. These assumptions result in the following equation for wind speed correction:

$$u_{z,\mathrm{act}} = \frac{\ln \dfrac{100 - d_{\mathrm{gr}}}{z_{om,\mathrm{gr}}}}{\ln \dfrac{z_{m,mea} - d_{\mathrm{gr}}}{z_{om,\mathrm{gr}}}} \frac{\ln \dfrac{z_{m,\mathrm{act}} - d_{\mathrm{act}}}{z_{om,\mathrm{act}}}}{\ln \dfrac{100 - d_{act}}{z_{om,\mathrm{act}}}} u_{z,\mathrm{mea}} \tag{7.32}$$

where $z_{m,\mathrm{act}}$ is the actual crop height with a minimum value of 2 m (m), $z_{m,\mathrm{mea}}$ is the altitude of wind speed measurement above the mean soil surface (m), $u_{z,\mathrm{act}}$ is the wind speed at the altitude of $z_{m,\mathrm{act}}$ (m s^{-1}), d_{act} and d_{gr} are zero plane displacement of actual crop and grass (m), and $z_{om,\mathrm{act}}$ and $z_{om,\mathrm{gr}}$ are roughness parameter for momentum actual crop and grass (m).

7.4 Actual soil evaporation

The actual soil evaporation (E_a, cm d^{-1}) is firstly estimated with one of three optional approaches which are proposed by Black et al. (1969) and Allen et al. (1998), respectively. The former two approaches apply the empirical exponential functions that have been adopted by the SWAP and described in Kroes and van Dam (2003), in which the E_a is only related to the rainfall and time but not to the surface soil moisture. Thus, this implies that the calibrated functions would still result in an under- or over-estimation of soil evaporation when different upward flux from lower soil layers exists, such as under conditions of shallow water tables (Xu et al., 2015). The third approach uses a mechanism two-stage method, with the equation of $E_{a1}=K_{rd}E_p$. The evaporation reduction coefficient K_{rd} is calculated as a function of the soil water content of the surface layer, as follows:

$$\begin{cases} K_{rd} = 1 & \text{for energy stage} \\ K_{rd} = \dfrac{e^{f_k \cdot W_{relat}} - 1}{e^{f_k} - 1} & \text{for falling stage, } 0 \le K_r \le 1 \end{cases} \tag{7.33}$$

where f_k is a decline factor (-) and W_{relat} (-) is the relative water content of the soil layer through which water moves to the evaporating soil surface layer (i.e., topsoil layer with an effective thickness of $DZ_{e,top}$). When unknown, a value of 10-15 cm is recommended for $DZ_{e,top}$ (cm). The depth of the surface evaporation layer may also expand to a maximum depth of about 40 cm (Raes et al., 2012). The third approach is more general and recommended especially for the shallow water table conditions. W_{relat} is calculated as:

$$w_{relat} = \frac{\theta_{el} - 0.5 \cdot \theta_{WP,el}}{f_{rde} \cdot \theta_{FC,el} - 0.5 \cdot \theta_{WP,el}} \tag{7.34}$$

where θ_{el}, $\theta_{FC,el}$ and $\theta_{WP,el}$ are the soil water content, field capacity, and wilting point of the surface evaporation layer (cm^3 cm^3), respectively, which are all calculated from the soil water pressure head, and f_{rde} is the empirical fraction coefficient (with default value = 0.8).

In addition, the effect of mulches on soil evaporation can be considered to further limit the evaporation loss in the AHC. It is described by two factors, i.e., the fraction of soil surface covered by mulch (CF_{mul}, -) and the adjusted factor of mulch material (f_{mm}, -) (Allen et al., 1998). The E_{a1} is then adjusted by multiplying by a mulching reduction coefficient ($K_{rd,mul}$, -) that is calculated as follows:

$$K_{rd,mul} = 1 - CF_{mul} \cdot f_{mm} \tag{7.35}$$

where CF_{mul} varies from 0 to 100%, and f_{mm} varies between 0.5 for mulches of plant material and is close to 1.0 for plastic mulches. The adjustment is not applied when standing pond water remains on the soil surface. Finally, the AHC determines the actual evaporation rate (E_a) by taking the minimum value of E_p and E_a.

7.5 Hargreaves ET$_o$ equation

When the data of solar radiation, relative humidity, and/or wind speed are missing, the Hargreaves equation is alternatively employed to calculate the reference evapotranspiration:

$$ET_{o,hg} = K_{Harg} \cdot \left(\frac{T_{\max} + T_{\min}}{2} + 17.8 \right) \left(T_{\max} - T_{\min} \right)^{0.5} \frac{R_{a,\mathrm{d}}}{\lambda_w \rho_l} \tag{7.36}$$

where $ET_{o,hg}$ is the reference evapotranspiration rate with Hargreaves equation (mm d^{-1}), K_{Harg} is the empirical Hargreaves coefficient (-) that is commonly equal to 0.0023, T_{max} is the maximum air temperature during the 24-hour period (°C), and T_{min} is the minimum air temperature during the 24-hour period (°C). It is suggested that the Hargreaves equation (Eq. 7.36) should be verified in each new region by comparing with estimates by the FAO Penman-Monteith equation at weather stations where solar radiation, air temperature, humidity, and wind speed are measured (Allen et al., 1998).

7.6 Canopy resistance correction: VPD and CO$_2$

(1) VPD correction

When calculating actual evapotranspiration, the canopy resistance term is modified to reflect the impact of high vapor pressure deficit on leaf conductance (Stockle et al., 1992). For a plant species, a threshold vapor pressure deficit is defined at which the plant's leaf conductance begins to drop in response to the vapor pressure deficit. The adjusted leaf conductance is calculated:

$$g_l = \begin{cases} g_{l,mx} \cdot \left[1 - \Delta g_{l,dcl} \left(vpd - vpd_{thr} \right) \right] & \text{if } vpd > vpd_{thr} \\ g_{l,mx} & \text{if } vpd \le vpd_{thr} \end{cases} \tag{7.37}$$

where g_l is the conductance of a single leaf (m s^{-1}), $g_{l,mx}$ is the maximum conductance of a single leaf (m s^{-1}), $\Delta g_{l,dcl}$ is the rate of decline in leaf conductance per unit increase in vapor pressure deficit (m s^{-1} kPa^{-1}), vpd is the vapor pressure deficit (kPa), and vpd_{thr} is the threshold vapor pressure deficit above which a plant will exhibit reduced leaf conductance (kPa).

Similarly, r_s is the reciprocal of the canopy conductance, and it is corrected as:

$$r_s = \begin{cases} r_s \big/ \left[1 - \Delta g_{l,dcl}\left(vpd - vpd_{thr}\right)\right] & \text{if } vpd > vpd_{thr} \\ r_s & \text{if } vpd \le vpd_{thr} \end{cases} \tag{7.38}$$

The $\Delta g_{l,dcl}$ is calculated by solving Eq. (7.37) using measured values for stomatal conductance at two different vapor pressure deficits:

$$\Delta g_{l,dcl} = \frac{\left(1 - fr_{g,mx}\right)}{vpd_{fr} - vpd_{thr}} \tag{7.39}$$

where $fr_{g,mx}$ (-) is the fraction of g_l to the maximum stomatal conductance ($g_{l,mx}$), achieved at the vapor pressure deficit vpd_{fr}. The threshold vapor pressure deficit is assumed to be 1.0 kPa for all plant species.

(2) CO_2 correction

For climate change simulation, the canopy resistance term can be modified to reflect the impact of change in CO_2 concentration on leaf conductance. The influence of increasing CO_2 concentrations on leaf conductance was reviewed by some previous researchers (Morison, 1987; Li et al., 2019). Such as, Morison found that at CO_2 concentrations between 330 and 660 ppmv, a doubling in CO_2 concentration resulted in a 40% reduction in leaf conductance. Within the specified range, the reduction in conductance is linear (Morison and Gifford, 1983); meanwhile, the non-linear relationship is more commonly observed as referring to much subsequent research (Li et al., 2019). Despite that, in AHC, we still adopt a simple linear equation to describe the modification to the leaf conductance term for simulating carbon dioxide concentration effects on evapotranspiration, as follows:

$$g_{l,\mathrm{CO_2}} = g_l \cdot \left[(1 + p_{sc}) - p_{sc} \cdot \frac{CO_2}{330} \right], \qquad 330 \le CO_2 \le 660 \tag{7.40}$$

where $g_{l,CO2}$ is the leaf conductance modified to reflect CO_2 effects (m s^{-1}), CO_2 is the concentration of carbon dioxide in the atmosphere (ppmv), and p_{sc} is the percentage decrease in leaf stomatal conductance (from 330 to 660 ppmv) which is a user-specified parameter (-). Therefore, the r_s which is the reciprocal of the canopy conductance is corrected as:

$$r_{s,\mathrm{CO_2}} = r_s \Big/ \left[(1 + p_{sc}) - p_{sc} \cdot CO_2 / 330 \right] \tag{7.41}$$

8. Crop growth and yield (EPIC)

The crop growth module in the AHC (called EPIC_CGM) is mainly based on the concept of the EPIC crop growth model (Williams et al., 1989), also considering the effects of salinity stress and dormancy. This module is developed because its biological process is relatively simple but well applicable to the field/regional issues compared with some other detailed models. EPIC_CGM considers leaf area development, light interception, and the conversion of intercepted light into biomass and yield, together with the effects of temperature, water, and salt/nitrogen stress. Biomass is computed from the solar radiation intercepted by the crop leaf area, which is estimated with Beer's law (Monsi and Saeki, 1953). The potential increase in biomass on a given day is estimated as a function of the plant radiation-use efficiency (RUE) with consideration of environmental stress (e.g., from soil water, salinity or nitrogen, and temperature). The RUE is estimated using the approach proposed by Stockle et al. (1992).

LAI is computed for the various phenological development stages considering heat unit accumulation. LAI represents the level of canopy cover and is estimated as a function of heat units, crop stress, and development stages. Crop height is estimated like that in the EPIC model (Williams et al., 1989). The fraction of total biomass partitioned to the root system is 30-50% in seedlings and reduces to 5-20% in mature plants (Jones, 1985). AHC decreases the fraction of total biomass in roots (fr_{root}) piecewise linearly from a larger value (e.g., 0.4) at emergence to a smaller value at maturity (e.g., 0.2). The root depth generally increases rapidly from planting to a specific maximum depth by early mid-season (Borg and Grimes, 1986). The maximum root depth is required and could be specified according to the field observations or the previous literature. The vertical root distribution in the soil profile is assumed to be a piecewise linear function of the root depth.

The actual crop yield is calculated using the harvest index concept following the EPIC model procedures (Williams et al., 1989), i.e., as a function of the above-ground biomass and environmental stress (from soil temperature, soil salinity, fertilizers, etc.). The detailed calculation procedure is provided in the following parts.

8.1 Phenological development

The phonological development stage is controlled by heat unit accumulation. The heat unit is computed by using the equation:

$$HU_i = \left(\frac{T_{\max,i} + T_{\min,i}}{2} \right) - T_b \qquad HU_i \geq 0 \tag{8.1}$$

where *HU* is the heat unit (°C) on day *i* (i.e., equaling to average daily temperature minus base temperature of air), $T_{\max}$, and $T_{\min}$ are the maximum and minimum air temperature (°C), respectively, and T_b is the crop-specific base temperature (no growth occurs at or below T_b). The heat unit index (*HUI*, -), representing the fraction of potential heat units accumulated in a given day in the growing season, is calculated as follows:

$$HUI_i = \frac{\sum_{k=1}^{i} HU_k}{PHU} \tag{8.2}$$

where *PHU* is the total potential heat units required for crop maturation (°C). The value of *HUI* ranges from 0 at planting to 1 at physiological maturity.

8.2 Crop growth and stress

Biomass: The solar radiation intercepted by crop leaf area is estimated with Beer's law (Monsi and Saeki, 1953):

$$PAR_{day} = 0.5 \cdot RA_{day} \cdot [1 - \exp(-k_l \cdot LAI)] \tag{8.3}$$

where PAR_{day} is the intercepted photosynthetic active radiation (PAR) on a given day (MJ m^{-2}), RA_{day} and $0.5{\cdot}RA_{day}$ are respectively the total incident solar radiation (MJ m^{-2}) and incident photosynthetic active radiation, and k_l is the light extinction coefficient for PAR (-). The k_l varied with foliage characteristics, sun angle, row spacing and direction, and latitude. The default value of k_l used in the AHC (=0.65) is representative of crops with narrow row spacing, and a smaller value (0.4-0.6) might be suitable for wide row spacing and for tropical areas with higher sun angles (Williams et al., 1989). Note that the extinction coefficient for global solar radiation (k_{gr}) is about 2/3 of the extinction coefficient calculated for PAR (k_l) (de Wit et al., 2020), due to the absorption of near-infrared radiation by the canopy being less efficient.

The potential increase in biomass on a given day is estimated according to Monteith and Moss (1977), as follows:

$$\Delta B_a = \Delta B_p \cdot REG = RUE \cdot PAR_{day} \cdot REG \tag{8.4}$$

where ΔB_a and ΔB_p are the daily actual and potential increase in total biomass (kg ha^{-1}), respectively, RUE is the potential (unstressed) plant radiation-use efficiency for converting energy to biomass (including roots) (kg ha^{-1})$\cdot$(MJ m^{-2})$^{-1}$) (but not CO_2), and REG is the crop growth-regulating factor (-). Subsequently, the daily actual increase in aboveground biomass and root biomass (ΔB_{AG} and ΔB_{root}, kg ha^{-1}) is respectively calculated as:

$$\Delta B_{AG} = \Delta B_A \left(1 - fr_{root}\right) \tag{8.5}$$

$$\Delta B_{root} = \Delta B_A fr_{root} \tag{8.6}$$

where fr_{root} is the fraction of total biomass partitioned to the root system (-). The biomass for plant, aboveground part, and root, above on a given day (d), is respectively calculated as:

$$B_a = \sum_{i=1}^{d} \Delta B_{a,i} \tag{8.7}$$

$$B_{AG} = \sum_{i=1}^{d} \Delta B_{AG,i} \tag{8.8}$$

$$B_{root} = \sum_{i=1}^{d} \Delta B_{root,i} \tag{8.9}$$

where B_a, B_{AG} and B_{root} are the total plant biomass, aboveground biomass and root biomass on a given day (kg ha^{-1}), respectively, and i is the day number (kg ha^{-1}).

The REG value is equal to the minimum stress factor of water or water-salt stress, nitrogen stress, temperature stress, and temperature stress. The crop stress factors including water-salt stress (α_{WSC}, -), air temperature stress (α_{TC}, -), and nitrogen stress (α_{NC}, -) are calculated using Eq.(8.10), Eq. (8.11) and Eqs. (8.12-8.13), respectively.

$$\alpha_{WSC} = \frac{T_a}{T_P} \tag{8.10}$$

$$\alpha_{TC} = \sin\left(\frac{\pi}{2}\left(\frac{T_{avg} - T_b}{T_{opt} - T_b}\right)\right) \tag{8.11}$$

$$\alpha_{NC} = \frac{N_{scale,i}}{N_{scale,i} + \exp\left(3.52 - 0.026 N_{scale,i}\right)} \tag{8.12}$$

$$N_{scale,\,i} = 200\left[\frac{\sum_{k=1}^{i} UN_k}{(C_{NB})_i B_i} - 0.5\right] \tag{8.13}$$

where T_{opt} is the optimal temperature for crop growth (°C), and N_{scale} is a scaling factor (-) for the N stress factor.

In addition, RUE is sensitive to the atmospheric CO_2 level and affected by vapor pressure deficit (VPD, kPa) (the value of VPD is saturation vapor pressure at mean air temperature minus vapor pressure at mean air temperature). The equations of Stockle et al. (1992) are applied to adjust RUE for considering the effects of CO_2, as follows:

$$RUE^{*} = \frac{100 CO_2}{CO_2 + \exp\left[bc_1 - bc_2 \cdot (CO_2)\right]} \tag{8.14}$$

where bc_1 and bc_2 are crop shape factors. Furthermore, the VPD correction for adjusting RUE is accomplished in the equation:

$$\begin{aligned} RUE^{'} &= RUE^{*} - \Delta RUE_{slope}(VPD - VPD_{thr}) && \text{if } VPD > VPD_{thr} \\ RUE^{'} &= RUE^{*} && \text{if } VPD \le VPD_{thr} \end{aligned} \tag{8.15}$$

where ΔRUE_{slope} is the rate of decline in RUE per unit increase in VPD ((kg ha^{-1})$\cdot$(MJ m^{-2})$^{-1}\cdot$ kPa^{-1}), and VPD_{thr} is the threshold VPD for the crop (usually 0.5-1.0 kPa, with default value = 1.0 kPa in AHC). Finally, the RUE' is substituted into Eq. (8.4) to estimate the corrected biomass. The value of ΔRUE_{slope} varies among species, but a value of 6 to 8 is suggested as an approximation of most crops.

Leaf Area Index: LAI represents the level of canopy cover, and is estimated as a function of heat unit, crop stress, and development stages using the equations:

$$LAI_i = LAI_{i-1} + \Delta LAI \tag{8.16}$$

$$\Delta LAI = (HUF_i - HUF_{i-1}) LAI_{mx} (1 - \exp(5(LAI_{i-1} - LAI_{mx}))) \sqrt{REG_i} \tag{8.17}$$

where subscript i is the day number, Δ is the daily change, subscript mx refers to the maximum value possible for the crop, and HUF is the heat unit factor (-) (the fraction of the plant's maximum leaf area index corresponding to a given fraction of potential heat units for the plant). The HUF is calculated as follows:

$$HUF_i = \frac{HUI_i}{HUI_i + \exp(ah_1 - ah_2 \cdot HUI_i)} \tag{8.18}$$

where ah_1 and ah_2 are the shape coefficients for the corresponding crop. From the start of leaf decline till to the end of the growing season, the LAI is estimated as follows:

$$LAI_i = LAI_c \left(\frac{1 - HUI_i}{1 - HUI_c}\right)^{adl_j} \tag{8.19}$$

where adl_j is a parameter governing the rate of decline in LAI for crop j (-), and the subscript c represents the day number of the year when LAI starts declining.

Crop height: Crop height has a significant effect on the aerodynamic resistance in ET calculation. It can be directly specified as a function of the development stage, or estimated as follows:

$$h_{\mathrm{c},i} = h_{\mathrm{c},i-1} + \Delta h_{\mathrm{c},i} \tag{8.20}$$

$$\Delta h_{\mathrm{c},i} = h_{\mathrm{c,mx}} \left(\sqrt{HUF_i} - \sqrt{HUF_{i-1}}\right) \tag{8.21}$$

where $h_{\mathrm{c},i}$ is the canopy height on day i (cm), and $h_{\mathrm{c,mx}}$ is the maximum canopy height (cm).

Root system: The fraction of total biomass partitioned to the root system is 30-50% in seedlings and reduced to 5-20% in mature plants (Jones, 1985). This model decreases this fraction of total biomass in roots (fr_{root}, -) linearly from rb_1 (0.3-0.5) at emergence to rb_2 (0.05-0.2) at maturity, using the equation:

$$fr_{\text{root}} = rb_1 - rb_2 HUI_i \tag{8.22}$$

where rb_1 and rb_2 are crop parameters with typical values of 0.4 and 0.2, respectively. Root depth normally increases rapidly from the seeding depth to a specific maximum depth before maturity (Borg and Grimes, 1986). The daily increase in root depth is simulated as a function of heat unit index and potential root zone depth, as follows:

$$RD_i = RD_{i-1} + \Delta RD_i \qquad \text{if } RD_i \le RD_{\text{mx}} \tag{8.23}$$

$$\Delta RD_i = 2.5 RD_{\text{mx}} (HUI_i - HUI_{i-1}) \tag{8.24}$$

where RD_i is the root depth on day i (cm) and RD_{mx} is the maximum root depth (cm). The vertical root distribution in the soil profile is assumed to be a piecewise linear function of root depth.

Dormancy: The AHC assumes crops can go dormant in the winter season if the dormancy option is activated. During dormancy, transpiration is stopped and crops do not grow. The dates of the beginning and end of dormancy are required to be specified by users. It is suggested that the dates should be determined based on the field observation or temperature. Such as in North China Plain, winter wheat will generally enter dormancy if the 5-day sliding average temperature drops below 0 °C for the first time in the winter season, and will come out of dormancy when that temperature is above 0 °C in the next spring (Ma et al., 2005).

8.3 Crop yield

The actual crop yield is calculated using the harvest index concept:

$$Y_a = HI \cdot B_{AG} \tag{8.25}$$

where Y_a is the amount of the crop removed from the field (kg ha^{-1}), HI is the harvest index (-), and B_{AG} is the actual aboveground biomass (kg ha^{-1}). The harvest index increases from 0 at planting to HI at maturity. For non-stressed conditions, the potential harvest index (HIP, -) on day i of the plant's growing season is calculated with the equations:

$$HIP_i = HI_{pot} \cdot (\sum_{k=1}^{i} \Delta HUFH_k) \tag{8.26}$$

$$HUFH_i = \frac{100HUI_i}{100HUI_i + \exp\left(11.1 - 10HUI_i\right)} \tag{8.27}$$

where HI_{pot} is the potential harvest index at maturity for a specific crop, and $HUFH$ is the heat unit factor affecting the harvest index. Then, the HIP is corrected considering the effects of water and salinity stress, as follows:

$$HIA^{*} = \left(HIP - HI_{\min}\right)\frac{WDF}{WDF + \exp\left(6.13 - 0.0883WDF\right)} + HI_{\min} \tag{8.28}$$

$$WDF = 100\left(\frac{\sum_{i=1}^{m} T_a}{\sum_{i=1}^{m} T_p}\right) \tag{8.29}$$

where HIA^{*} is the actual harvest index (-), HI_{min} is the lower limit of harvest index (-) for the plant in drought conditions (under water-salt stress) and represents the minimal harvest index allowed for the plant, and WDF is the water deficiency factor (-) equaling to the ratio of actual crop transpiration to potential crop transpiration (Eq. 8.29). Finally, HIA^{*} is substituted into Eq. (8.25) to estimate the corrected crop yield.

8.4 Root respiration module

Root respiration (RR) is composed of two parts: root tissue respiration (R_{ts}) and rhizodeposit respiration (R_{rz}). R_{ts} is the actual respiration by roots to obtain energy for the maintenance of the metabolism and concentration gradient in cells (maintenance respiration), growth, and active uptake of nutrients. R_{rz} is the respiration by heterotrophic microorganisms decomposing organic substances released by living roots (rhizodeposits) (Kuzyakov and

Larionova, 2005). Thus, the root-derived CO_2 efflux (in contrast to SOM-derived CO_2) from the soil system (frequently termed rhizosphere respiration) could be divided into actual root tissue respiration (R_{ts}, respiration by autotrophs) and rhizomicrobial respiration (R_{rz}, respiration by heterotrophs). *RR* is modeled in relation to the root growth and specific root respiration rates. The amount of CO_2 released by root respiration (R_{ts}) was modeled as a function of the standing root tissue biomass plus a modification for soil temperature (Morell et al., 2012). Osman (1971) reported rates of wheat root respiration at 10, 20, and 30 °C and at three different stages of plant growth (Eq. 8.30). The Q_{10} (-) value, which defines the temperature dependence or sensitivity of root tissue respiration to temperature variation, is obtained by fitting lines to his data as follows:

$$Q_{10,i} = \begin{cases} 0 & T \le 0 \\ Q_{12,i} T / 12 & 0 < T < 12 \\ Q_{12,i} + (T - 12)/4 & T \ge 12 \end{cases} \tag{8.30}$$

where Q_{12} is a parameter (-) for the temperature sensitivity of root respiration on the i^{th} day, and its value changes with the growth stage (e.g. decreasing from 5.0 to 1.6 for the stages from emergence to harvest for barley, as suggested by Osman (1971)).

Root tissue respiration R_{ts} (mg CO_2-C cm^{-2} d^{-1}) is then described by:

$$R_{ts,i} = 10^{-5} R_m \varphi_r Q_{10,i} R_{F,i} \tag{8.31}$$

where R_m is the root mass in each layer (kg ha^{-1}), φ_r is a constant reflecting the average rate of respiration in mg CO_2-C d^{-1} per g dry matter and takes the value of 26.18 based on Osman (1971), and R_F is the proportion of photosynthate that is diverted to roots following Keulen and Seligman (1987).

In crops and grasses (such as wheat, ryegrass, barley, buckwheat, maize, meadow fescue, and prairie grasses), R_{ts} amounts on average to 48±5% and R_{rz} to 52±5% of root-derived CO_2 (Kuzyakov and Larionova, 2005). In AHC, microbial respiration is obtained the soil organic matter turnover module (section 6.1) directly, which includes microbial maintenance respiration, decomposition of SOM, and decomposition of rhizodeposits (Morell et al., 2012). This simplication implies the calculation of R_{rz} has already been included in the simulation of the soil carbon and nitrogen turnover process.

9. Numerical solution of water flow, solute and heat transport

9.1 Numerical solution of Richards equation

9.1.1 Space and time discretization

The continuity equation for water in the soil is given in the modified form of Richards Equation (Eq. 2.7). The numerical solution of this equation is not easy due to its hyperbolic nature, the high non-linearity of the soil hydraulic funcitons, freezing-thawing effects that relate the ice content with the temperature, and the rapid changes of surface boundary conditions. This water flow equation can be commonly solved using the implicit, backward, finite difference scheme with explicit linearization. In AHC, the numerical discretization of Eq. 2.7 can be expressed in finite difference forms as:

$$\begin{aligned}&\left(\theta_i^{j+1}-\theta_i^{j}\right)+\frac{\rho_i}{\rho_l}\left(\alpha_i^{j+1}-\alpha_i^{j}\right)=\\&\frac{\Delta t}{\Delta z_i}\left(K_{i-1/2}^{j+1}\frac{h_{i-1}^{j+1}-h_i^{j+1}}{\Delta z_u}+K_{i-1/2}^{j+1}-K_{i+1/2}^{j+1}\frac{h_i^{j+1}-h_{i+1}^{j+1}}{\Delta z_L}-K_{i+1/2}^{j+1}\right)-\frac{\Delta t}{\Delta z_i}\frac{1}{\rho_l}\left(q_{v,i-1/2}^{j+1}-q_{v,i+1/2}^{j+1}\right)-\Delta t S_i^{j}\end{aligned} \quad (9.1)$$

where $\Delta t = t^{j+1}-t^{j}$, $\Delta z_u = z_{i-1}-z_i$, $\Delta z_L = z_i - z_{i-1}$, Δz_i is the compartment thickness; the subscripts *j* and *j*+1 indicate values at *t* and *t*+Δ*t*, respectively, and *i* is the node number. Notice also that the sink term, *S*, is evaluated at the previous time level. Fig. 9.1 shows the symbols in the space-time domain.

Note that computing the internodal hydraulic conductivity appropriately is an important prerequisite for obtaining reliable soil water flow predictions. In AHC, the internodal hydraulic conductivity (e.g., $K_{i\text{-}1/2}$) can be optionally calculated as:

Arithmetic mean: $K_{i-1/2}=\frac{1}{2}\left(K_{i-1}+K_i\right)$ (9.2)

Harmonic mean: $K_{i-1/2}=\frac{2K_{i-1}K_i}{K_{i-1}+K}$ (9.3)

Geometric mean: $K_{i-1/2}=\left(K_{i-1}\right)^{1/2}\left(K_i\right)^{1/2}$ (9.4)

$$\text{Weighted arithmetic mean: } K_{i-1/2} = \frac{\Delta z_{i-1} K_{i-1} + \Delta z_i K_i}{\Delta z_{i-1} + \Delta z_i} \tag{9.5}$$

$$\text{Weighted harmonic mean: } K_{i-1/2} = \frac{\Delta z_{i-1} + \Delta z_i}{\dfrac{\Delta z_{i-1}}{K_{i-1}} + \dfrac{\Delta z_i}{K_i}} \tag{9.6}$$

$$\text{Weighted geometric mean: } K_{i-1/2} = \left[\left(K_{i-1} \right)^{\Delta z_{i-1}} \left(K_i \right)^{\Delta z_i} \right]^{\frac{1}{\Delta z_{i-1} + \Delta z_i}} \tag{9.7}$$

The geometric mean may cause the fluxes to be less sensitive to changes in nodal distance, such as, Haverkamp and Vauclin (1979) find that the accuracy of the model using this method is negligibly influenced by the change of space increments; however, it may severely underestimate the water fluxes when simulating infiltration in dry soils or high evaporation from wet soils. The simple arithmetic mean may cause some physical inconsistencies (Brunone et al., 2003). For instance, the arithmetic mean may overestimate the soil water fluxes in case of infiltration and evaporation events, or cause errors for layered soil conditions with coarse node division. However, if the nodal distances are very fine (e.g., in the order of 1 cm), the arithmetic mean can be close to the theoretically correct solution (van Dam and Feddes, 2000).

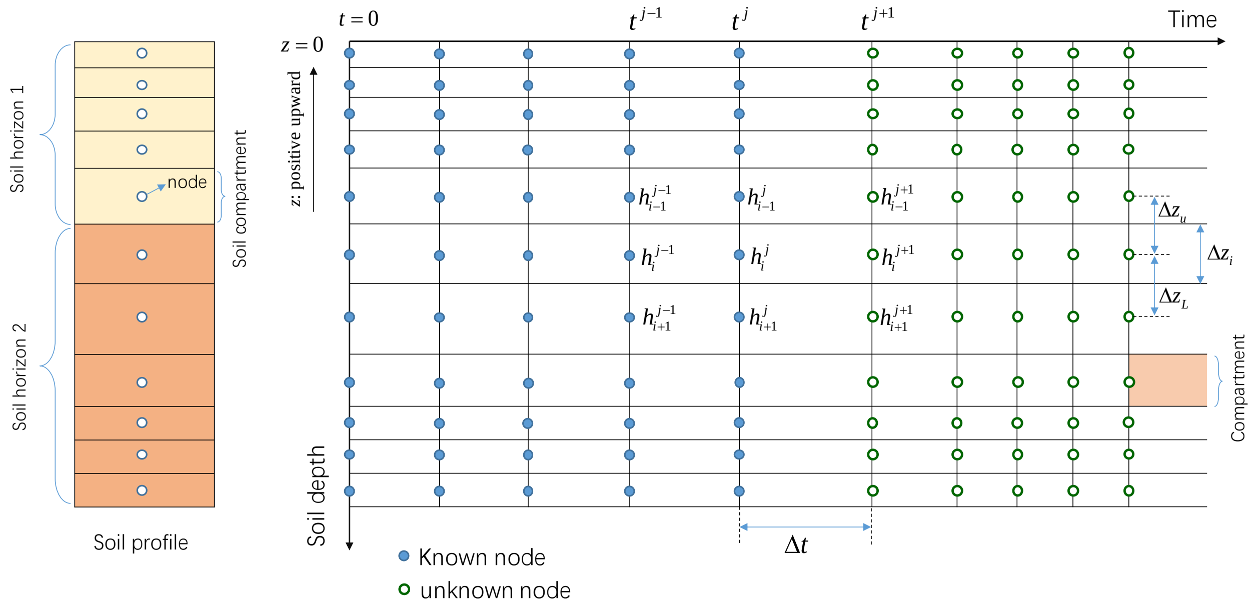


Figure 9.1 Spatial and temporal discretization with the time (t) and depth (z) for the solution of the Richards equation.

For any Euler method other than the explicit forward method, nonlinear algebraic equations result and some linearization and/or iteration procedure must be used to solve the discrete equations. Standard iteration techniques include Picard and Newton methods (Celia et al., 1990). In AHC, the above equation (Eq. 9.1) is solved using the Newton-Raphson iterative technique (N-RIT) in the following parts. In addition, AHC also provides a modified Picard iteration method (Celia et al., 1990) to solve Eq. 9.1 (see details in Annex C), and the Picard iteration is an extension of the fixed-point iteration method. The N-RIT finds the solution x, to a non-linear equation of which $f(x)$=0. At some value x^0, $f(x)$ will have a value F^0 usually other than zero. The difference between x^0 and the solution, x, may be approximated by:

$$F^0 + \frac{dF}{dx}\left(x^1 - x^0\right) = 0 \quad \text{or} \quad \frac{dF}{dx}\left(x^1 - x^0\right) = -F^0 \quad \text{or} \quad \frac{dF}{dx}\left(x^0 - x^1\right) = F^0 \qquad (9.8)$$

where x^1 is the approximation to x after the first iteration. A second approximation, x^2, is found by evaluating Eq. (9.8) at x^1. Iteration continues until:

$$\left|x^n - x^{n-1}\right| < \varepsilon_{NT} \qquad (9.9)$$

where ε_{NT} is an error tolerance.

The modified Richards equation contains more than one unknown, i.e., the state variable, x_i at $t+\Delta t$, and usually two adjacent nodes, $x_{i\text{-}1}$ and x_{i+1}. The Newton-Raphson expression in this case is (with $\Delta x = x^n - x^{n-1}$):

$$\frac{\partial F_i}{\partial x_{i-1}}\Delta x_{i-1} + \frac{\partial F_i}{\partial x_i}\Delta x_i + \frac{\partial F_i}{\partial x_{i+1}}\Delta x_{i+1} = -F_i \qquad (9.10)$$

By defining A_i, B_i and C_i as the respective partial derivatives, and D_i as $-F_i$, this equation may be expressed as:

$$A_i\Delta x_{i-1} + B_i\Delta x_i + C_i\Delta x_{i+1} = D_i \qquad (9.11)$$

Application of N-RIT to Eq. (9.1) for each node results in a tridiagonal matrix:

$$\begin{bmatrix} B_1 & C_1 & & & & & \\ A_2 & B_2 & C_2 & & & & \\ & A_3 & B_3 & C_3 & & & \\ & & & & & & \\ & & & & A_{N-1} & B_{N-1} & C_{N-1} \\ & & & & & A_N & B_N \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \Delta x_3 \\ \\ \Delta x_{N-1} \\ \Delta x_N \end{bmatrix} = \begin{bmatrix} D_1 \\ D_2 \\ D_3 \\ \\ D_{N-1} \\ D_N \end{bmatrix} \tag{9.12}$$

AHC provides two kinds of algorithms for N-RIT: one is an ordinary single-variable algorithm and the other is a primary variable switching algorithm. The former one uses the soil water pressure head (h) as the state variable x, which is described in detail in this section. The latter one uses h and ice content (α) as the switching state variable (see Annex D), which is dependent on whether the node is frozen.

Application of N-RIT to Eq. (9.1) gives the coefficients for the ***intermediate nodes*** (i=2~n-1), as follows:

$$A_i = -\frac{\Delta t}{\Delta z_i \Delta z_u} K_{i-1/2}^{j+1,p-1} + \frac{\Delta t}{\Delta z_i \rho_l} \frac{\partial q_{v,i-1/2}^{j+1,p-1}}{\partial h_{i-1}} \tag{9.13}$$

$$B_i = C_i^{j+1,p-1} + \frac{\Delta t}{\Delta z_i \Delta z_u} K_{i-1/2}^{j+1,p-1} + \frac{\Delta t}{\Delta z_i \Delta z_L} K_{i+1/2}^{j+1,p-1} + \frac{\Delta t}{\Delta z_i \rho_l} \left(\frac{\partial q_{v,i-1/2}^{j+1,p-1}}{\partial h_i^{j+1}} - \frac{\partial q_{v,i+1/2}^{j+1,p-1}}{\partial h_i^{j+1}} \right) \tag{9.14}$$

$$C_i = -\frac{\Delta t}{\Delta z_i \Delta z_L} K_{i+1/2}^{j+1,p-1} - \frac{\Delta t}{\Delta z_i \rho_l} \frac{\partial q_{v,i+1/2}^{j+1,p-1}}{\partial h_{i+1}} \tag{9.15}$$

$$\begin{aligned} D_i = & -\left(\theta_i^{j+1,p-1} - \theta_i^{j}\right) - \frac{\rho_i}{\rho_l}\left(\alpha_i^{j+1,p-1} - \alpha_i^{j}\right) - \frac{\Delta t}{\Delta z_i}\left(q_{i-1/2}^{j+1,p-1} - q_{i+1/2}^{j+1,p-1}\right) \\ & - \frac{\Delta t}{\Delta z_i \rho_l}\left(q_{v,i-1/2}^{j+1,p-1} - q_{v,i+1/2}^{j+1,p-1}\right) - \Delta t S_i^{j} \end{aligned} \tag{9.16}$$

where the superscript p is the iteration level and $C_i^{j+1,p-1}$ is the water capacity evaluated at the h value of the previous iteration. The terms $q_{i-1/2}^{j+1,p-1}$ and $q_{i+1/2}^{j+1,p-1}$ are as follows:

$$q_{i-1/2}^{j+1,p-1} = -\left(K_{i-1/2}^{j+1,p-1} \frac{h_{i-1}^{j+1,p-1} - h_i^{j+1,p-1}}{\Delta z_u} + K_{i-1/2}^{j+1,p-1} \right) \tag{9.17}$$

$$q_{i+1/2}^{j+1,p-1} = -\left(K_{i+1/2}^{j+1,p-1} \frac{h_i^{j+1,p-1} - h_{i+1}^{j+1,p-1}}{\Delta z_L} - K_{i+1/2}^{j+1,p-1} \right) \tag{9.18}$$

The terms $q_{v,i-1/2}^{j+1,p-1}$ and $q_{v,i+1/2}^{j+1,p-1}$ are as follows:

$$q_{v,i-1/2}^{j+1,p-1} = -\left(D_{v,i-1/2}^{j+1,p-1} h_{r,i}^{j+1,p-1} s_{v,i}^{j+1,p-1} \frac{T_{i-1}^{j+1,p-1} - T_i^{j+1,p-1}}{\Delta z_u} \right) - D_{v,i-1/2}^{j+1,p-1} \rho_{vs,i}^{j+1,p-1} \zeta_i^{j+1,p-1} \frac{h_{r,i-1}^{j+1,p-1} - h_{r,i}^{j+1,p-1}}{\Delta z_u} \tag{9.19}$$

$$q_{v,i+1/2}^{j+1,p-1} = -\left(D_{v,i+1/2}^{j+1,p-1} h_{r,i}^{j+1,p-1} s_{v,i}^{j+1,p-1} \frac{T_i^{j+1,p-1} - T_{i+1}^{j+1,p-1}}{\Delta z_L} \right) - D_{v,i+1/2}^{j+1,p-1} \rho_{vs,i}^{j+1,p-1} \zeta_i^{j+1,p-1} \frac{h_{r,i}^{j+1,p-1} - h_{r,i+1}^{j+1,p-1}}{\Delta z_L} \tag{9.20}$$

In addition, we have:

$$\frac{\partial \left(q_{v,i-1/2}^{j+1} - q_{v,i+1/2}^{j+1} \right)}{\partial h_{i-1}} = \frac{\partial q_{v,i-1/2}^{j+1}}{\partial h_{i-1}} = -\left(D_{v,i-1/2}^{j+1} \rho_{vs,i}^{j+1} \frac{1}{\Delta z_u} \right) h_{r,i-1}^{j+1} \left(\frac{Mg}{RT_k} \right) \tag{9.21}$$

$$\frac{\partial \left(q_{v,i-1/2}^{j+1,p} - q_{v,i+1/2}^{j+1,p} \right)}{\partial h_i} = \left(D_{v,i-1/2}^{j+1} \rho_{vs,i}^{j+1} \frac{1}{\Delta z_u} + D_{v,i+1/2}^{j+1} \rho_{vs,i}^{j+1} \frac{1}{\Delta z_L} \right) h_{r,i}^{j+1} \left(\frac{Mg}{RT_k} \right) \tag{9.22}$$

$$\frac{\partial \left(q_{v,i-1/2}^{j+1} - q_{v,i+1/2}^{j+1} \right)}{\partial h_{i+1}} = -\frac{\partial q_{v,i+1/2}^{j+1}}{\partial h_{i+1}} = -\left(D_{v,i+1/2}^{j+1} \rho_{vs,i}^{j+1} \frac{1}{\Delta z_L} \right) h_{r,i+1}^{j+1} \left(\frac{Mg}{RT_k} \right) \tag{9.23}$$

9.1.2 Treatment of top boundary conditions

(a) Flux boundary condition (top node, i=1)

Transforming the right-hand side of Eq. (9.1) yields:

$$\begin{aligned} &\left(\theta_1^{j+1} - \theta_1^j \right) + \frac{\rho_i}{\rho_l} \left(\alpha_i^{j+1} - \alpha_i^j \right) = \\ &\frac{\Delta t}{\Delta z_1} \left(-q_{sur} - K_{i+1/2}^{j+1} \frac{h_1^{j+1} - h_2^{j+1}}{\Delta z_L} - K_{i+1/2}^{j+1} \right) - \frac{\Delta t}{\Delta z_i} \left(0 - q_{v,i+1/2}^{j+1} \right) - \Delta t S_1^j \end{aligned} \tag{9.24}$$

Application of N-RIT to this equation and rearrangement give the coefficients:

$$B_1 = C_1^{j+1,p-1} + \frac{\Delta t}{\Delta z_1 \Delta z_L} K_{1+1/2}^{j+1,p-1} - \frac{\Delta t}{\Delta z_1 \rho_l} \frac{\partial q_{v,1+1/2}^{j+1,p-1}}{\partial h_1^{j+1}} \tag{9.25}$$

$$C_1 = -\frac{\Delta t}{\Delta z_i \Delta z_L} K_{1+1/2}^{j+1,p-1} - \frac{\Delta t}{\Delta z_i \rho_l} \frac{\partial q_{v,1+1/2}^{j+1,p-1}}{\partial h_2} \tag{9.26}$$

$$D_1 = -\left(\theta_1^{j+1,p-1} - \theta_1^{\,j}\right) - \frac{\rho_1}{\rho_l}\left(\alpha_1^{\,j+1,p-1} - \alpha_1^{\,j}\right) + \frac{\Delta t}{\Delta z_1}\left(-q_{sur} + q_{1+1/2}^{j+1,p-1}\right) - \frac{\Delta t}{\Delta z_1}\left(0 - q_{v,1+1/2}^{j+1,p-1}\right) - \Delta t S_1^{\,j} \tag{9.27}$$

(b) Third-type boundary condition (i=1)

Transforming the right-hand side of Eq. (9.1) yields:

$$\left(\theta_1^{\,j+1} - \theta_1^{\,j}\right) + \frac{\rho_i}{\rho_l}\left(\alpha_i^{\,j+1} - \alpha_i^{\,j}\right) = \frac{\Delta t}{\Delta z_1}\left(K_{1-1/2}^{j+1}\frac{h_{sur} - h_1^{j+1}}{\Delta z_u} + K_{1-1/2}^{j+1} - K_{1+1/2}^{j+1}\frac{h_1^{j+1} - h_2^{j+1}}{\Delta z_L} - K_{1+1/2}^{j+1}\right) - \frac{\Delta t}{\Delta z_1}\left(0 - q_{v,1+1/2}^{j+1}\right) - \Delta t S_1^{\,j} \tag{9.28}$$

Application of N-RIT to this equation and rearrangement give the coefficients:

$$B_1 = C_1^{j+1,p-1} + \frac{\Delta t}{\Delta z_1 \Delta z_u}K_{1-1/2}^{j+1,p-1} + \frac{\Delta t}{\Delta z_1 \Delta z_L}K_{1+1/2}^{j+1,p-1} - \frac{\Delta t}{\Delta z_1 \rho_l}\frac{\partial q_{v,1+1/2}^{j+1,p-1}}{\partial h_1^{j+1}} \tag{9.29}$$

$$C_1 = -\frac{\Delta t}{\Delta z_1 \Delta z_L}K_{1+1/2}^{j+1,p-1} - \frac{\Delta t}{\Delta z_i \rho_l}\frac{\partial q_{v,1+1/2}^{j+1,p-1}}{\partial h_2} \tag{9.30}$$

$$D_1 = -\left(\theta_1^{j+1,p-1} - \theta_1^{\,j}\right) - \frac{\rho_i}{\rho_l}\left(\alpha_1^{\,j+1,p-1} - \alpha_1^{\,j}\right) + \frac{\Delta t}{\Delta z_1}\left(-q_{1-1/2}^{j+1,p-1} + q_{1+1/2}^{j+1,p-1}\right) - \frac{\Delta t}{\Delta z_1}\left(0 - q_{v,1+1/2}^{j+1,p-1}\right) - \Delta t S_1^{\,j} \tag{9.31}$$

where the term $q_{1-1/2}^{j+1,p-1}$ is as follows:

$$q_{1-1/2}^{j+1,p-1} = -\left(K_{1-1/2}^{j+1,p-1}\frac{h_{sur} - h_1^{j+1,p-1}}{\Delta z_u} + K_{1-1/2}^{j+1,p-1}\right) \tag{9.32}$$

9.1.3 Treatment of bottom boundary conditions

(a) Pressure head boundary condition (i=N)

If a first-type (Dirichlet) boundary condition is specified at the bottom of the soil profile, then the term B_N is equal to unity, A_N reduces to zero, and D_N is equal to zero. Thus, the calculated pressure head for node 1 or N is always equal to the prescribed pressure head, h_0.

(b) Flux boundary condition (i=N)

Transforming the right-hand side of Eq. (9.1) yields:

$$\left(\theta_N{}^{j+1}-\theta_N{}^{j}\right)+\frac{\rho_i}{\rho_l}\left(\alpha_N{}^{j+1}-\alpha_N{}^{j}\right)=$$
$$\frac{\Delta t}{\Delta z_N}\left(K_{N-1/2}^{j+1}\frac{h_{N-1}^{j+1}-h_N^{j+1}}{\Delta z_u}+K_{N-1/2}^{j+1}+q_{bot}\right)-\frac{\Delta t}{\Delta z_N}\left(q_{v,N-1/2}^{j+1}-0\right)-\Delta t S_N^{j} \tag{9.33}$$

Application of N-RIT to this equation and rearrangement give the coefficients:

$$A_N=-\frac{\Delta t}{\Delta z_N \Delta z_N}K_{N-1/2}^{j+1,p-1}+\frac{\Delta t}{\Delta z_N \rho_l}\frac{\partial q_{v,N-1/2}^{j+1,p-1}}{\partial h_{N-1}} \tag{9.34}$$

$$B_N=C_N^{j+1,p-1}+\frac{\Delta t}{\Delta z_N \Delta z_u}K_{N-1/2}^{j+1,p-1}+\frac{\Delta t}{\Delta z_N \rho_l}\left(\frac{\partial q_{v,N-1/2}^{j+1,p-1}}{\partial h_N^{j+1}}\right) \tag{9.35}$$

$$D_N=-\left(\theta_N{}^{j+1,p-1}-\theta_N{}^{j}\right)-\frac{\rho_i}{\rho_l}\left(\alpha_N{}^{j+1,p}-\alpha_N{}^{j}\right)+\frac{\Delta t}{\Delta z_N}\left(-q_{N-1/2}^{j+1,p-1}+q_{bot}\right)$$
$$-\frac{\Delta t}{\Delta z_N}\left(q_{v,N-1/2}^{j+1,p-1}-0\right)-\Delta t S_N^{j} \tag{9.36}$$

(c) Third-type boundary condition (i=N)

Transforming the right-hand side of Eq. (9.1) yields:

$$\left(\theta_N{}^{j+1}-\theta_N{}^{j}\right)+\frac{\rho_i}{\rho_l}\left(\alpha_N{}^{j+1}-\alpha_N{}^{j}\right)=$$
$$\frac{\Delta t}{\Delta z_N}\left(K_{N-1/2}^{j+1}\frac{h_{N-1}^{j+1}-h_N^{j+1}}{\Delta z_u}+K_{N-1/2}^{j+1}-K_{N+1/2}^{j+1}\frac{h_N^{j+1}-h_{bot}}{\Delta z_L}-K_{N+1/2}^{j+1}\right)-\frac{\Delta t}{\Delta z_N}\left(q_{v,N-1/2}^{j+1}-0\right)-\Delta t S_N^{j} \tag{9.37}$$

Application of N-RIT to this equation and rearrangement give the coefficients:

$$A_N=-\frac{\Delta t}{\Delta z_N \Delta z_u}K_{N-1/2}^{j+1,p-1}+\frac{\Delta t}{\Delta z_N \rho_l}\frac{\partial q_{v,N-1/2}^{j+1,p-1}}{\partial h_{N-1}} \tag{9.38}$$

$$B_N=C_N^{j+1,p-1}+\frac{\Delta t}{\Delta z_N \Delta z_u}K_{N-1/2}^{j+1,p-1}+\frac{\Delta t}{\Delta z_N \Delta z_L}K_{N+1/2}^{j+1,p-1}+\frac{\Delta t}{\Delta z_N \rho_l}\frac{\partial q_{v,N-1/2}^{j+1,p-1}}{\partial h_N^{j+1}} \tag{9.39}$$

$$D_N=-\left(\theta_N{}^{j+1,p-1}-\theta_N{}^{j}\right)-\frac{\rho_i}{\rho_l}\left(\alpha_N{}^{j+1,p}-\alpha_N{}^{j}\right)+\frac{\Delta t}{\Delta z_N}\left(-q_{N-1/2}^{j+1,p-1}+q_{N+1/2}^{j+1,p-1}\right)$$
$$-\frac{\Delta t}{\Delta z_N}\left(q_{v,N-1/2}^{j+1,p-1}-0\right)-\Delta t S_N^{j} \tag{9.40}$$

where the term $q_{N+1/2}^{j+1,p-1}$ is as follows:

$$q_{N+1/2}^{j+1,p-1} = -\left(K_{N+1/2}^{j+1,p-1} \frac{h_N^{j+1,p-1} - h_{bot}}{\Delta z_L} + K_{N+1/2}^{j+1,p-1} \right) \tag{9.41}$$

9.2 Numerical solution of advection-dispersion equation (ADE) for solute transport

When considering the action/transformation term, the ADE (i.e. Eq. 4.10) for the mobile regions is written as to:

$$\frac{\partial(\theta c + \rho_b Q)}{\partial t} = -\frac{\partial(qc)}{\partial z} + \frac{\partial\left[\theta(D_{dif} + D_{dis})\frac{\partial c}{\partial z}\right]}{\partial z} - \mu\left(\theta c + \rho_b Q\right) - K_r Sc \tag{9.42}$$

$$Q = \frac{k_s c^{\beta}}{1+\eta c^{\beta}} \cdot f_m \tag{9.43}$$

Rearrangement of the above equation results in:

$$\frac{\partial(\theta c)}{\partial t} + R_1 \frac{\partial c}{\partial t} = -\frac{\partial(qc)}{\partial z} + \frac{\partial\left[\theta(D_{dif} + D_{dis})\frac{\partial c}{\partial z}\right]}{\partial z} - \mu\left(\theta c + \rho_b Q\right) - K_r Sc \tag{9.44}$$

$$R_1(c) = \frac{f_m \rho_b k_s \beta}{\left(1+\eta c^{\beta}\right)^2} c^{\beta-1} \tag{9.45}$$

where Q is the absorbed solute content (g g^{-1}) in mobile region corresponding to the solute concentration c. Notice that R_1 is the coefficient related to the retardation factor, and f_m equals the mobile fraction of the soil volume (cm^3 cm^{-3}) through which flow actually occurs (f_m=1 means no immobile zone). The mobile/immobile correction is needed as only the soil volume fraction f_m is mobile.

The solute ADE (Eq. 9.44) is numerically solved using the fully implicit or Crank-Nicholson finite difference scheme. An iterative procedure is necessary when a nonlinear reaction between the solid and liquid phases is considered. In this case, the parameter R_1 is a nonlinear function of the independent variable c. On the other hand, there is no need for an iterative solution process for solutions with non-adsorption (i.e., R_1=0) or undergoing only linear sorption reactions. In case of linear sorption, the R_1 is a constant term ($R_1 = f_m \rho_b k_s$), independent of c.

9.2.1 Equation discretization: Non-linear adsorption

For non-linear adsorption, the discrete form of transformed ADE (Eq. 9.44) is:

$$
\begin{aligned}
&\frac{\theta_i^{j+1} c_i^{j+1} - \theta_i^j c_i^j}{\Delta t} + \left[\varepsilon R_{1i}^{j+1} + (1-\varepsilon) R_{1i}^j \right] \frac{c_i^{j+1} - c_i^j}{\Delta t} = -\varepsilon q_{i-1/2}^{j+1} \left(\frac{\Delta z_{i-1}}{2\Delta z_i \Delta z_u} c_{i-1}^{j+1} + \frac{\Delta z_i}{2\Delta z_i \Delta z_u} c_i^{j+1} \right) \\
&+\varepsilon q_{i+1/2}^{j+1} \left(\frac{\Delta z_i}{2\Delta z_i \Delta z_\ell} c_i^{j+1} + \frac{\Delta z_{i+1}}{2\Delta z_i \Delta z_\ell} c_{i+1}^{j+1} \right) - (1-\varepsilon) \left(\frac{q_{i-1/2}^j c_{i-1/2}^j - q_{i+1/2}^j c_{i+1/2}^j}{\Delta z_i} \right) \\
&+ \frac{\varepsilon}{\Delta z_i} \left[\frac{\theta_{i-1/2}^{j+1} D_{i-1/2}^{j+1} \left(c_{i-1}^{j+1} - c_i^{j+1} \right)}{\Delta z_u} - \frac{\theta_{i+1/2}^{j+1} D_{i+1/2}^{j+1} \left(c_i^{j+1} - c_{i+1}^{j+1} \right)}{\Delta z_\ell} \right] \\
&+ \frac{(1-\varepsilon)}{\Delta z_i} \left[\frac{\theta_{i-1/2}^j D_{i-1/2}^j \left(c_{i-1}^j - c_i^j \right)}{\Delta z_u} - \frac{\theta_{i+1/2}^j D_{i+1/2}^j \left(c_i^j - c_{i+1}^j \right)}{\Delta z_\ell} \right] - \varepsilon \mu_i^{j+1} \theta_i^{j+1} c_i^{j+1} - (1-\varepsilon) \mu_i^j \theta_i^j c_i^j \\
&- \varepsilon \mu_i^{j+1} \rho_b Q_i^{j+1} - (1-\varepsilon) \mu_i^j \rho_b Q_i^j - \varepsilon K_r S_i c_i^{j+1} - (1-\varepsilon) K_r S_i c_i^j
\end{aligned}
\tag{9.46}
$$

where ε=1.0 is for the implicit scheme and ε=0.5 is for the Crank-Nicholson scheme. In addition, ε=0 is for the explicit scheme, which may not be stable enough in numerical solution and thus not adopted in AHC. Because the time step should meet the stability criterion for ensuring stability (van Genuchten and Wierenga, 1974). When the size of the time step exceeds a limit and the stability criterion is not satisfied, the numerical errors in the solution are amplified as time marches forward, leading to an invalid or unstable solution (Zheng and Bennett, 2002).

Application of Eq. (9.44) to each node results in a tridiagonal matrix:

$$
\begin{bmatrix}
\beta_1 & \gamma_1 & & & & \\
\alpha_2 & \beta_2 & \gamma_2 & & & \\
 & \alpha_3 & \beta_3 & \gamma_3 & & \\
 & & & & & \\
 & & & \alpha_{N-1} & \beta_{N-1} & \gamma_{N-1} \\
 & & & & \alpha_N & \beta_N
\end{bmatrix}
\begin{bmatrix}
c_1^{j+1} \\ c_2^{j+1} \\ c_3^{j+1} \\ \\ c_{N-1}^{j+1} \\ c_N^{j+1}
\end{bmatrix}
=
\begin{bmatrix}
f_1 \\ f_2 \\ f_3 \\ \\ f_{N-1} \\ f_N
\end{bmatrix}
\tag{9.47}
$$

(1) Intermediate nodes

Rearrangement of Eq. (9.46) results in the coefficients for the intermediate nodes:

$$\alpha_i = +\frac{\varepsilon q_{i-1/2}^{j+1}\Delta z_{i-1}}{2\Delta z_i\Delta z_u} - \frac{\varepsilon\theta_{i-1/2}^{j+1}D_{i-1/2}^{j+1}}{\Delta z_i\Delta z_u} \tag{9.48}$$

$$\begin{aligned}\beta_i = {} & \frac{\theta_i^{j+1}}{\Delta t} + \frac{\varepsilon R_{1i}^{j+1} + (1-\varepsilon)R_{1i}^{j}}{\Delta t} + \frac{\varepsilon q_{i-1/2}^{j+1}}{2\Delta z_u} - \frac{\varepsilon q_{i+1/2}^{j+1}}{2\Delta z_\ell} + \frac{\varepsilon\theta_{i-1/2}^{j+1}D_{i-1/2}^{j+1}}{\Delta z_i\Delta z_u} + \frac{\varepsilon\theta_{i+1/2}^{j+1}D_{i+1/2}^{j+1}}{\Delta z_i\Delta z_\ell} \\ & + \varepsilon\mu_i^{j+1}\theta_i^{j+1} + \varepsilon K_r S_i^{j}\end{aligned} \tag{9.49}$$

$$\gamma_i = -\frac{\varepsilon q_{i+1/2}^{j+1}\Delta z_{i+1}}{2\Delta z_i\Delta z_\ell} - \frac{\varepsilon\theta_{i+1/2}^{j+1}D_{i+1/2}^{j+1}}{\Delta z_i\Delta z_\ell} \tag{9.50}$$

$$\begin{aligned}f_i = {} & \frac{\theta_i^{j}c_i^{j}}{\Delta t} + \left[\varepsilon R_{1i}^{j+1} + (1-\varepsilon)R_{1i}^{j}\right]\frac{c_i^{j}}{\Delta t} - (1-\varepsilon)\frac{q_{i-1/2}^{j}c_{i-1/2}^{j} - q_{i+1/2}^{j}c_{i+1/2}^{j}}{\Delta z_i} \\ & + (1-\varepsilon)\left[\frac{\theta_{i-1/2}^{j}D_{i-1/2}^{j}\left(c_{i-1}^{j} - c_i^{j}\right)}{\Delta z_i\Delta z_u} - \frac{\theta_{i+1/2}^{j}D_{i+1/2}^{j}\left(c_i^{j} - c_{i+1}^{j}\right)}{\Delta z_i\Delta z_\ell}\right] \\ & - (1-\varepsilon)\mu_i^{j}\theta_i^{j}c_i^{j} - \varepsilon\mu_i^{j+1}\rho_b Q_i^{j+1} - (1-\varepsilon)\mu_i^{j}\rho_b Q_i^{j} - (1-\varepsilon)K_r S_i\, c_i^{j}\end{aligned} \tag{9.51}$$

(2) Tope node: third-type boundary condition

The right-hand side of Eq. (9.46) transforms to:

$$\begin{aligned} & -\varepsilon\left(\frac{Q_{ct}^{j+1}}{\Delta z_i}\right) - \varepsilon q_{1½}^{j+1}\left(\frac{c_1^{j+1}}{2\Delta z_\ell} + \frac{\Delta z_2}{2\Delta z_1\Delta z_\ell}c_2^{j+1}\right) - (1-\varepsilon)\left(\frac{Q_{ct}^{j} - q_{1½}^{j}c_{1½}^{j}}{\Delta z_1}\right) \\ & -\varepsilon\left[\frac{\theta_{1½}^{j+1}D_{1½}^{j+1}\left(c_1^{j+1} - c_2^{j+1}\right)}{\Delta z_1\Delta z_\ell}\right] - (1-\varepsilon)\left[\frac{\theta_{1½}^{j}D_{1½}^{j}\left(c_1^{j} - c_2^{j}\right)}{\Delta z_1\Delta z_\ell}\right] - \varepsilon\mu_1^{j+1}\theta_1^{j+1}c_1^{j+1} - (1-\varepsilon)\mu_1^{j}\theta_1^{j}c_1^{j} \\ & -\varepsilon\mu_1^{j+1}\rho_b Q_1^{j+1} - (1-\varepsilon)\mu_1^{j}\rho_b Q_1^{j} - \varepsilon K_r S_1 c_1^{j+1} - (1-\varepsilon)K_r S_1 c_1^{j}\end{aligned} \tag{9.52}$$

Rearrangement gives the coefficients:

$$\begin{aligned}\beta_1 = {} & \frac{\theta_1^{j+1}}{\Delta t} + \frac{\varepsilon R_{1,1}^{j+1} + (1-\varepsilon)R_{1,1}^{j}}{\Delta t} - \frac{\varepsilon q_{1½}^{j+1}}{2\Delta z_\ell} + \frac{\varepsilon\theta_{1½}^{j+1}D_{1½}^{j+1}}{\Delta z_1\Delta z_\ell} \\ & + \varepsilon\mu_1^{j+1}\theta_1^{j+1} + \varepsilon K_r S_1\end{aligned} \tag{9.53}$$

$$\gamma_1 = -\frac{\varepsilon q_{1½}^{j+1}\Delta z_2}{2\Delta z_1\Delta z_\ell} - \frac{\varepsilon\theta_{1½}^{j+1}D_{1½}^{j+1}}{\Delta z_1\Delta z_\ell} \tag{9.54}$$

$$f_1 = \frac{\theta_1^j c_1^j}{\Delta t} + \left[\varepsilon R_{1,1}^{j+1} + (1-\varepsilon) R_{1,1}^j \right] \frac{c_i^j}{\Delta t} - \varepsilon \frac{Q_{ct}^{j+1}}{\Delta z_1} - (1-\varepsilon)\left(\frac{Q_{ct}^{j+1} - q_{1\frac{1}{2}}^j c_{1\frac{1}{2}}^j}{\Delta z_i} \right)$$
$$- (1-\varepsilon)\left[\frac{\theta_{1\frac{1}{2}}^j D_{1\frac{1}{2}}^j \left(c_1^j - c_2^j \right)}{\Delta z_i \Delta z_\ell} \right] - (1-\varepsilon)\mu_1^j \theta_1^j c_1^j - \varepsilon \mu_1^{j+1} \rho_b Q_1^{j+1} \tag{9.55}$$
$$- (1-\varepsilon)\mu_1^j \rho_b Q_1^j - (1-\varepsilon) K_r S_1 c_1^j$$

(3) Bottom node

(a) Concentration boundary condition

If a first-type (Dirichlet) boundary condition is specified at the bottom of the soil profile, then the term B_N is equal to unity, α_N reduces to zero, and D_1 or D_N is equal to the prescribed solute concentration, c_0.

(b) third-type boundary condition

The right-hand side of Eq. (9.46) transforms to:

$$-\varepsilon q_{N-\frac{1}{2}}^{j+1} \left(\frac{\Delta z_{N-1}}{2\Delta z_N \Delta z_u} c_{N-1}^{j+1} + \frac{1}{2\Delta z_u} c_N^{j+1} \right) + \varepsilon \left(\frac{Q_{cb}^{j+1}}{\Delta z_N} \right) - (1-\varepsilon)\left(\frac{q_{N-\frac{1}{2}}^j c_{N-\frac{1}{2}}^j - Q_{cb}^j}{\Delta z_N} \right) +$$
$$\varepsilon \left[\frac{\theta_{N-\frac{1}{2}}^{j+1} D_{N-\frac{1}{2}}^{j+1} \left(c_{N-1}^{j+1} - c_N^{j+1} \right)}{\Delta z_N \Delta z_u} \right] - (1-\varepsilon)\left[\frac{\theta_{N-\frac{1}{2}}^j D_{N-\frac{1}{2}}^j \left(c_{N-1}^j - c_N^j \right)}{\Delta z_N \Delta z_u} \right] - \varepsilon \mu_N^{j+1} \theta_N^{j+1} c_N^{j+1} - \tag{9.56}$$
$$(1-\varepsilon)\mu_N^j \theta_N^j c_N^j - \varepsilon \mu_N^{j+1} \rho_b Q_N^{j+1} - (1-\varepsilon)\mu_N^j \rho_b Q_N^j - \varepsilon K_r S_N c_N^{j+1} - (1-\varepsilon) K_r S_N c_N^j$$

Rearrangement gives the coefficients:

$$\alpha_N = \frac{\varepsilon q_{N-\frac{1}{2}}^{j+1} \Delta z_{N-1}}{2\Delta z_N \Delta z_u} - \frac{\varepsilon \theta_{N-\frac{1}{2}}^{j+1} D_{N-\frac{1}{2}}^{j+1}}{\Delta z_N \Delta z_u} \tag{9.57}$$

$$\beta_N = \frac{\theta_N^{j+1}}{\Delta t} + \frac{\varepsilon R_{1,N}^{j+1} + (1-\varepsilon) R_{1,N}^j}{\Delta t} + \frac{\varepsilon q_{N-\frac{1}{2}}^{j+1}}{2\Delta z_u} + \frac{\varepsilon \theta_{N-\frac{1}{2}}^{j+1} D_{N-\frac{1}{2}}^{j+1}}{\Delta z_N \Delta z_u} \tag{9.58}$$
$$+ \varepsilon \mu_N^{j+1} \theta_N^{j+1} + \varepsilon K_r S_N^{j+1}$$

$$f_N = \frac{\theta_N^j c_N^j}{\Delta t} + \left[\varepsilon R_{1,N}^{j+1} + (1-\varepsilon) R_{1,N}^j \right] \frac{c_N^j}{\Delta t} + \varepsilon \frac{Q_{cb}^{j+1}}{\Delta z_N} - (1-\varepsilon)\left(\frac{q_{N-\frac{1}{2}}^j c_{N-\frac{1}{2}}^j - Q_{ct}^{j+1}}{\Delta z_N} \right)$$
$$+ (1-\varepsilon)\left[\frac{\theta_{N-\frac{1}{2}}^j D_{N-\frac{1}{2}}^j \left(c_{N-1}^j - c_N^j \right)}{\Delta z_N \Delta z_u} \right] - (1-\varepsilon)\mu_N^j \theta_N^j c_N^j - \varepsilon \mu_N^{j+1} \rho_b Q_N^{j+1} \tag{9.59}$$
$$- (1-\varepsilon)\mu_N^j \rho_b Q_N^j - (1-\varepsilon) K_r S_N c_N^j$$

9.2.2 Equation discretization: Non-adsorption and linear adsorption

For linear adsorption, the discrete form of transformed ADE (Eq. 9.44) is:

$$
\begin{aligned}
&\frac{\theta_i^{j+1}c_i^{j+1}-\theta_i^{j}c_i^{j}}{\Delta t}+\left[\varepsilon R_{1i}^{j+1}+(1-\varepsilon)R_{1i}^{j}\right]\frac{c_i^{j+1}-c_i^{j}}{\Delta t}=-\varepsilon q_{i-1/2}^{j+1}\left(\frac{\Delta z_{i-1}}{2\Delta z_i\Delta z_u}c_{i-1}^{j+1}+\frac{\Delta z_i}{2\Delta z_i\Delta z_u}c_i^{j+1}\right)\\
&+\varepsilon q_{i+1/2}^{j+1}\left(\frac{\Delta z_i}{2\Delta z_i\Delta z_\ell}c_i^{j+1}+\frac{\Delta z_{i+1}}{2\Delta z_i\Delta z_\ell}c_{i+1}^{j+1}\right)-(1-\varepsilon)\left(\frac{q_{i-1/2}^{j}c_{i-1/2}^{j}-q_{i+1/2}^{j}c_{i+1/2}^{j}}{\Delta z_i}\right)\\
&+\frac{\varepsilon}{\Delta z_i}\left[\frac{\theta_{i-1/2}^{j+1}D_{i-1/2}^{j+1}\left(c_{i-1}^{j+1}-c_i^{j+1}\right)}{\Delta z_u}-\frac{\theta_{i+1/2}^{j+1}D_{i+1/2}^{j+1}\left(c_i^{j+1}-c_{i+1}^{j+1}\right)}{\Delta z_\ell}\right]\\
&+\frac{(1-\varepsilon)}{\Delta z_i}\left[\frac{\theta_{i-1/2}^{j}D_{i-1/2}^{j}\left(c_{i-1}^{j}-c_i^{j}\right)}{\Delta z_u}-\frac{\theta_{i+1/2}^{j}D_{i+1/2}^{j}\left(c_i^{j}-c_{i+1}^{j}\right)}{\Delta z_\ell}\right]-\varepsilon\mu_i^{j+1}\theta_i^{j+1}c_i^{j+1}-(1-\varepsilon)\mu_i^{j}\theta_i^{j}c_i^{j}\\
&-\varepsilon\mu_i^{j+1}\rho_b k_s c_i^{j+1} f_m-(1-\varepsilon)\mu_i^{j}\rho_b k_s c_i^{j} f_m-\varepsilon K_r S_i\, c_i^{j+1}-(1-\varepsilon)K_r S_i\, c_i^{j}
\end{aligned}
\tag{9.60}
$$

If the parameter k_s is set to zero, it becomes the equation for non-adsorption conditions. There is no need for an iterative solution process for solutions with non-adsorption or undergoing only linear sorption reactions. R_1 equals 0 for non-adsorption, and R_1 is a constant term ($R_1=f_m\rho_b k_s$) independent of c for linear sorption reactions.

Rearrangement of Eq. (9.60) results in the coefficients (noted as α_i', γ_i', β_i' and f_i') for all nodes:

$$\alpha_i' = \alpha_i \tag{9.61}$$

$$\beta_i' = \beta_i + \varepsilon f_m \mu_i^{j+1}\rho_b k_s \tag{9.62}$$

$$\gamma_i' = \gamma_i \tag{9.63}$$

$$f_i' = f_i + \varepsilon\mu_i^{j+1}\rho_b Q_i^{j+1} \tag{9.64}$$

9.2.3 Immobile solute transport

After completing the numerical solution of the solute ADE equation for the mobile zone, the solute concentration in the immobile zone (c_{im}) is calculated based on solving Eq. (4.21) with the explicit scheme. In AHC, the numerical discretization of Eq. (4.21) can be expressed in finite difference forms as:

$$\frac{\left(1-f_{\mathrm{m}}^{j+1}\right)\left(\theta_{\mathrm{im}}^{j+1}c_{\mathrm{im}}^{j+1}+\rho_b\frac{k_s\left(c_{\mathrm{im}}^{j+1}\right)^{\beta}}{1+\eta\left(c_{\mathrm{im}}^{j+1}\right)^{\beta}}\right)-\left(1-f_{\mathrm{m}}^{j}\right)\left(\theta_{\mathrm{im}}^{j}c_{\mathrm{im}}^{j}+\rho_b Q_{\mathrm{im}}^{j}\right)}{\Delta t}=-\left(1-f_{\mathrm{m}}^{j+1}\right)\left(\mu_w\theta_{im}c_{\mathrm{im}}^{j}+\mu_s\rho_b Q_{\mathrm{im}}^{j}\right)+G_c \qquad (9.65)$$

It can be further rewritten as:

$$\begin{aligned}\left(1-f_{\mathrm{m}}^{j+1}\right)\left(\theta_{\mathrm{im}}^{j+1}c_{\mathrm{im}}^{j+1}+\rho_b\frac{k_s\left(c_{\mathrm{im}}^{j+1}\right)^{\beta}}{1+\eta\left(c_{\mathrm{im}}^{j+1}\right)^{\beta}}\right)&=\left(1-f_{\mathrm{m}}^{j}\right)\left(\theta_{\mathrm{im}}^{j}c_{\mathrm{im}}^{j}+\rho_b Q_{\mathrm{im}}^{j}\right)\\&\quad-\left(1-f_{\mathrm{m}}^{j+1}\right)\left(\mu_w\theta_{im}c_{\mathrm{im}}^{j}+\mu_s\rho_b Q_{\mathrm{im}}^{j}\right)\Delta t+G_c\Delta t\end{aligned} \qquad (9.66)$$

With $X_{\mathrm{im}}{}^{j}=\left(1-f_{\mathrm{m}}^{j}\right)\left(\theta_{\mathrm{im}}^{j}c_{\mathrm{im}}^{j}+\rho_b Q_{\mathrm{im}}^{j}\right)$, we obtain:

$$\left(1-f_{\mathrm{m}}^{j+1}\right)\left(\theta_{\mathrm{im}}^{j+1}c_{\mathrm{im}}^{j+1}+\rho_b\frac{k_s\left(c_{\mathrm{im}}^{j+1}\right)^{\beta}}{1+\eta\left(c_{\mathrm{im}}^{j+1}\right)^{\beta}}\right)=X_{\mathrm{im}}{}^{j}-\left(1-f_{\mathrm{m}}^{j+1}\right)\left(\mu_w\theta_{im}c_{\mathrm{im}}^{j}+\mu_s\rho_b Q_{\mathrm{im}}^{j}\right)\Delta t+G_c\Delta t \qquad (9.67)$$

where X_{im} is the solutes present per volume of soil in the immobile region (g cm^{-3}). For non-adsorption and linear adsorption, the c_{im} can be directly calculated as:

$$c_{\mathrm{im}}^{j+1}=\frac{X_{\mathrm{im}}{}^{j}-\left(1-f_{\mathrm{m}}^{j+1}\right)\left(\mu_w\theta_{im}c_{\mathrm{im}}^{j}+\mu_s\rho_b Q_{\mathrm{im}}^{j}\right)\Delta t+G_c\Delta t}{\left(1-f_{\mathrm{m}}^{j+1}\right)\left(\theta_{\mathrm{im}}^{j+1}+\rho_b k_s\right)} \qquad (9.68)$$

For non-linear adsorption, c_{im} is obtained by the Newton iteration method (see Annex E).

9.3 Numerical solution of advection-dispersion equation (ADE) for heat transport

9.3.1 Equation discretization

The energy balance equation which includes terms for convection heat transport by liquid and vapor for a layer of freezing soil is given in Eq. (5.7). The discrete form (or finite-difference expression) of Eq. (5.7) can be written as:

$$\begin{aligned}&\frac{C_s^{j+1}T_i^{j+1}-C_s^{j}T_i^{j}}{\Delta t}-L_f\rho_i\frac{\alpha_i^{j+1}-\alpha_i^{j}}{\Delta t}+L_v\frac{\partial\rho_v}{\partial T}\frac{T_i^{j+1}-T_i^{j}}{\Delta t}+L_v\frac{q_{v,i-1/2}^{j+1}-q_{v,i+1/2}^{j+1}}{\Delta z_i}=\\&\frac{1}{\Delta z_i}\left(\lambda_{i-1/2}^{j+1}\frac{T_{i-1}^{j+1}-T_i^{j+1}}{\Delta z_u}-\lambda_{i+1/2}^{j+1}\frac{T_i^{j+1}-T_{i+1}^{j+1}}{\Delta z_L}\right)-\rho_l C_l\frac{q_{l,i-1/2}^{j+1}\left(T_{i-1}^{j+1}+T_i^{j+1}\right)/2}{\Delta z_i}+\rho_l C_l\frac{q_{l,i+1/2}^{j+1}\left(T_i^{j+1}+T_{i+1}^{j+1}\right)/2}{\Delta z_i}\\&-c_v\frac{q_{v,i-1/2}^{j+1}\left(T_{i-1}^{j+1}+T_i^{j+1}\right)/2}{\Delta z_i}+c_v\frac{q_{v,i+1/2}^{j+1}\left(T_i^{j+1}+T_{i+1}^{j+1}\right)/2}{\Delta z_i}-S_{h,i}-\rho_l C_l S_{w,i}T_{w,i}^{j}\end{aligned} \qquad (9.69)$$

When ice is present in node i, the Richards equation can be written as:

$$\frac{\partial \theta_l}{\partial t} + \frac{\rho_i}{\rho_l}\frac{\partial \alpha}{\partial t} = -\frac{\partial q_l}{\partial z} - S_w \tag{9.70}$$

Then, the second term on the left side of Eq. (9.58) can be transformed as follows:

$$L_f \rho_i \frac{\partial \alpha}{\partial t} = -L_f \rho_l \left(\frac{\partial \theta_l}{\partial t} + \frac{\partial q_l}{\partial z} + S_w \right) \tag{9.71}$$

It can be further rewritten as:

$$\begin{aligned} L_f \rho_i \frac{\partial \alpha}{\partial t} &= -L_f \rho_l \left(\frac{\partial \theta_l}{\partial t} + \frac{\partial q_l}{\partial z} + S_w \right) \\ &= -L_f \rho_l \left(\frac{\partial \theta_l}{\partial T}\frac{\partial T}{\partial t} + \frac{\partial q_l}{\partial z} + S_w \right) \\ &= -L_f \rho_l \left(C_T \frac{\partial T}{\partial t} + \frac{\partial q_l}{\partial z} + S_w \right) \end{aligned} \tag{9.72}$$

where C_T is the derivation of θ_l with respect to T, which is calculated by Eq. A-18 (see the derivation in Annex A). In addition, the derivate of q_l with respective to z can be expressed as:

$$\frac{\partial q_l}{\partial z} = \frac{\partial}{\partial z}\left[-K\left(\frac{\partial h}{\partial z} + 1 \right) \right] = \frac{\partial}{\partial z}\left[-K\left(\frac{\partial h}{\partial \theta_u}\frac{\partial \theta_u}{\partial T}\frac{\partial T}{\partial z} + 1 \right) \right] = -\frac{\partial}{\partial z}\left[K\frac{C_T}{C_w}\frac{\partial T}{\partial z} + K \right] \tag{9.73}$$

where C_w is the specific soil moisture capacity, differentiating θ with respect to h (see Eq. 2.13).

Eq. (9.72) can be written into the finite-difference form as:

$$L_f \rho_i \frac{\Delta \alpha_i}{\Delta t} = -L_f \rho_l \left[\frac{C_{T,i}}{\Delta t}\left(T_i^{j+1} - T_i^{j} \right) + \frac{q_{i-1/2}^{j+1} - q_{i+1/2}^{j+1}}{\Delta z} \right] \tag{9.74}$$

According to Eq. (9.73), the finite-difference form of the third term in Eq. (9.74) can be written as:

$$\frac{q_{i-1/2}^{j+1} - q_{i+1/2}^{j+1}}{\Delta z_i} = -\frac{1}{\Delta z_i}\left[K_{i-1/2}^{j+1}\left(\frac{C_T}{C_h} \right)_i \frac{T_{i-1}^{j+1} - T_i^{j+1}}{\Delta z_u} + K_{i-1/2}^{j+1} - K_{i+1/2}^{j+1}\left(\frac{C_T}{C_h} \right)_i \frac{T_i^{j+1} - T_{i+1}^{j+1}}{\Delta z_L} - K_{i+1/2}^{j+1} \right] \tag{9.75}$$

Thus, the partial derivative of $L_f \rho_i \frac{\Delta \alpha}{\Delta t}$ with respect to T_i^{j+1} is written as:

$$\frac{\partial}{\partial T_i^{j+1}}\left(L_f\rho_i\frac{\alpha_i^{j+1}-\alpha_i^j}{\Delta t}\right)=-L_f\rho_l\left[\frac{C_{T,i}^{j+1/2}}{\Delta t}+\frac{1}{\Delta z_i\Delta z_u}K_{i-1/2}^{j+1}\left(\frac{C_T}{C_h}\right)_i+\frac{1}{\Delta z_i\Delta z_L}K_{i+1/2}^{j+1}\left(\frac{C_T}{C_h}\right)_i\right] \tag{9.76}$$

Then the application of N-RIT to Eq. (9.69) and rearrangement give the coefficients for the ***intermediate nodes***:

$$A_i=-\frac{\Delta t}{\Delta z_i\Delta z_u}\lambda_{i-1/2}^{j+1}+\frac{\Delta t^j}{2\Delta z_i}\rho_l c_l q_{i-1/2}^{j+1}+\frac{\Delta t}{2\Delta z_i}c_v q_{v,i-1/2}^{j+1} \tag{9.77}$$

$$\begin{aligned}B_i=C_{s,i}^{j+1}&+\frac{\Delta t}{\Delta z_i\Delta z_u}\lambda_{i-1/2}^{j+1}+\frac{\Delta t}{\Delta z_i\Delta z_L}\lambda_{i+1/2}^{j+1}+\frac{\Delta t}{2\Delta z_i}\rho_l c_l q_{i-1/2}^{j+1}-\frac{\Delta t}{2\Delta z_i}\rho_l c_l q_{i+1/2}^{j+1}\\&+L_f\rho_l C_{T,i}^{j+1/2}+\frac{\Delta t}{\Delta z_i\Delta z_u}\rho_l L_f K_{i-1/2}^{j+1}\left(\frac{C_T}{C_h}\right)_i+\frac{\Delta t}{\Delta z_i\Delta z_L}\rho_l L_f K_{i+1/2}^{j+1}\left(\frac{C_T}{C_h}\right)_i\\&+L_v\theta_v\left(\frac{\partial\rho_v}{\partial T}\right)_i+\frac{\Delta t}{2\Delta z_i}c_v q_{v,i-1/2}^{j+1}-\frac{\Delta t}{2\Delta z_i}c_v q_{v,i+1/2}^{j+1}\end{aligned} \tag{9.78}$$

$$C_i=-\frac{\Delta t}{\Delta z_i\Delta z_L}\lambda_{i+1/2}^{j+1}-\frac{\Delta t}{2\Delta z_i}\rho_l c_l q_{i+1/2}^{j+1}-\frac{\Delta t}{2\Delta z_i}c_v q_{v,i+1/2}^{j+1} \tag{9.79}$$

$$\begin{aligned}D_i=&-\left(C_s^{j+1}T_i^{j+1}-C_s^jT_i^j\right)+L_f\rho_l\left(\alpha_i^{j+1}-\alpha_i^j\right)-L_v\theta_v\frac{\partial\rho_v}{\partial T}\left(T_i^{j+1}-T_i^j\right)-L_v\frac{\Delta t}{\Delta z_i}\left(q_{v,i-1/2}^{j+1}-q_{v,i+1/2}^{j+1}\right)\\&+\frac{\Delta t}{\Delta z_i}\left(\lambda_{i-1/2}^{j+1}\frac{T_{i-1}^{j+1}-T_i^{j+1}}{\Delta z_u}-\lambda_{i+1/2}^{j+1}\frac{T_i^{j+1}-T_{i+1}^{j+1}}{\Delta z_L}\right)\\&-\Delta t\rho_l C_l\frac{q_{l,i-1/2}^{j+1}\left(T_{i-1}^{j+1}+T_i^{j+1}\right)/2}{\Delta z_i}+\Delta t\rho_l C_l\frac{q_{l,i+1/2}^{j+1}\left(T_i^{j+1}+T_{i+1}^{j+1}\right)/2}{\Delta z_i}-\Delta tS_{h,i}-\Delta t\rho_l C_l S_{w,i}T_w^j\\&-\Delta tc_v\frac{q_{v,i-1/2}^{j+1}\left(T_{i-1}^{j+1}+T_i^{j+1}\right)/2}{\Delta z_i}+\Delta tc_v\frac{q_{v,i+1/2}^{j+1}\left(T_i^{j+1}+T_{i+1}^{j+1}\right)/2}{\Delta z_i}\end{aligned} \tag{9.80}$$

9.3.2 Treatment of boundary conditions

(1) Tope node

For the third-type boundary condition, Eq. (9.69) can be transformed into (i=1):

$$\begin{aligned}&\frac{C_s^{j+1}T_i^{j+1}-C_s^jT_i^j}{\Delta t}-L_f\rho_i\frac{\alpha_i^{j+1}-\alpha_i^j}{\Delta t}+L_v\frac{\partial\rho_v}{\partial T}\frac{T_i^{j+1}-T_i^j}{\Delta t}+L_v\frac{0-q_{v,i+1/2}^{j+1}}{\Delta z_i}=\\&\frac{1}{\Delta z_i}\left(\lambda_{i-1/2}^{j+1}\frac{T_{avg}-T_i^{j+1}}{\Delta z_u}-\lambda_{i+1/2}^{j+1}\frac{T_i^{j+1}-T_{i+1}^{j+1}}{\Delta z_L}\right)-\Delta t\rho_l C_l\frac{q_{l,top}^{j+1}T_{top}^j}{\Delta z_i}+\rho_l C_l\frac{q_{l,i+1/2}^{j+1}\left(T_i^{j+1}+T_{i+1}^{j+1}\right)/2}{\Delta z_i}\\&-0+c_v\frac{q_{v,i+1/2}^{j+1}\left(T_i^{j+1}+T_{i+1}^{j+1}\right)/2}{\Delta z_i}-S_{h,i}-\rho_l C_l S_{w,i}T_{w,i}^j\end{aligned} \tag{9.81}$$

Then applying the N-RIT gives the coefficients (i=1):

$$
\begin{aligned}
B_1 = {} & C_{s,i}^{j+1} + \frac{\Delta t^j}{\Delta z_i \Delta z_u}\lambda_{i-1/2}^{j+1} + \frac{\Delta t^j}{\Delta z_i \Delta z_L}\lambda_{i+1/2}^{j+1} - \frac{\Delta t^j}{2\Delta z_i}\rho_l c_l q_{i+1/2}^{j+1} \\
& + L_f \rho_l C_{T,i}^{j+1/2} + \frac{\Delta t^j}{\Delta z_i \Delta z_L}\rho_l L_f \left(\frac{C_T}{C_h}\right)_i K_{i+1/2}^{j+1}
\end{aligned}
\tag{9.82}
$$

$$
C_i = -\frac{\Delta t^j}{\Delta z_i \Delta z_L}\lambda_{i+1/2}^{j+1} - \frac{\Delta t^j}{2\Delta z_i}\rho_l c_l q_{i+1/2}^{j+1} - \frac{\Delta t}{2\Delta z_i} c_v q_{v,i+1/2}^{j+1}
\tag{9.83}
$$

$$
\begin{aligned}
D_i = {} & -\left(C_s^{j+1}T_i^{j+1} - C_s^j T_i^j\right) + L_f \rho_l \left(\alpha_i^{j+1} - \alpha_i^j\right) - L_v \theta_v \frac{\partial \rho_v}{\partial T}\left(T_i^{j+1} - T_i^j\right) - L_v \frac{\Delta t}{\Delta z_i}\left(0 - q_{v,i+1/2}^{j+1}\right) \\
& + \frac{\Delta t}{\Delta z_i}\left(\lambda_{i-1/2}^{j+1}\frac{T_{avg} - T_i^{j+1}}{\Delta z_u} - \lambda_{i+1/2}^{j+1}\frac{T_i^{j+1} - T_{i+1}^{j+1}}{\Delta z_L}\right) \\
& - \Delta t \rho_l C_l \frac{q_{l,top}^{j+1} T_{top}^j}{\Delta z_i} + \Delta t \rho_l C_l \frac{q_{l,i+1/2}^{j+1}\left(T_i^{j+1} + T_{i+1}^{j+1}\right)/2}{\Delta z_i} - \Delta t S_{h,i} - \Delta t \rho_l C_l S_{w,i} T_{w,i}^j \\
& - 0 + \Delta t c_v \frac{q_{v,i+1/2}^{j+1}\left(T_i^{j+1} + T_{i+1}^{j+1}\right)/2}{\Delta z_i}
\end{aligned}
\tag{9.84}
$$

(2) Bottom node

If considering both the heat flux and the effects of bottom temperature (i.e. the third-type boundary condition), Eq. (9.69) can be transformed into (i=N):

$$
\begin{aligned}
& \frac{C_s^{j+1}T_i^{j+1} - C_s^j T_i^j}{\Delta t} - L_f \rho_i \frac{\alpha_i^{j+1} - \alpha_i^j}{\Delta t} + L_v \frac{\partial \rho_v}{\partial T}\frac{T_i^{j+1} - T_i^j}{\Delta t} + L_v \frac{q_{v,i-1/2}^{j+1} - 0}{\Delta z_i} = \\
& \frac{1}{\Delta z_i}\left(\lambda_{i-1/2}^{j+1}\frac{T_{i-1}^{j+1} - T_i^{j+1}}{\Delta z_u} - \lambda_{i+1/2}^{j+1}\frac{T_i^{j+1} - T_{bot.soil}}{\Delta z_L}\right) - \rho_L C_l \frac{q_{l,i-1/2}^{j+1}\left(T_{i-1}^{j+1} + T_i^{j+1}\right)/2}{\Delta z_i} + \rho_L C_l \frac{q_{bot}^{j+1} T_{bot}}{\Delta z_i} \\
& - c_v \frac{q_{v,i-1/2}^{j+1}\left(T_{i-1}^{j+1} + T_i^{j+1}\right)/2}{\Delta z_i} + 0 - S_{h,i} - \rho_L C_l S_{w,i} T_{w,i}^j
\end{aligned}
\tag{9.85}
$$

Then applying the N-RIT gives the coefficients:

$$
A_N = -\frac{\Delta t}{\Delta z_N \Delta z_u}\lambda_{N-1/2}^{j+1} + \frac{\Delta t}{2\Delta z_N}\rho_l c_l q_{N-1/2}^{j+1} + \frac{\Delta t}{2\Delta z_N} c_v q_{v,N-1/2}^{j+1}
\tag{9.86}
$$

$$
\begin{aligned}
B_N = {} & C_{s,N}^{j+1} + \frac{\Delta t}{\Delta z_N \Delta z_u}\lambda_{N-1/2}^{j+1} + \frac{\Delta t}{2\Delta z_N}\rho_l c_l q_{N-1/2}^{j+1} + \frac{\Delta t}{\Delta z_N \Delta z_L}\lambda_{N+1/2}^{j+1} \\
& + \frac{\Delta t}{\Delta z_N \Delta z_L}\rho_l L_f K_{N-1/2}^{j+1}\left(\frac{C_T}{C_h}\right)_N + L_v \theta_v \left(\frac{\partial \rho_v}{\partial T}\right)_N + \frac{\Delta t}{2\Delta z_N} c_v q_{v,N-1/2}^{j+1}
\end{aligned}
\tag{9.87}
$$

$$
\begin{aligned}
D_N = & -\left(C_s^{j+1}T_N^{j+1} - C_s^{j}T_N^{j}\right) + L_f\rho_l\left(\alpha_N^{j+1} - \alpha_N^{j}\right) - L_v\theta_v\frac{\partial\rho_v}{\partial T}\left(T_N^{j+1} - T_N^{j}\right) - L_v\frac{\Delta t}{\Delta z_N}\left(q_{v,N-1/2}^{j+1} - 0\right) \\
& + \frac{\Delta t}{\Delta z_N}\left(\lambda_{N-1/2}^{j+1}\frac{T_{N-1}^{j+1} - T_N^{j+1}}{\Delta z_u} - \lambda_{N+1/2}^{j+1}\frac{T_N^{j+1} - T_{bot.soil}}{\Delta z_L}\right) \\
& - \Delta t\rho_l C_l\frac{q_{l,N-1/2}^{j+1}\left(T_{N-1}^{j+1} + T_N^{j+1}\right)/2}{\Delta z_N} + \Delta t\rho_l C_l\frac{q_{bot}^{j+1}T_{bot}}{\Delta z_N} - \Delta t S_{h,N} - \Delta t\rho_l C_l S_{w,N} T_{w,N}^{j} \\
& - \Delta t c_v\frac{q_{v,N-1/2}^{j+1}\left(T_{N-1}^{j+1} + T_N^{j+1}\right)/2}{\Delta z_N} + 0
\end{aligned}
\tag{9.88}
$$

In AHC, we provide two options to treat the heat bottom BCs for actual simulation: (1) **zero-heat flux BC**, in which $q_{h,cv}$ = 0 and $q_{h,dv}$= 0 are set at the bottom of soil profile (i.e., $-\lambda_{N+1/2}^{j+1}\left(T_N^{j+1} - T_{bot.soil}\right)/\Delta z_L = 0$ and $q_{bot}^{j+1}T_{bot} = 0$ in Eq. (9.69). BC is suitable to the conditions in which there is no (or very few) heat exchange between the bottom soil and the outside. This BC requires no extra input data; (2) **temperature-dependent BC**, in which AHC only assumes the term $q_{bot}^{j+1}T_{bot} = 0$. The heat bottom BC becomes only dependent on soil temperature ($T_{bot,soil}$) at the bottom of the soil profile. The observation data of $T_{bot,soil}$ is required to define this BC. Note that the temperature-dependent BC is recommended preferentially in actual field conditions if the observed $T_{bot,soil}$ data is available. In actuality, the temperature value and its changes in deep soils are relatively stable compared to shallow soils. If there is no observed data on $T_{bot,soil}$, we could set a soil profile with enough depth, and thus the $T_{bot,soil}$ could be empirically specified based on the observations from the past or nearby areas. Moreover, the zero-heat flux BC is applicable when the bottom is in an adiabatic state or the bottom heat flux is extremely small. It may also be able to approximately define the bottom BC (with deep soil profile) for actual field conditions if no observed data is available.

9.3.3 Treatment of calculation stability

(1) Freezing condition for heat transport

When the freezing point is reached, the apparent heat capacity (C_a, J cm^{-3} °C^{-1}) will include the latent heat (stored in the ice) as an additional term in the soil heat capacity C_s and is calculated as:

$$C_a = C_s - L_f \rho_i \frac{d\alpha}{dT} = C_s + \rho_l L_f C_T \tag{9.89}$$

The apparent heat capacity C_a increases by several orders (e.g., about two for silty clay and five for sand) of magnitude for soils when reaching freezing point, compared to the value of only C_s. The significant changes in the C_a for temperatures above and below the freezing point can lead to significant numerical oscillations. Let us assume a situation without flow, take a soil slightly above zero temperature, and remove a certain amount of energy. During the first numerical iteration at a particular time-step, the C_a only includes the first term (i.e., C_s) and thus the decrease in temperature of the soil is evaluated by only C_s (a relatively small number). This can lead to a considerable decrease in temperature below zero and corresponding freezing of soil water. Thus, the decrease of soil temperature is obviously overestimated since the apparent heat capacity will increase several orders of magnitude at the point where soil water starts freezing. This may result in oscillations in calculating temperatures between negative and positive values within a timestep (Flerchinger, 2000; Hansson et al., 2004).

Thus, a method based on available energy for a phase change is adopted to aovid the above-mentioned oscillation or instability caused by a phase change (from liquid water to solid ice), referring to Flerchinger (2000). For the iterative calculation of water and heat equations, the soil temperature is calculated using heat ADE equation, and then ice content of each frozen layer can be determined. When the temperature decreases from above to below freezing point, the new temperature should be adjusted as follows:

$$T_r = T_{fr} - \frac{E_{na}}{C_a} \tag{9.90}$$

where T_r is the revised soil temperature (°C), E_{na} is the available energy for freezing (J cm^{-3}), and T_{fr} is the freezing point temperature (°C). E_{na} is calculated as:

$$E_{na} = C_s\left(T_{fr} - T\right) \tag{9.91}$$

T_{fr} is lower than 0°C for soils and can be derived from Eq. (5.13), as follows:

$$T_{fr} = T_k \cdot \frac{\varphi}{\left(10^5\, L_f / g - \varphi\right)} \tag{9.92}$$

The above adjustment method is found to produce smooth calculation and without undesired numerical oscillations, which also allows for larger time-steps and reduces computation time for iterative calculation of soil water and heat equations.

(2) Limitation of net water flux when reaching full saturation

Under soil freezing conditions, it is possible that the net soil water flux into a node (or compartment) is greater than the available porosity in numerical calculation due to the presence of ice, especially when the soil moisture content is high in the profile. This may cause an error in that the sum of the soil water content (θ) and the ice content (α) is greater than the saturated soil water content (θ_s). This would make the model become unstable in soil freezing simulation when the medium reaches full saturation (Šimůnek et al., 2016). To solve this problem, AHC limits net water flux into a node to be less than the available porosity (Φ_{av}) of that node.

The specific procedure starts from the bottom node and upwards to the top node, such as that for node i: (1) in case of the water flux (Q_i) through the top of the compartment moving downward into compartment i (positive upward), if the net water flux (i.e., Q_{i+1}-Q_i) beyond the Φ_{av}, the Q_i is adjusted into $Q_i' = Q_{i+1} - \Phi_{\text{av}}$ and the $K_{i-1/2}$ (i.e., mean hydraulic conductivity between the node i and the one above i-1) is set to zero at this iteration level. Meanwhile, if Q_i' becomes positive, it will be set to zero and Q_{i+1} will be updated into $Q_{i+1}' = \Phi_{\text{av}}$ (positive); (2)

in case of the Q_i moving upward out of the compartment *i* (positive), if the net water flux beyond the Φ_{av}, the Q_{i+1} will be reduced into $Q'_{i+1} = Q_i + \Phi_{av}$ and $K_{i+1/2}$ is set to zero at this iteration level. In this way, AHC could effectively ensure the relationship of $\theta+\alpha \leqslant \theta_s$ for any node, and maintain the stability of numerical calculations for full saturation state.

9.4 Numerical solution strategy: water-heat-solute calculation

9.4.1 Iterative scheme

AHC applies a method of simultaneous solution of water flow and heat transport equations, referring to the iteration approach of Flerchinger and Saxton (1989). It can effectively achieve numerical stability in water-heat coupling calculation. The variables to be solved for the water flow equation (Eq. 2.7) are the pressure head water content h when using single-variable algorithm, or both h and the ice content α when using primary switching algorithm. The heat transport equation (Eq. 5.7) has one unknown of soil Temperature T. The solution for each time step involves alternating back and forth between a Newton-Raphson iteration for the water flow equation and one for the heat transport equation. An iteration is started to be conducted for the water flow equation, and θ and α for the end of the time step are updated. This is followed by an iteration for the heat transport equations, where T is calculated and updated; meantime, the ice content α is subsequently determined based on the T, h and osmotic head, according to Eq. (5.13). Upon completion of the iteration for the heat transport equation and ice freezing, the solution reverts back to an iteration for the water flow equation with the updated values (i.e., T, θ and α). Iteration continue until all subsequent iterations of the both water flow and heat transport equations for each soil compartment/node are within a prescribed tolerance (Flerchinger, 2000). This means that the water flow and heat transport equations are solved simultaneously. After iterations for the water flow and heat transport equations have reached convergence, solute transport is computed using the liquid fluxes from the water balance calculations.

9.4.2 Convergence and time Control

The head difference $\left|h_i^{j+1,p} - h_i^{j+1,p-1}\right|$ and water content difference $\left|\theta_i^{j+1,p} - \theta_i^{j+1,p-1}\right|$ in the iterative solution of Eq. (9.1) are both used as convergence criterium in AHC, under non-frozen conditions. The ice content difference $\left|\alpha_i^{j+1,p} - \alpha_i^{j+1,p-1}\right|$ will be also included as another convergence criterium if simulating the soil freezing and thawing with the switching algorithm. Compare to a criterium based on h, the θ-criterium is automatically more sensitive in pressure head ranges with a large differential soil water capacity (C_w), while it allows less iterations at low h-values where θ hardly changes The θ-criterium has higher efficiency for a large number of infiltration problems, and moreover it is also found to be more robust when the soil hydraulic characteristics were extremely non-linear (Huang et al., 1996). In addition, if the soil gets saturated at the node considered, θ becomes constant and the convergence criterium is switched to maximum differences of h.

The soil temperature difference $\left|T_i^{j+1,p} - T_i^{j+1,p-1}\right|$ is used as convergence criterium in the iterative solution of Eq. (9.69). The solution concentration difference $\left|c_i^{j+1,p} - c_i^{j+1,p-1}\right|$ is used as convergence criterium in the iterative solution of Eq. (9.46) under non-linear adsorption condition.

The optimal time step should minimize the computational effort of a simulation while the numerical solution still meets the convergence criteria. A minimum and a maximum time step ($\triangle t_{min}$ and $\triangle t_{max}$, d) are specified by users in AHC. Time discretization starts with a prescribed initial time-step, taking as $\Delta t = \sqrt{\Delta t_{min} \Delta t_{max}}$. The time increment is automatically adjusted at each time level based on the number of iterations needed to reach convergence (N_{it}) in the previous time step. N_{it} is related to the water flow equation and water-heat transport equations, respectively, under non-frozen and frozen conditions. The detailed treatment is according to the following rules:

(a) Time increment Δt is always confined to the range $\Delta t_{min} < \Delta t < \Delta t_{max}$.

(b) If, during a particular time step, the iteration number $N_{it} < 2$, the Δt for the next time step is increased by multiplying the time step with a predetermined constant > 1 (e.g., 1.25); if $2 < N_{it} < 4$, the Δt for next time step maintains the same; if $N_{it} > 4$, the Δt for next time step is decreased and multiplied by a constant <1 (e.g., 0.8).

(c) If, during a particular time step, the iteration number N_{it} becomes greater than a prescribed maximum (e.g., 10-50), the iterative process for this time level is terminated. The time step is subsequently reset to $\Delta t/3$, and the iterative process starts again.

Exceptions to the above procedure occur, when the upper boundary flux changes from evaporation to intensive rainfall (> 1.0 cm d^{-1}), in which case Δt is reset to Δt_{min}, and at the end of a day, in which case Δt_{min} is set equal to the remaining time in the day. In addition, Δt_{max} and Δt_{min} are speficied by users under non-frozen conditions, such as with default values of $\Delta t_{max} = 0.2$ and $\Delta t_{min} = 10^{-5}$; however, they will be automatically adjusted to be smaller (e.g., $\Delta t_{max} = 0.02$ and $\Delta t_{min} = 10^{-7}$) if soil is under frozen conditions, vice versa. This treatment could maintain the numerical stability and calculation efficiency. Moreeover, the Grid Peclet number and Courant number are used together to ensure the numerical stability of solving solute ADE.

10. Model parameterization

10.1 Sensitivity-Analysis Process

The LH-OAT method is adopted in the AHC to perform the parameter sensitivity analysis. It has a sampling strategy that is a combination of Latin Hypercube (LH) and One-factor-At-a-Time (OAT) sampling. LH-OAT allows performing GSA for a long list of parameters with less computational cost (van Griensven et al., 2006). The calculation procedures are presented in detail in Fig. 10.1, with the parameter number being equal to M as an example. LH-OAT starts with dividing the parameter range into N equal intervals and taking N LH sampling points for each parameter (with each interval to be sampled only once, randomly generated). Thus, N parameter groups are generated. After that, each of the parameter groups is varied for M times by changing each of the M parameters one at a time, using the OAT design. Thus, it only requires a total of N×(M+1) model runs. A global sensitivity index (i.e., S_p) for the j^{th} parameter (i.e., P), corresponding to a criterion, is calculated by averaging the partial effects ($S_{p,i}$) for all LH points (Eq. 10.1 and Fig. 10.1).

$$S_p = \frac{1}{N}\sum_{i=1}^{N}\left| \frac{\dfrac{M(e_{1,i},\ldots,e_{j,i}(1+f_j),\ldots,e_{M,i}) - M(e_{1,i},\ldots,e_{j,i},\ldots,e_{M,i})}{\left[M(e_{1,i},\ldots,e_{j,i}(1+f_j),\ldots,e_{M,i}) + M(e_{1,j},\ldots,e_{j,i},\ldots,e_{M,i})\right]/2}}{f_j} \right| \tag{10.1}$$

where M(·) refers to the model functions, f_i is the fraction by which the parameter e_i is changed (a predefined constant), and j refers to a LH point. GSA result is obtained according to S_i value, with the largest value being given rank 1 and the smallest value being given a rank equal to the total number of parameters analyzed. The default value of f_i is set to 5%.

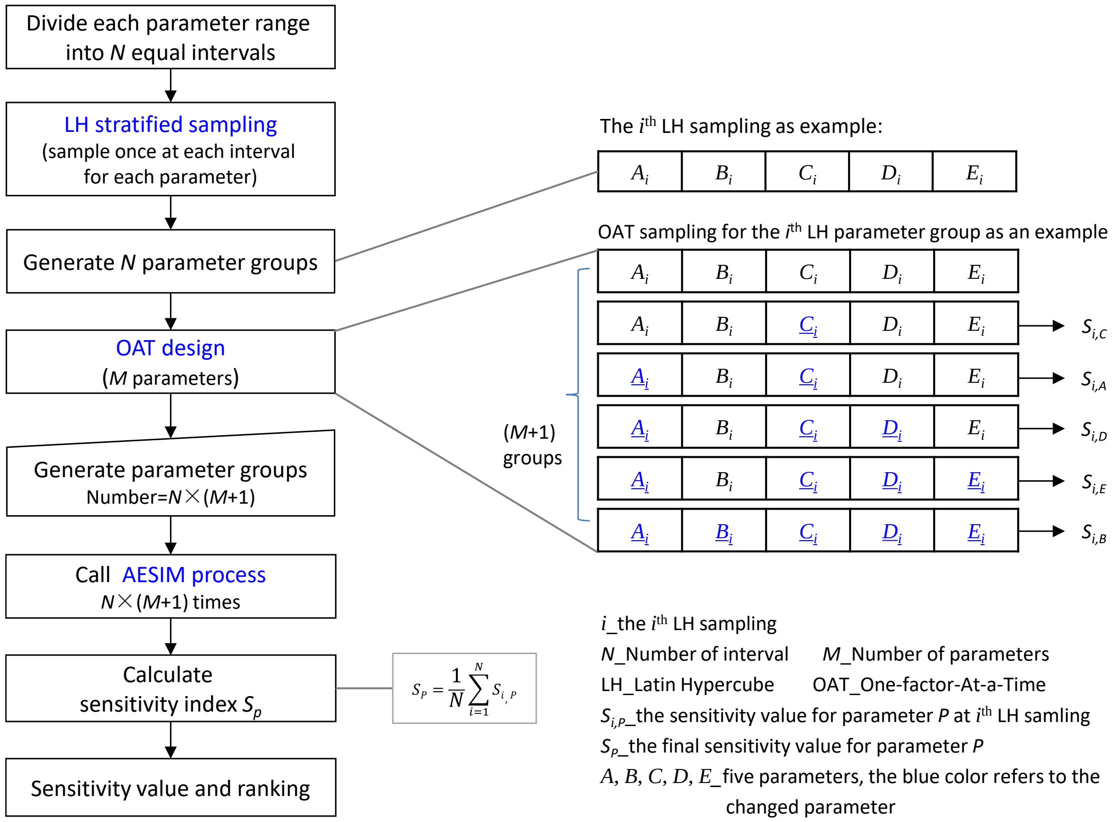


Figure 10.1 The flow chart of the GSA (Global-Sensitivity-Analysis) Process based on the LH-OAT method.

where M(·) refers to the model functions, $e_{j,i}$ is the value for the j^{th} parameter P at the i^{th} LH point sampling, and f_j is the fraction by which the parameter $e_{j,i}$ is changed. The default value of f_i is set to 5%. The final global sensitivity value for a parameter could then be estimated by the weighted average of the S_p values under multicriteria conditions. GSA result is obtained according to the S_p value, with the largest value being given the rank of 1 and the smallest value being given a rank equal to the total number of parameters analyzed Xu et al. (2016).

The above LH-OAT sampling method was applied and embedded into AHC as a new module for GSA. A total of 316 parameters are involved in the SA database, which could be optionally selected in the analysis. They could be arranged into four main groups: those relative to soil water flow, those related to solute transport and transformation, those related to plant growth, and those describing stress factors of water/salt/nitrogen. Their ranges are required and often determined based on the existing literature. The model outputs could be optionally selected as objective variables, including actual evapotranspiration (ET_a), LAI, bottom flux, dry aboveground biomass (D-AGB), and layer-specific soil moisture and solute concentration/content. The parameters with higher sensitivity and uncertainty would be selected for further calibration. More description of calculation procedures and parameter setting of the GSA Process can be found inXu et al. (2016).

10.2 Parameter-Estimation Process

The genetic algorithm (GA) is a global and robust method searching for the optimum solution to complex problems using the precept of natural selection (Holland, 1975; Goldberg, 1989). The calculation flow chart of the GA and its coupling with the AESIM Process is shown in Fig. 10.2. The GA consists of three basic operators of selection, crossover, and mutation. It represents a solution using strings (referred to as chromosomes) of variables (represented as genes) in a search problem. The search will start by initializing a population of chromosomes. Each chromosome is evaluated on its performance using a fitness function. On the basis of the performance, the chromosomes are selected into the mating pool to form offspring through the crossover and mutation of genes. Selection, crossover, and mutation are repeated for many generations to reproduce the chromosome that fits the environment best. The best chromosome would represent the optimal or near-optimal solution to the search problem. The modified-MGA has been adopted to calibrate the model parameters by minimizing the error between simulated and observed data in the GPE Process. The uniqueness of modified-MGA is the ability to restart when the chromosomes of the micropopulation are nearly 90% similar in structure (Ines and Droogers, 2002; Shin et al., 2012).

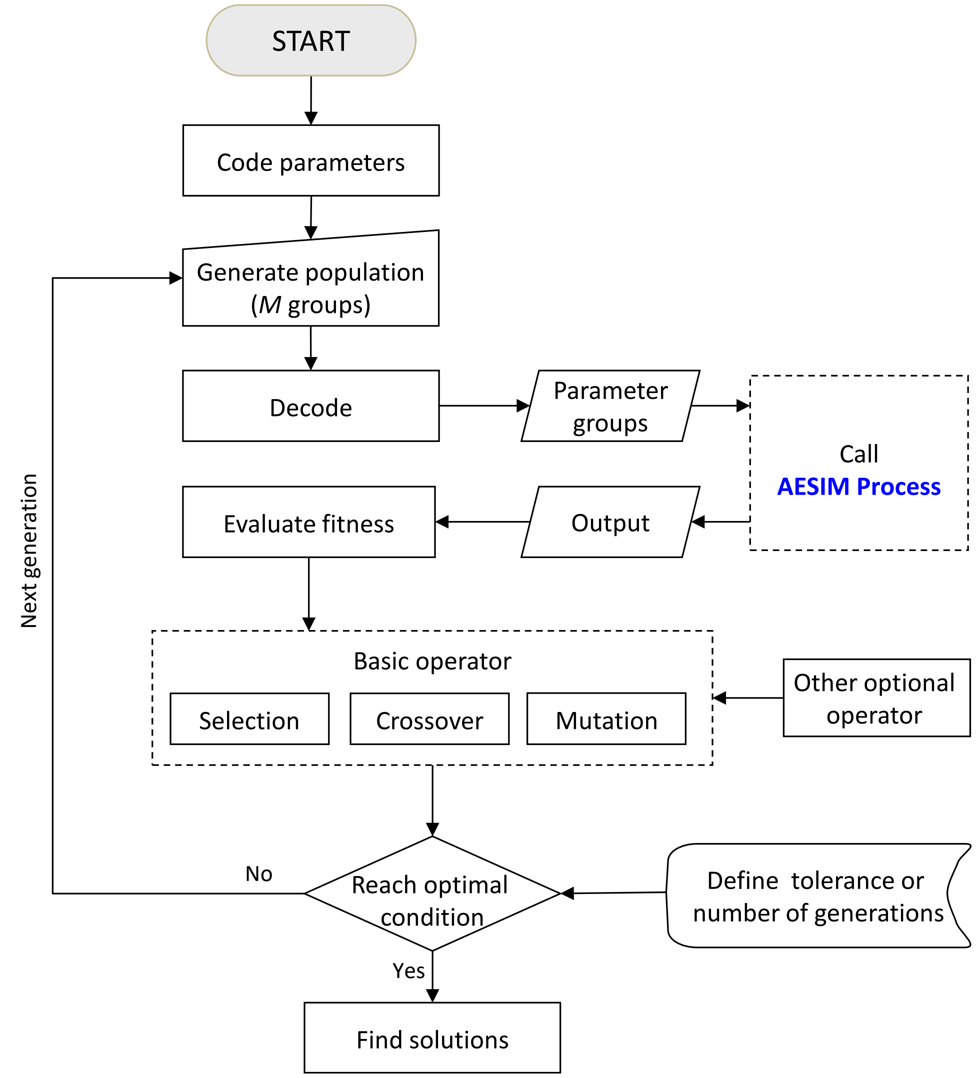


Figure 10.2 The flow chart of the GPE (Global-Parameter-Estimation) Process based on modified-MGA approach.

The modified-MGA code (Carroll, 1998; Ines and Droogers, 2002) is adapted and embedded into the AHC. The parameters set with four groups are the same as those in the GSA Process. The highly sensitive parameters selected by the LH-OAT method were adopted to be calibrated. Similar to the sensitivity analysis, the model outputs could be optionally selected as objective variables according to the observation data collected. On the basis of these objective variables, the fitness function was constructed using the weighted average method, as follows:

$$fitness(j) = \sum_{k=1}^{m} \omega_k \left(\frac{1}{\frac{1}{n_k} \sum_{i=1}^{n_k} \left| x_{sim} - x_{obs} \right|_i} \right) \tag{10.2}$$

where *fitness*(j) is the fitness value of the j^{th} chromosome, x_{sim} and x_{obs} are the normalized values of simulated and observed objective variables, respectively (-), the subscript i donates the i^{th} observation, the n_k is the number of observations for the k^{th} objective variable, m is the number of total selected objective variables, and ω_k is the weighting factor for the k^{th} objective variable (-) with $\sum_{k=1}^{m}\omega_k$ equals to one.

When GA approaches the solution, the fitness values of most chromosomes in a population may be close. To avoid the non-uniqueness and instability in the inverse solution, a kind of uncertainty bounds to the solution was created analogous to that used by Abbaspour et al. (2004) or Shin et al. (2012). The inverse modeling was conducted with a multi-population generated by various random generator seeds. The average fitness of all chromosomes was calculated and classified as above or below average at the end of searching. The above-average solutions were considered as the most probable solutions, which were used respectively to run the model. Then, the 95 percent confidence interval (95PCI) of outputs with the probable solutions was calculated for each observation time. The upper and lower 95PCI was the range of output, which may represent some modeling uncertainties.

It should be mentioned that the number of parameters is recommended not to be more than 15 to avoid the problem of non-uniqueness. The parameters related to plant growth and stress factors could generally be determined from previous literature and specific experiments. Additionally, the parameters related to solute transformation are generally difficult to estimate inversely, and thus, they were suggested to be determined or manually calibrated on the basis of previous experimental or modeling studies, while the parameters related to soils (mainly soil water flow and solute transport) are difficult to determine at each point of interest and are responsible for a large part of the spatial variability of the output variables. Thus, we recommend that users be concerned about these soil parameters in the GPE Process. Similar to the sensitivity analysis, the model outputs can be optionally selected as objective variables according to the observation data collected. On the basis of these objective variables, the fitness function is constructed using the weighted average method. A more detailed description of the calculation procedures and parameter setting of the GSA Process can be found in Xu et al. (2016).

11. Management aspects

A dynamic model can be applied in various ways to analyze field management aspects of irrigation, drainage and fertilizer strategies, in order to develop optimal application strategies. The main objective of optimal strategies is to maximize net return; other objectives may include: minimizing irrigation and fertilizer costs, maximizing yield, optimally distributing a limited water and fertilizer supply, and minimizing water pollution. In addition, irrigation may cause waterlogging problems in humid zones and salinity problems in arid zones. Artificial drainage is thus required to be implemented to build favorable conditions of moisture and salinity in the root zone if natural drainage is not sufficient to discharge excessive rainfall or leach salt (Kroes and van Dam, 2003). AHC can be used to support the design of irrigation, drainage and fertilization practices in the field.

11.1 Field Irrigation

The appropriate management objectives depend on available water amount and irrigation costs. In many cases, the optimal approach is to produce near-maximum yield across all irrgable areas. In this case, the prime objective is to prevent crop water stress throughout the growing season. If water supplies do not allow irrigation for maximum yield, or if irrigation costs are too high, resulting in economically optimum irrigation being lower than yield-maximizing irrigation levels, deficit irrigation must be implemented. In this situation, the goal of irrigation management is to maximize the economic returns to water, typically involving three decision criteria: irrigated area; which crops to plant; and how to distribute the available supply over the irrigable area throughout the season. A module for irrigation scheduling is modified from SWAP v2.0 (van Dam et al., 1997) and incorporated into AHC, which has been tested and applied in Qi et al. (2024).

11.1.1 Irrigation scheduling options

In AHC irrigations can be prescribed at fixed times or scheduled according to the controlled criteria of irrigation timing and application depth. Also, a combination of irrigation prescription and scheduling can be applied. Irrigation time can be determined by different trigger conditions, and then irrigation depth is calculated by defined scheduling criteria. A specified combination of timing and depth criteria is valid from a user-defined date in the cropping season until the end of crop growth. Both timing and depth criteria can be defined as a function of the crop development stage.

11.1.2 Timing criteria

Six different timing criteria can be selected to generate an irrigation schedule:

(1) *Allowable daily stress*

Irrigation is applied whenever the actual transpiration rate T_a drops below a predetermined fraction f_1 (-) of the potential transpiration rate T_p:

$$T_{\mathrm{a}} \leq f_1 T_p \tag{11.1}$$

The option is relevant for sub-optimal (deficit) irrigation when the water supply is limited.

(2) *Allowable depletion of readily available water in the root zone*

Irrigation is applied whenever the water depletion in the root zone is larger than a fraction f_2 (-) of the readily available water amount:

$$U_{FC}\text{-}U_a \geq f_2\left(U_{FC}\text{-}U_{h3}\right) \tag{11.2}$$

where U_a is the actual water storage in the root zone (cm), U_{FC} (cm) is the root zone water storage at h equaling to field capacity (e.g. h=-100 cm), and U_{h3} is the root water storage at h = h_3, where root water extraction starts being reduced due to drought stress (Fig. 2.1). A larger value of f_2 means that more soil water is allowed to be depleted and less irrigation frequency is required. U_a is calculated by integrating numerically the water content in the rooting layer. This option is useful for optimal scheduling where irrigation is always secured before conditions of soil moisture stress occur. For deficit irrigation purposes, stress can be allowed by specifying f_2 >1. In addition, it should be noted that the use of matric head values to approximate the field capacity in soils varies widely, and the values for different soils are found to vary from -50 to less than -350 cm in previous literature. Customarily, the coarser the texture of the soil, the higher the matric head for field capacity. In general, a commonly used approximation of field capacity is the water content at a matric head value of about -100 cm (-10 kPa) for coarse-textured soils, a value of about -330 cm (-33 kPa) for fine-texture soils, and even a value of about -500 cm (-50 kPa) with the clayed soils (Romano and Santini, 2002).

(3) *Allowable depletion of totally available water in the root zone*

Irrigation is applied whenever the depletion is larger than a fraction of f_3 (-) of the total available water amount between field capacity and permanent wilting point:

$$U_{FC} - U_a \geq f_3\left(U_{FC} - U_{h4}\right) \tag{11.3}$$

where U_{h4} is the root water storage at $h = h_4$, the pressure head at which root water extraction is reduced to zero (Fig. 2.1). Note that a larger value of f_3 means that more soil water is allowed to be depleted and less irrigation frequency is required.

(4) *Allowable depletion amount of water in the root zone*

Irrigation is applied whenever a predetermined water amount, ΔU_{max} (cm), is extracted below field capacity:

$$U_a \leq U_{FC} - \Delta U_{\max} \tag{11.4}$$

This option is useful in case of high frequency irrigation systems (drip).

(5) *Critical pressure head or moisture content at sensor depth*

Irrigation is applied whenever moisture content or pressure head at certain depth in the root zone drops below a prescribed threshold value θ_{min} (cm^3 cm^{-3}), or h_{min} (cm):

$$\theta_{sensor} \leq \theta_{\min} \quad \text{or} \quad h_{sensor} \leq h_{\min} \tag{11.5}$$

This option may be used to verify field experiments or to simulate irrigation with automated systems.

(6) *Critical soil moisture and ponding water depth in planned wetting zone*

Irrigation time can be determined by using different trigger conditions or their combination, including the lower limit options of critical soil water content and critical ponded water depth. Irrigation is applied whenever moisture content in the wetting zone (θ_{wz} cm^3 cm^{-3}) drops below a prescribed threshold value θ_{min2} (cm^3 cm^{-3}), or surface ponding water depth drops lower than a prescribed threshold value PD_{min} (cm):

$$\theta_{wz} \le \theta_{\min 2} \quad \text{or} \quad h_{pond} \le PD_{\min} \tag{11.6}$$

This option is designed for rice irrigation scheduling, which is particularly suitable for some water-saving irrigation regimes (e.g. shallow-wet irrigation).

11.1.3 Application depth criteria

Three irrigation depth criteria can be selected:

(a) *Back to field capacity (+/- specified amount)*

The soil water content in the wetting layer is brought back to field capacity. The depth of the wetting layer could be set with three options: (1) equal to rooting depth; (2) limited by user-specified wetting depth, i.e. minimum value of rooting depth and user-specified wetting layer depth; and (3) equal to the user-specified depth of the planned wetting layer. An additional irrigation amount can be defined for leach salts, while the user may define a smaller irrigation amount when rainfall is expected. This option is useful in the case of sprinkler irrigation and micro irrigation systems, which allow variation of irrigation application depth.

(b) *Fixed irrigation depth*

A specified amount of water is applied. This option is applied to most gravity systems, which allow little variation in application depth.

(c) *Back to desired ponding depth*

The irrigation depth is determined based on the desired ponded water depth after irrigation. This option is applied to irrigation scheduling for rice. The irrigation application depth (IRAD, cm) is calculated as follows:

$$IRAD = x(t) + \sum_{i=1}^{NC} \left[\Delta z_i \cdot \left(\theta_{s,i} - \theta_i \right) \right] \tag{11.7}$$

where NC is the number of soil compartments above the plow sole layer, Δd is the thickness of the i^{th} soil compartment (cm), t is the time (d), and x (cm) is the user-specified value of ponded water depth after irrigation (i.e., the upper limit of ponded water depth).

11.2 Design of field drainage

Drainage design can be evaluated using the equations of Hooghoudt and Ernst (section 3.3). Using these formulae, one may analyze the impact of various physical parameters (soil, crop, and climate) on drain spacing and drain depth. Combining options for irrigation (section 11.1) and salinity (chapter 4), one may analyze the relation between irrigation and field-scale soil salinity (Kroes and van Dam, 2003). More examples are extensively elaborated by Ritzema (1994).

11.3 Field fertilization

Rational fertilization aims to minimize potential crop nutrient deficits in order to maximize yield, while avoiding over-fertilization that could lead to nutrient loss and pollution. In AHC fertilization can be prescribed at fixed times and given amount by users, or automatically scheduled accrdong to the controlled criteria. Nitrogen fertilization is applied whenever the daily nitrogen stress (α_{NC}) is less than a nitrogen stress threshold (NS_{cri}, -):

$$\alpha_{\mathrm{NC}} \le NS_{\mathrm{cri}} \tag{11.8}$$

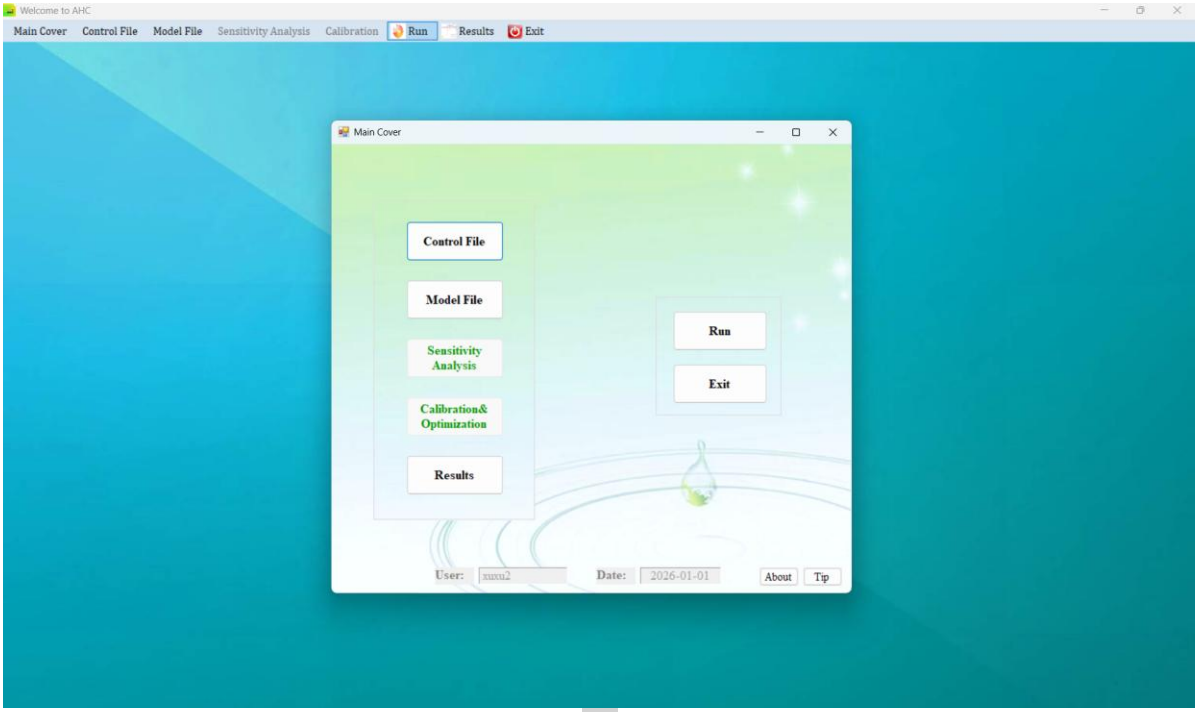
NS_{cri} should be specified by users within a typical range 0.7-0.95. However, two additional conditions are used to determine whether automatic fertilization is triggered, including: (1) if the user has already specified the fixed fertilization for the day, the automatic fertilization calculation will be skipped; and (2) the interval between two consecutive fertilization events must be greater than the minimum allowable interval. Additionally, if there is no corresponding irrigation or rainfall on the day of fertilization, the model will consider applying an extra irrigation amount (set by the user, typically as 2.0 cm) to complement the normal fertilization. The user also needs to specify single application rate of pure nitrgon, the fertilization depth, fraction of ammonia nitrogen, fertilization efficiency, and urea hydrolysis status.

12. Graphical user interface and application description

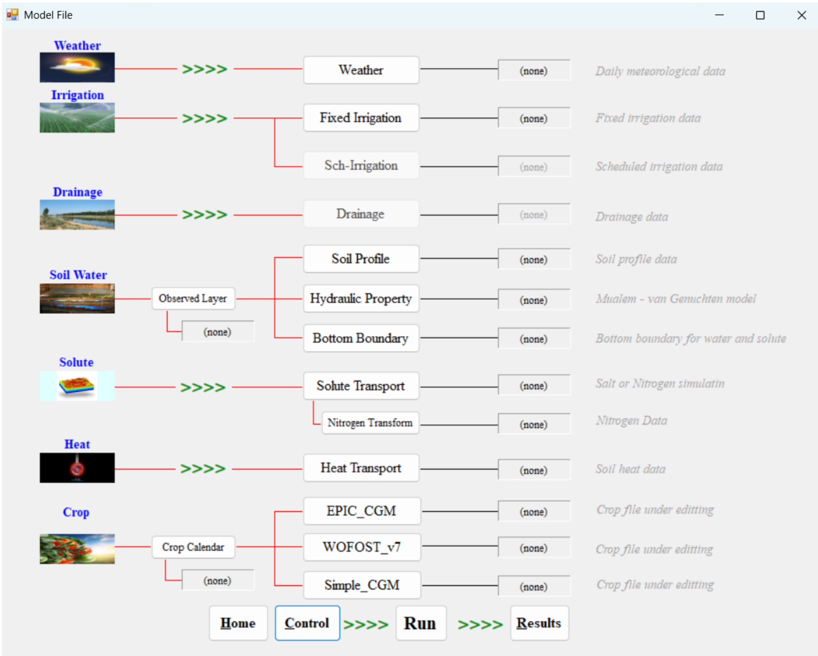
12.1 Graphical user interface

The GUI for the AHC is developed using VB.NET, an object-oriented programming language, using the Microsoft.NET Framework 4.5 in the Microsoft Visual Studio 2022 environment. The GUI allows the user to easily manipulate input files, run the model, and view the output. A main window is designed as a guide for operating the AHC, including launching the windows for data input (i.e., for control file, model files, sensitivity analysis, and calibration), result display, model run, and exit (Fig. 12.1a). The control file is used for managing the file path, data file connection, function selection, simulation period, output settings, etc. In this input window, the simulation can also be customized to invoke only those modules of interest for a specified application (e.g., solute or heat simulation can be disabled if not desired), which can produce more efficient input operation. All data input related to the AESIM Process is integrated and shown in the input window of the model files (Fig. 12.1b). It could make the data input more intuitive and clearer. The input window for the sensitivity analysis and calibration is designed specifically for the data input of GSA and GPE processes, respectively. In the result display, various simulation results (e.g., soil water content, soil solution concentration/content, leaf area index, and biomass) can be presented and conveniently compared with the observation data in graphics windows (Fig. 12.1c, d), which can help the user calibrate the model. Meanwhile, the statistical indicators (i.e., MRE, RMSE, NSE, R^2, and *b*) for model performance evaluation are also calculated and provided for users.

The efficiency and stability of the GUI are also repeatedly tested in several different cases. In addition, with a slight adaptation, the AHC model has been closely linked to GIS software (i.e., ArcInfo 10) for regional applications in a distributed manner. A basic graphical user interface has been constructed using the Visual Basic language.



(a)



(b)

Figure 12.1 Graphical user interface for the AHC – representative windows for model input and output: (a) the window for control file; (b) the window for the AESIM Process; (c) and (d) are part of the display windows for the soil water and solute output and the crop modeling output, respectively.

In addition, the GUI provides calculation and statistical functions for simulation accuracy indicators. The mean relative error (MRE), root mean square error (RMSE), Nash and Sutcliffe model efficiency (NSE), coefficient of determination (R^2), and coefficient of regression (b) are used to evaluate the model performance for soil water content, salinity concentration, and crop growth indicators. Their definitions are expressed as follows:

$$\mathrm{MRE} = \frac{1}{N}\sum_{i=1}^{N}\frac{(P_i - O_i)}{O_i}\times 100\% \tag{12.1}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(P_i - O_i)^2} \tag{12.2}$$

$$\mathrm{NSE} = 1 - \frac{\sum_{i=1}^{N}(P_i - O_i)^2}{\sum_{i=1}^{N}\left(O_i - \overline{O}\right)^2} \tag{12.3}$$

$$R^2 = \left[\frac{\sum_{i=1}^{N}\left(O_i - \overline{O}\right)\left(P_i - \overline{P}\right)}{\left[\sum_{i=1}^{N}\left(O_i - \overline{O}\right)^2\right]^{0.5}\left[\sum_{i=1}^{N}\left(P_i - \overline{P}\right)^2\right]^{0.5}}\right]^2 \tag{12.4}$$

$$b = \frac{\sum_{i=1}^{N} O_i \times P_i}{\sum_{i=1}^{N} O_i^2} \tag{12.5}$$

where N is the total number of observations, P_i and O_i are respectively the i^{th} model predicted and observed values ($i = 1, 2, \ldots, N$), and $\overline{P}$ and $\overline{O}$ are the predicted and observed mean values, respectively. b and R^2 values close to 1 and MRE and RMSE values close to zero indicate good model predictions. NSE=1.0 represents a perfect fit, and negative NSE values indicate that the mean observed value is a better predictor than the simulated value (Moriasi et al., 2007).

12.2 Application description with field cases

The AHC model has been evaluated and tested with many experimental datasets by comparing the simulated results with the observed data. The test cases could provide a comprehensive presentation of the latest release of the AHC. In this report, we provide several different examples to help users use the model more conveniently. Users can find specific data and examples in the files we provide, and each example focuses on a specific issue. Example 1 considers the simulation of soil water and crop growth processes. Example 2 serves to apply the model in simulating soil water, salinity, and crop growth during the crop season. Example 3 further tests the water-heat-salinity simulation over an entire agricultural year (agri-year), considering the soil frozen and thaw process. Example 4 considers the nitrogen simulation in a wheat-maize rotation field with two consecutive years. Example 5 shows the joint simulation of water, salinity, and nitrogen in a spring wheat field, also including the functions of designing irrigation schedules. Example 6 demonstrates the application of AHC for nitrogen and greenhouse gas (GHG) simulation. The above six examples are designed to help users understand the model capabilities and how to build models to achieve practical applications for different goals.

(1) Example 1 – soil water dynamics and seed maize growth

In the water case, we simulate the soil water dynamics and crop growth for a maize field, located at the Yingke Irrigation District (YID) (38°50'-38°58' N, 100°17'-100°34' E) of middle reaches of the Heihe River basin, northwest China. The data is collected from the field of seed maize (*Zea mays L*, Jinkai No. 3). Crop growth is mainly affected by the arid climate and soil moisture conditions, with almost no salinity stress under deep water table depths (over 10 m). The fertilization practices are set according to local farmers, usually with sufficient or even excessive fertilizer application. Therefore, only the options for soil water and crop growth simulation are activated, while salinity and nitrogen simulation are not included in this case.

The simulation period is from April 20th to September 30th of 2013, covering the whole crop growth period. Daily weather data from Zhangye weather station are used. The soil profile is divided into two layers: a topsoil (from the ground surface to 100-140 cm depth) and a subsoil (from the bottom of the topsoil to 200 cm depth). Simulation depth is further discretized into 84 compartments, with thickness varying from 1 cm to 5 cm from the surface to deep soil.

The upper boundary condition is determined by the actual evaporation and transpiration rates, and the irrigation and precipitation fluxes. The bottom boundary condition is defined as free drainage. The initial conditions of soil water content are specified according to the measured values of collected soil samples. The observation data includes layered soil water contents, LAI, dry above-ground biomass (D-AGB), and crop yield, which are used to calibrate the model.

All input and output files involved in this example can be found in folder "Yingke_water-corn_2013". Some more descriptions about the model setup and calibration can also refer to the previous two papers (Jiang et al., 2015; Liu et al., 2020). Moreover, the example for designing irrigation schedules (irrigation depth and timing) is also included in this case, which is applied to show how to use the functions of irrigation scheduling in AHC (in section 11.1).

(2) Example 2 – soil water-salt transport and maize growth during growth season

In this salinity case, we simulate the soil water-salt transport and crop growth in a maize field during the crop season in 2013 (without soil-frozen conditions). The field data is collected from the spring maize field with plastic mulching, located at the Yangchang canal command area (YCA) of the Hetao Irrigation District (Hetao), Inner Mongolia Autonomous Region, northwest China (40°48'40"N, 107°05'15"E, 1039 m altitude). The field condition is representative of that in the upper Yellow River basin, i.e., affected by arid climate, salinity condition under shallow water tables, and agronomy condition under plastic film mulching.

A soil profile with a 300-cm depth was specified with four horizons according to the soil survey. The domain was further discretized into 300 compartments with a uniform thickness of 1 cm. The simulation period was from early May to late September, covering the maize growth period. The maximum root depth was set at 90 cm.

For soil water flow, the upper boundary condition was determined by the actual evaporation and transpiration rates, and the irrigation and precipitation fluxes. A time-variable pressure head is selected as the bottom boundary condition, which is defined by the observed groundwater depths (GWDs). For solute transport, the upper and lower boundary conditions are defined by a concentration flux considering the observed salinity concentration of irrigation water and groundwater, respectively. The initial conditions of the soil water content and salinity concentration were set based on field measurements.

The observation data of layered soil water contents and salinity concentration, LAI, dry above-ground biomass (D-AGB), and crop yield, are used to calibrate the model. In addition, the GSA Process and GPE Process are used in conjunction to determine the parameters for model calibration in this case. The parameters are initially selected for sensitivity analysis using the GSA Process. Then, the measured soil water contents, salinity concentrations, and LAI (as mentioned above) are used to calibrate parameters automatically using the GPE Process. All input and output files involved in this example are provided in the folder "Hetao_salt-maize_2013-SA". More detailed descriptions of the model setup, results, and discussion can be found in our previous paper (Xu et al., 2018).

(3) Example 3 – heat-water-salt transport and crop growth with frozen conditions over a whole agr-year

Moreover, we provide a case simulation for continuous, coupled simulation of heat-water-salt transport and crop growth processes over an agricultural year (agr-year, from October 1st, 2012 to September 30th, 2013). This case is used to show how to simulate the soil freeze and thaw processes, with a detailed mechanistic approach. It can be seen as an extension and supplementary example to the salinity case 1 for soil frozen conditions. The detailed descriptions about the model setup and calibration can be found in Liu et al. (2025), where a uniform spacing of 1 cm discretization is adopted. To improve the calculation efficiency, the domain is discretized into 60 compartments with a uniform thickness of 5 cm in the example given here. All input and output files involved in this example are provided in the folder "YCA_Sunflower_Frozen-5cm_20-21".

(4) Example 4 – nitrogen simulation under a wheat-maize rotation

In the nitrogen case, we simulate soil water and nitrogen dynamics and crop growth in the experimental fields with a winter wheat-summer maize rotation system. The experimental data were obtained from the fields from October 2010 to September 2012 in the Shandong Agricultural University Agricultural Experiment Station, Tai'an City, Shandong Province, northern China (36°09′37″N, 117°09′18″E, 130 m altitude). The experimental site is situated in an important food production base – NCP, where the nitrogen fertilizer is often overused by farmers to obtain high yields. The related data is available from the previous study of Liu (2014).

The simulation period is from October 3rd, 2010 to September 30th, 2012, covering two seasons of the rotation of wheat and maize. A soil profile with a 300 cm depth is specified to have five material horizons. For depths below 160 cm, the soil properties are initially assumed to be the same as the last observed layer. In vertical discretization, the soil domain is discretized into 200 compartments with uniform 1-cm thickness for the first 100 cm depth and 2-cm thickness for 100-300 cm.

For soil water flow, the upper boundary condition is determined by the actual evaporation and transpiration rates and the irrigation and precipitation fluxes. The surface mulch by wheat straw is considered to estimate E_a during the maize season. At the bottom, a free drainage boundary condition is used to account for the deep GWD. For nitrogen transport, the upper and lower boundary conditions are defined as the third-type boundary conditions by specifying a concentration flux using the concentration of NO_3-N and ammonia-nitrogen (NH_4-N) in irrigation water, rainfall, and groundwater. For heat transport, the daily average air temperature is used to define top boundary conditions. If there is rainfall or irrigation, a third-type boundary condition is used to prescribe the top heat flux. The third-type boundary condition is also applied at the bottom to specify the heat flux. The initial conditions of soil water content, nitrogen concentration, and temperature are set based on field observations. The crop dormancy is considered for wheat growth in the winter season.

In addition, the GSA Process and GPE Process are used in conjunction to determine the parameters for model calibration in this case. The measured soil water contents, NO_3-N contents, LAI, and D-AGB (as mentioned above) are used for parameter estimation through the GPE Process. A two-year simulation is included in this example. All input and output files involved in this example are provided in the folder "Tai-an_N-rotation-GA". More detailed descriptions of the model setup, results, and discussion can be found in our previous paper (Xu et al., 2018).

(5) Example 5 – salinity and nitrogen joint simulation at a maize field

In the salinity-nitrogen case, we simulate the soil water-salt-nitrogen dynamics and crop growth during 2022. The field data is collected from the maize field with plastic mulching, located at the Hetao Experimental Station of China Agricultural University, Inner Mongolia, Northwest China (41° 09' N, 107° 39' E, 1039 m altitude). This region has an arid to semi-arid continental climate. The field condition was representative of that in the upper Yellow River basin, i.e., affected by arid climate, salinity under shallow water tables, and under plastic film mulching.

The drip irrigation under plastic mulching (referred to as DI-M) is used for maize cultivation, using the Kehe 699 variety. The soil is sandy loam, with a silty loam layer at deeper depths. The maize is planted with a wide-narrow row pattern, and the spaces of the wide row and narrow row are 70 cm and 40 cm, respectively. The soil surface is covered by transparent plastic mulching, with a mulching fraction $f_{r\ \mathrm{mulch}} = 0.6$. The planting density is about 60,000 plants per hectare. DI-M is triggered by a tensiometer located at a depth of 20 cm with soil matric potential ($h_{20\mathrm{cm}}$) of -15 kPa, and the irrigation water is pumped from groundwater.

A total of 200 kg ha^{-1} nitrogen (N) fertilization is applied for DI-M. The base fertilizer is applied prior to sowing, which consists of phosphorus fertilizer (diammonium phosphate, P=46%, N=18%; both 150 kg ha^{-1} for BI-M and DI-M), and potassium fertilizer (potassium oxide, K=50%; 60 kg ha^{-1} and 75 kg ha^{-1} for BI-M and DI-M, respectively). In addition, lignite carbon-based organic fertilizer (4,500 kg ha^{-1}) is also applied as base fertilizer in 2022, which is mainly used to partly replace the inorganic N fertilizer. The organic fertilizer is manually spread before sowing and mixed into deep soil (20 cm) using a rototiller. Urea (N=46%) is used as a topdressing fertilizer, in which 40% and 60% are applied at the jointing stage and the heading and filling stages, respectively. For DI-M, it is dissolved in water and applied to the root zone using the drip irrigation system. Other cultivation practices, including pests, crop diseases, and weed control follow local farmer practices in the region.

The simulation period is from the beginning of May to the end of September, covering the whole maize growth season. A soil profile of 300 cm depth is specified during the model simulation of soil water flow and solute (salt and N) transport. The soil profile is generalized with four horizontal layers according to the soil horizon observation. The vertical one-dimensional soil domain is further discretized into 300 compartments (i.e., nodes), with a uniform thickness of 1 cm. The simulation period is from the beginning of May to the end of September, covering the whole maize growth season.

For soil water flow, the upper boundary condition is determined by irrigation, rainfall, and actual daily soil evaporation. The lower boundary condition uses a variable head boundary, determined by the monitored daily GWD. For solute transport, concentration fluxes are used to define the upper and lower boundary conditions, considering TDS, NH_4-N, and NO_3-N concentrations observed in irrigation water and groundwater, respectively. Furthermore, the NH_4-N and NO_3-N contents in precipitation are also considered. The initial conditions of soil water content, salinity concentration, and NH_4-N and NO_3-N contents are determined by field observations before the beginning of the growth season.

Soil water content, salinity concentration (i.e., salt content of the soil solution, g L^{-1}), ammonia nitrogen (NH_4-N) and nitrate nitrogen (NO_3-N) contents, groundwater depth (GWD), and crop growth indicators are monitored from planting to harvest during the crop growth seasons in 2022. The data is used to evaluate the model performance. All input and output files involved in this example are provided in the folder “Heji_N1-salt-2022maize-drip”. More detailed descriptions of the model setup, results, and discussion can be found in our previous paper (Qi et al., 2024).

(6) Example 6 – nitrogen and GHG simulation at wheat field

This case mainly presents the simulation of nitrogen dynamics and green house gas (GHG) in a spring wheat field during crop season of 2019. The field data is collected from the wheat field, located at the Hetao Irrigation District of the upper Yellow River basin, Northwest China (41° 09' N, 107° 39' E, 1042 m altitude). This region has an arid to semi-arid continental climate. Unlike salt-tolerant crops such as sunflower, wheat is not a salt-tolerant crop. The wheat is planted in the experiment field with non-salinized soil (with SSC<1.5 g kg^{-1} at 0-20 cm surface soils). The salt simulation is thus not considered in this case.

The soil water flow and solute transport were calculated for soil layers from 0 to 300 cm. The vertical 1-D soil profile was uniformly discretized into 301 nodes with 1 cm spacing and generalized to 4 layers based on the soil particle data of the test field. The simulation period was from early March to late July, including the entire growth season of spring wheat.

For soil water flow, the flux boundary and variable pressure water head boundary were selected as the upper boundary and lower boundary conditions, respectively. The upper boundary, i.e., flux boundary was determined by the daily precipitation and irrigation and the daily actual evaporation and actual transpiration. The lower boundary, i.e., variable pressure water head boundary was determined by the monitored daily groundwater depth. For solute transport, the concentration flux boundary was selected as the upper and lower boundary conditions, which were determined by specifying the NH_4^+-N and NO_3^--N concentrations of precipitation, irrigation water, and groundwater. The field test data, including soil water, NH_4^+-N, and NO_3^--N contents, were used to determine the initial conditions.

Before running the simulation, the soil hydraulic parameters, solute (NH_4^+-N and NO_3^--N) transport parameters, C/N turnover parameters, water stress parameters, and crop-growth parameters were initially specified according to studies by Ren et al. (2016) and Xu et al. (2018). The soil water contents, soil NH_4^+-N and NO_3^--N contents, soil CO_2, N_2O, and NH_3 fluxes, inorganic nitrogen leaching, crop-growth indicators (plant height, LAI, and aboveground dry biomass), and grain yield were used to calibrate and evaluate the model. All input and output files involved in this example are provided in the folder "Fenzidi_CN-gas_wheat_I1NI-2019-LD". More detailed descriptions of the model setup, results, and discussion can be found in our previous paper (Li et al., 2023JoH).

Acknowledgments

We gratefully acknowledge the support from the National Science Foundation of China (grant numbers: 51209204, 51679235, 52022108, and 52379053). We are especially grateful to the graduate students (Fengtian NENG, Jing FU, Huimin MENG, Zhi QI, Yihao XUN, Shibo DING, Yue Li, Minghuan LIU, and Yao JIANG) who actively worked with us, especially in model testing, application, and improvement.

References

Appendix 1: Notation（List of all variables)

h	soil water pressure head (cm)
H	soil hydraulic head including both suction and gravitational components (cm)
$K(h)$	unsaturated soil hydraulic conductivity (cm d^{-1})
K_s	saturated soil hydraulic conductivity (cm d^{-1})
q	soil water flux density (positive upward) (cm d^{-1})

Annex A Derivation of various partial derivatives

(1) Derivation of C_{so}

The solutes may dissolve in the soil water and be absorbed by organic matter or clay minerals. Hence, the total solutes can be expressed as:

$$X = \theta c + \rho_b Q \tag{A-1}$$

where X is the total solutes present per volume of soil (g cm^{-3}), Q is solute amount absorbed per volume of soil (g g^{-1}), c is solute concentration in soil water (g cm^{-3}), ρ_b is the soil bulk density (g cm^{-3}), and ρ_l is the liquid water bulk density (g cm^{-3}).

We assume a linear adsorption relationship between Q and c such that

$$Q = k_d \cdot c \tag{A-2}$$

where k_d is the empirical coefficient relating the solution and adsorbed concentrations (cm^3 g^{-1}). Hence, Eq. (A-1) becomes:

$$X = \theta c + \rho_b k_d c \tag{A-3}$$

$$c = \frac{X}{\theta + \rho_b k_d} \tag{A-4}$$

$$c = \frac{\frac{X}{\rho_b}}{\frac{\theta}{\rho_b} + k_d} \tag{A-5}$$

Setting $S = X/\rho_b$, we get

$$c = \frac{S}{k_d + \frac{\theta}{\rho_b}} \tag{A-6}$$

with S being the total solutes present per mass of soil (g g^{-1}).

We differentiate c with respect to θ, to get

$$\frac{\partial c}{\partial \theta} = -\left(k_d + \frac{\theta}{\rho_b}\right)^{-2} \frac{1}{\rho_b} S \tag{A-7}$$

In dilute solutions, the osmotic pressure is proportional to the concentration of the solution and to its temperature according to the following equation (Hillel, 1998):

$$h_s = -\frac{i_v c_M R T_k}{10 \cdot \rho_l g} = -\frac{i_v R T_k}{10 \cdot \rho_l g}\frac{c}{M} \tag{A-8}$$

where h_s is the osmotic potential of soil solution (cm), i_v is the van't hoff factor, c_M is the molar concentration of the solute in the solution (mol cm^{-3}), T_k is the temperature of the soil in Kelvin (K) with T_k=T+273.16, g is gravitational acceleration (9.81 m s^{-2}), and M_s is the molecular weight of the specific solute (g mol^{-1}). When the simulated solute consists of multiple salt ions, the user may need to specify an equivalent representative value for M_s.

We further differentiate h_s with respect to θ, to get:

$$\frac{\partial h_s}{\partial \theta} = \frac{\partial\left(\dfrac{-i_v c R T_k}{10 \cdot M \rho_l g}\right)}{\partial \theta} = \frac{-i_v R T_k}{10 \cdot M \rho_l g}\frac{\partial c}{\partial \theta} = \frac{i_v R T_k}{10 \cdot M \rho_l g}\left(k_d + \frac{\theta}{\rho_b}\right)^{-2}\frac{1}{\rho_b} S \tag{A-9}$$

Here we note a new variable C_{so} (cm^{-1}) equaling to $\frac{\partial \theta}{\partial h_s}$, with its reciprocal being $\frac{1}{C_{so}} = \frac{\partial h_s}{\partial \theta}$. C_{so} will be used in calculating the allowed maximum liquid water content at the freezing point (see Eq. B-6 in Annex B)

(2) Derivation of C_T

According to the freezing point depression equation (Eq. 5.13), the soil matric potential could be expressed as:

$$\varphi = h + h_s = \frac{100 L_f}{g}\left(\frac{T}{T_k}\right) \tag{A-10}$$

$$h = \frac{100 L_f}{g}\left(\frac{T}{T_k}\right) - h_s \tag{A-11}$$

where φ is the total potential of the soil water with ice present (cm), and T is the actual soil temperature (^{o}C).

We differentiate h with respect to T, to get:

$$\begin{aligned}\frac{\partial h}{\partial T} &= 100\left(\frac{L_f}{g}\frac{T_k - T}{T_k^2} + 10^{-3}\frac{c_M R}{\rho_l g} + 10^{-3}\frac{RT_k}{\rho_l g}\frac{\partial c}{\partial T}\right)\\ &= 100\left(\frac{L_f}{g}\frac{T_f}{T_k^2} + 10^{-3}\frac{c_M R}{\rho_l g} + 10^{-3}\frac{RT_k}{\rho_l g}\frac{\partial c}{\partial \theta}\ \frac{\partial \theta}{\partial T}\right)\end{aligned} \tag{A-12}$$

We define C_T as the derivate of θ_l with respect to T, and then obtain:

$$C_T = \frac{\partial \theta_l}{\partial T} = \frac{\partial \theta_l}{\partial h}\frac{\partial h}{\partial T} = C_w\frac{\partial h}{\partial T} = C_w\left(\frac{L_f T_f}{g T_k^2} + 10^{-3}\frac{c_M R}{\rho_l g} + 10^{-3}\frac{RT_k}{\rho_l g}\frac{\partial c_M}{\partial \theta}\frac{\partial \theta_l}{\partial T}\right)100 \tag{A-13}$$

where C_T is the change of liquid soil water content when soil temperature rises or falls by unit degree (K^{-1}), and C_w is soil water capacity in unit of cm^{-1}. Rewriting Eq. (A-13), we obtain C_T as:

$$\frac{\partial \theta_l}{\partial T} = C_w\left(\frac{L_f T_f}{g T_k^2} + 10^{-3}\frac{c_M R}{\rho_l g} + 10^{-3}\frac{RT_k}{\rho_l g}\frac{\partial c_M}{\partial \theta}\frac{\partial \theta_l}{\partial T}\right)100 \tag{A-14}$$

$$\frac{\partial \theta_l}{\partial T}\left(\frac{1}{100 C_w} - 10^{-3}\frac{RT_k}{\rho_l g}\frac{\partial c_M}{\partial \theta}\right) = \frac{L_f T_f}{g T_k^2} + 10^{-3}\frac{c_M R}{\rho g} \tag{A-15}$$

$$\frac{\partial \theta_l}{\partial T} = \left(\frac{L_f T_f}{g T_k^2} + 10^{-3}\frac{c_M R}{\rho_l g}\right)\left(\frac{1}{100 C_w} - 10^{-3}\frac{RT_k}{\rho_l g}\frac{\partial c_M}{\partial \theta}\right)^{-1} \tag{A-16}$$

$$\text{i.e. } \frac{\partial \theta_l}{\partial T} = \left(\frac{L_f T_f}{g T_k^2} + 10^{-3}\frac{cR}{M\rho_l g}\right)\left(\frac{1}{100 C_w} - 10^{-3}\frac{RT_k}{M\rho_l g}\frac{\partial c}{\partial \theta}\right)^{-1} \tag{A-17}$$

When using L_{fw}=335 kJ kg^{-1} instead of above L_f=335000 J kg^{-1}, we can obtain the C_T expression used in AHC, as follows:

$$C_T = \frac{\partial \theta_l}{\partial T} = \left(\frac{10^3 L_{fw} T_f}{g T_k^2} + 10^{-3}\frac{cR}{M\rho_l g}\right)\left(\frac{10^{-2}}{C_w} - 10^{-3}\frac{RT_k}{M\rho_l g}\frac{\partial c}{\partial \theta}\right)^{-1} \quad \text{or}$$

$$C_T = \frac{\partial \theta_l}{\partial T} = \left(\frac{10^5 L_{fw} T_f}{g T_k^2} + 10^{-3}\frac{10^2 cR}{M\rho_l g}\right)\left(\frac{1}{C_w} - 10^{-3}\frac{10^2 RT_k}{M\rho_l g}\frac{\partial c}{\partial \theta}\right)^{-1} \tag{A-18}$$

Note that C_T will be used in solving heat transport equation (see Eqs. 9.72-9.73)

Annex B: Calculation of maximum liquid soil water content during freezing conditions

The total potential of the soil water with ice present is controlled by the vapor pressure over ice, and given by the freezing point depression equation (i.e., Eq. 5.13), as follows:

$$\varphi = h + h_s = \frac{100L_f}{g}\left(\frac{T}{T_k}\right) \qquad \text{(B-1)}$$

where L_f is the heat of fusion (335000 J kg^{-1}), and 100 is used for unit conversion.

According to Newton and Raphson iteration approach, setting $F = \varphi - h - h_s$, we obtain:

$$\frac{\partial F^j}{\partial \theta} \cdot \Delta\theta + F^{j-1} = 0 \qquad (\Delta\theta = \theta^{j+1} - \theta^j) \qquad \text{(B-2)}$$

$$\frac{\partial F}{\partial \theta} = -\frac{\partial h}{\partial \theta} - \frac{\partial h_s}{\partial \theta} \qquad \text{(B-3)}$$

$$\left(\frac{\partial h}{\partial \theta} + \frac{\partial h_s}{\partial \theta}\right)\left(\theta^{j+1} - \theta^j\right) = \varphi - h^j - h_s^j \qquad \text{(B-4)}$$

where θ^j is the allowed maximum (unfrozen) liquid water content at the freezing point (i.e. VL2 in source code) at the j^{th} iteration.

Introducing the relationships of $\frac{\partial h}{\partial \theta} = \frac{1}{C_w}$ and $\frac{\partial h_s}{\partial \theta} = \frac{1}{C_{so}}$, we get:

$$\left(\frac{1}{C_w} + \frac{1}{C_{so}}\right)\left(\theta^{j+1} - \theta^j\right) = \varphi - h^j - h_c^j \qquad \text{(B-5)}$$

$$\theta^{j+1} = \theta^j + \frac{\varphi - h^j - h_c^j}{\left(\frac{1}{C_w} + \frac{1}{C_{so}}\right)} \qquad \text{(B-6)}$$

where C_w and C_{so} are calculated by Eq. (2.13) and Eq. (A-9), respectively.

Annex C: Numerical solution of Richards equation using Picard iteration method

We use the Picard iteration to linearize the Richards equation (i.e., Eq. 2.5), as follows:

$$\begin{aligned}\frac{\theta_i^{j+1,p}-\theta_i^{j}}{\Delta t}&=\frac{\theta_i^{j+1,p}-\theta_i^{j+1,p-1}+\theta_i^{j+1,p-1}-\theta_i^{j}}{\Delta t}\\&=\frac{\theta_i^{j+1,p}-\theta_i^{j+1,p-1}}{\Delta t}+\frac{\theta_i^{j+1,p-1}-\theta_i^{j}}{\Delta t}\\&=\frac{C_i^{j+1,p-1}\left(h_i^{j+1,p}-h_i^{j+1,p-1}\right)}{\Delta t}+\frac{\theta_i^{j+1,p-1}-\theta_i^{j}}{\Delta t}\end{aligned}\tag{C-1}$$

The discrete form for the implicit scheme of Richards equation is:

$$\begin{aligned}&C_i^{j+1,p-1}\left(h_i^{j+1,p}-h_i^{j+1,p-1}\right)+\left(\theta_i^{j+1,p-1}-\theta_i^{j}\right)+\frac{\rho_i}{\rho_l}\left(\alpha_i^{j+1}-\alpha_i^{j}\right)=\\&\frac{\Delta t^{j}}{\Delta z_i}\left(K_{i-1/2}^{j+1,p-1}\frac{h_{i-1}^{j+1,p}-h_i^{j+1,p}}{\Delta z_u}+K_{i-1/2}^{j+1,p-1}-K_{i+1/2}^{j+1,p-1}\frac{h_i^{j+1,p}-h_{i+1}^{j+1,p}}{\Delta z_L}-K_{i+1/2}^{j+1,p-1}\right)-\Delta t^{j}S_i^{j}\\&-\frac{\Delta t}{\Delta z_i}\frac{1}{\rho_l}\left(q_{v,i-1/2}^{j+1}-q_{v,i+1/2}^{j+1}\right)\end{aligned}\tag{C-2}$$

where $\Delta t=t^{j+1}-t^{j}$, $\Delta z_u= z_{i-1}- z_i$, $\Delta z_l= z_i- z_{i+1}$ and Δz_i is the compartment thickness, the subscript j and $j+1$ indicate values at t and $t+\Delta t$, respectively, and i is node number. Application of Eq. (C-2) to each node results in a tridiagonal matrix:

$$\begin{bmatrix}\beta_1&\gamma_1&&&&&\\\alpha_2&\beta_2&\gamma_2&&&&\\&\alpha_3&\beta_3&\gamma_3&&&\\&&&&&&\\&&&&\alpha_{N-1}&\beta_{N-1}&\gamma_{N-1}\\&&&&&\alpha_N&\beta_N\end{bmatrix}\begin{bmatrix}h_1^{j+1,p}\\h_2^{j+1,p}\\h_3^{j+1,p}\\\\h_{N-1}^{j+1,p}\\h_N^{j+1,p}\end{bmatrix}=\begin{bmatrix}f_1\\f_2\\f_3\\\\f_{N-1}\\f_N\end{bmatrix}\tag{C-3}$$

Note that the Picard iteration is an extension of the fixed-point iteration method, particularly in the context of ordinary differential equations (ODEs). Meanwhile, the deferred correction method is used to handle the non-linear terms, such as for vapour flow term. That is, the nonlinear terms in the equation are treated as source/sink terms, calculated using the values from the previous iteration, and thus the linear equation is solved at each iteration. The Picard iterative method does not require derivative derivation and is easy to implement; however, it is more complicated to program in freeze-thaw simulations (when dealing with the limit of net water flux).

(1) Intermediate nodes

Rearrangement of Eq. (C-2) results in the coefficients:

$$\alpha_i = -\frac{\Delta t^j}{\Delta z_i \Delta z_u} K_{i-1/2}^{j+1,p-1} \tag{C-4}$$

$$\beta_i = C_i^{j+1,p-1} + \frac{\Delta t^j}{\Delta z_i \Delta z_u} K_{i-1/2}^{j+1,p-1} + \frac{\Delta t^j}{\Delta z_i \Delta z_L} K_{i+1/2}^{j+1,p-1} \tag{C-5}$$

$$\gamma_i = -\frac{\Delta t^j}{\Delta z_i \Delta z_L} K_{i+1/2}^{j+1,p-1} \tag{C-6}$$

$$\begin{aligned} f_i = {} & C_i^{j+1,p-1} h_i^{j+1,p-1} + \theta_i^{j} - \theta_i^{j+1,p-1} + \frac{\Delta t^j}{\Delta z_i}\left(K_{i-1/2}^{j+1,p-1} - K_{i+1/2}^{j+1,p-1}\right) - \Delta t^j S_i^j \\ & - \frac{\Delta t}{\Delta z_i}\frac{1}{\rho_l}\left(q_{v,i-1/2}^{j+1} - q_{v,i+1/2}^{j+1}\right) - \frac{\rho_i}{\rho_l}\left(\alpha_i^{j+1} - \alpha_i^{j}\right) \end{aligned} \tag{C-7}$$

(2) Top node

(a) Flux boundary condition

Transforming the right-hand side of Eq. (C-2) yields:

$$\begin{aligned} & C_1^{j+1,p-1}\left(h_1^{j+1,p} - h_1^{j+1,p-1}\right) + \left(\theta_1^{j+1,p-1} - \theta_1^{j}\right) + \frac{\rho_i}{\rho_l}\left(\alpha_i^{j+1} - \alpha_i^{j}\right) = \\ & \frac{\Delta t^j}{\Delta z_1}\left(-q_{sur} - K_{1+1/2}^{j+1,p-1}\frac{h_1^{j+1,p} - h_2^{j+1,p}}{\Delta z_L} - K_{i+1/2}^{j+1,p-1}\right) - \Delta t^j S_1^j \\ & - \frac{\Delta t}{\Delta z_i}\frac{1}{\rho_l}\left(0 - q_{v,i+1/2}^{j+1}\right) \end{aligned} \tag{C-8}$$

where q_{sur} is the soil water flux at the soil surface boundary. Rearrangement gives the coefficients:

$$\beta_1 = C_1^{j+1,p-1} + \frac{\Delta t^j}{\Delta z_1 \Delta z_L} K_{i+1/2}^{j+1,p-1} \tag{C-9}$$

$$\gamma_1 = -\frac{\Delta t^j}{\Delta z_1 \Delta z_L} K_{i+1/2}^{j+1,p-1} \tag{C-10}$$

$$\begin{aligned} f_1 = {} & C_1^{j+1,p-1} h_1^{j+1,p-1} + \theta_1^{j} - \theta_1^{j+1,p-1} - \frac{\Delta t^j}{\Delta z_1}\left(q_{surf} + K_{i+1/2}^{j+1,p-1}\right) - \Delta t^j S_1^j \\ & - \frac{\Delta t}{\Delta z_i}\frac{1}{\rho_l}\left(0 - q_{v,i+1/2}^{j+1}\right) - \frac{\rho_i}{\rho_l}\left(\alpha_i^{j+1} - \alpha_i^{j}\right) \end{aligned} \tag{C-11}$$

(b) Third-type boundary condition

Transforming the right-hand side of Eq. (C-2) yields:

$$\begin{aligned} & C_1^{j+1,p-1}\left(h_1^{j+1,p} - h_1^{j+1,p-1}\right) + \left(\theta_1^{j+1,p-1} - \theta_1^{j}\right) + \frac{\rho_i}{\rho_l}\left(\alpha_i^{j+1} - \alpha_i^{j}\right) = \\ & \frac{\Delta t^j}{\Delta z_1}\left(K_{1-1/2}^{j+1,p-1}\frac{h_{sur} - h_1^{j+1,p}}{\Delta z_u} + K_{1-1/2}^{j+1,p-1} - K_{1+1/2}^{j+1,p-1}\frac{h_1^{j+1,p} - h_2^{j+1,p}}{\Delta z_L} - K_{i+1/2}^{j+1,p-1}\right) - \Delta t^j S_1^j \\ & - \frac{\Delta t}{\Delta z_i}\frac{1}{\rho_l}\left(0 - q_{v,i+1/2}^{j+1}\right) \end{aligned} \tag{C-12}$$

where h_{sur} is the soil pressure head at the soil surface boundary. Rearrangement gives the coefficients:

$$\beta_1 = C_1^{j+1,p-1} + \frac{\Delta t^j}{\Delta z_1 \Delta z_u} K_{1-1/2}^{j+1,p-1} + \frac{\Delta t^j}{\Delta z_1 \Delta z_L} K_{1+1/2}^{j+1,p-1} \tag{C-13}$$

$$\gamma_1 = -\frac{\Delta t^j}{\Delta z_1 \Delta z_L} K_{i+1/2}^{j+1,p-1} \tag{C-14}$$

$$\begin{aligned} f_1 = {} & C_1^{j+1,p-1} h_1^{j+1,p-1} + \theta_1^{j} - \theta_1^{j+1,p-1} + \\ & \frac{\Delta t^j}{\Delta z_1}\left(K_{1-1/2}^{j+1,p-1} - K_{1+1/2}^{j+1,p-1}\right) + \frac{\Delta t^j}{\Delta z_1 \Delta z_u} K_{1-1/2}^{j+1,p-1} h_{sur} - \Delta t^j S_i^j \\ & - \frac{\Delta t}{\Delta z_i}\frac{1}{\rho_l}\left(0 - q_{v,i+1/2}^{j+1}\right) - \frac{\rho_i}{\rho_l}\left(\alpha_i^{j+1} - \alpha_i^{j}\right) \end{aligned} \tag{C-15}$$

(3) Bottom node

(a) Pressure head boundary condition

If a first-type (Dirichlet) boundary condition is specified at the bottom of the soil profile, then the term β_n is equal to unity, α_n reduce to zero, and f_n equal to the prescribed pressure head h_0 at the bottom node. Thus, the calculated pressure head h_n for the bottom node is always equal to h_0.

(b) Flux boundary condition

Transforming the right-hand side of Eq. (C-2) yields:

$$\begin{aligned}&C_N^{\,j+1,p-1}\left(h_N^{j+1,p}-h_N^{j+1,p-1}\right)+\left(\theta_N^{\,j+1,p-1}-\theta_N^{\,j}\right)+\frac{\rho_i}{\rho_l}\left(\alpha_N^{\,j+1}-\alpha_N^{\,j}\right)=\\&\frac{\Delta t^j}{\Delta z_N}\left(K_{N-1/2}^{j+1,p-1}\frac{h_{N-1}^{j+1,p}-h_N^{j+1,p}}{\Delta z_u}+K_{N-1/2}^{j+1,p-1}+q_{bot}\right)-\Delta t^j S_N^j\\&-\frac{\Delta t}{\Delta z_N}\frac{1}{\rho_l}\left(q_{v,N-1/2}^{j+1}-0\right)\end{aligned} \tag{C-16}$$

where q_{bot} is the soil water flux at the soil bottom boundary. Rearrangement gives the coefficients:

$$\alpha_N=-\frac{\Delta t^j}{\Delta z_N\Delta z_u}K_{N-1/2}^{j+1,p-1} \tag{C-17}$$

$$\beta_N=C_N^{j+1,p-1}+\frac{\Delta t^j}{\Delta z_N\Delta z_u}K_{N-1/2}^{j+1,p-1} \tag{C-18}$$

$$\begin{aligned}f_N=\;&C_N^{\,j+1,p-1}h_N^{j+1,p-1}+\theta_N^{\,j}-\theta_N^{\,j+1,p-1}+\frac{\Delta t^j}{\Delta z_N}\left(K_{N-1/2}^{j+1,p-1}+q_{bot}\right)-\Delta t^j S_N^j\\&-\frac{\Delta t}{\Delta z_N}\frac{1}{\rho_l}\left(q_{v,N-1/2}^{j+1}-0\right)-\frac{\rho_i}{\rho_l}\left(\alpha_N^{\,j+1}-\alpha_N^{\,j}\right)\end{aligned} \tag{C-19}$$

Annex D: Numerical solution of Richards equation using Newton-Raphson iterative technique (primary switching algorithm)

Application of Newton-Raphson iterative technique (N-RIT) to Eq. (9.1) for each node results in a tridiagonal matrix:

$$\begin{bmatrix} B_1 & C_1 & & & & & \\ A_2 & B_2 & C_2 & & & & \\ & A_3 & B_3 & C_3 & & & \\ & & & \ddots & & & \\ & & & & A_{N-1} & B_{N-1} & C_{N-1} \\ & & & & & A_N & B_N \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \Delta x_3 \\ \vdots \\ \Delta x_{N-1} \\ \Delta x_N \end{bmatrix} = \begin{bmatrix} D_1 \\ D_2 \\ D_3 \\ \vdots \\ D_{N-1} \\ D_N \end{bmatrix} \tag{D-1}$$

The state variable x_i could be the soil water pressure head/matric potential (h) or ice content (α), which is dependent on whether the node is frozen. This method could be called as the primary-variable switching algorithm, which uses h and α as the primary variable for non-ice nodes and ice nodes, respectively. When ice is not present in node i, the equation is solved for h_i at the current time-step (i.e. t+Δt), and α_i is equal to zero; otherwise, when the node is frozen, the equation is solved for α_i, and h_i is defined by temperature and kept unchanged in solving Eq. (9.12).

Application of N-RIT to Eq. (9.1) gives the coefficients for the ***intermediate nodes*** (i=2~N-1), as follows:

$$A_i = \begin{cases} 0 & \text{if node } i\text{-1 is frozen} \\ -\dfrac{\Delta t}{\Delta z_i \Delta z_u} K_{i-1/2}^{j+1,p-1} + \dfrac{\Delta t}{\Delta z_i \rho_l} \dfrac{\partial q_{v,i-1/2}^{j+1,p-1}}{\partial h_{i-1}} & \text{otherwise} \end{cases} \tag{D-2}$$

$$B_i = \begin{cases} C_i^{j+1,p-1} + \dfrac{\Delta t}{\Delta z_i \Delta z_u} K_{i-1/2}^{j+1,p-1} + \dfrac{\Delta t}{\Delta z_i \Delta z_L} K_{i+1/2}^{j+1,p-1} \\ \qquad + \dfrac{\Delta t}{\Delta z_i \rho_l}\left(\dfrac{\partial q_{v,i-1/2}^{j+1,p-1}}{\partial h_i^{j+1}} - \dfrac{\partial q_{v,i+1/2}^{j+1,p-1}}{\partial h_i^{j+1}} \right) & \text{otherwise} \\ \dfrac{\rho_i}{\rho_l} & \text{if } i \text{ is frozen} \end{cases} \tag{D-3}$$

$$C_i = \begin{cases} 0 & \text{if node } i+1 \text{ is frozen} \\ -\dfrac{\Delta t}{\Delta z_i \Delta z_L} K_{i+1/2}^{j+1,p-1} - \dfrac{\Delta t}{\Delta z_i \rho_l} \dfrac{\partial q_{v,i+1/2}^{j+1,p-1}}{\partial h_{i+1}} & \text{otherwise} \end{cases} \tag{D-4}$$

$$\begin{aligned} D_i = & -\left(\theta_i^{j+1,p-1} - \theta_i^{j}\right) - \frac{\rho_i}{\rho_l}\left(\alpha_i^{j+1,p-1} - \alpha_i^{j}\right) - \frac{\Delta t}{\Delta z_i}\left(q_{i-1/2}^{j+1,p-1} - q_{i+1/2}^{j+1,p-1}\right) \\ & - \frac{\Delta t}{\Delta z_i \rho_l}\left(q_{v,i-1/2}^{j+1,p-1} - q_{v,i+1/2}^{j+1,p-1}\right) - \Delta t S_i^{j} \end{aligned} \tag{D-5}$$

where the superscript p is the iteration level and $C_i^{j+1,p-1}$ is the water capacity evaluated at the h value of the previous iteration. The terms $q_{i-1/2}^{j+1,p-1}$ and $q_{i+1/2}^{j+1,p-1}$ are as follows:

$$q_{i-1/2}^{j+1,p-1} = -\left(K_{i-1/2}^{j+1,p-1} \frac{h_{i-1}^{j+1,p-1} - h_i^{j+1,p-1}}{\Delta z_u} + K_{i-1/2}^{j+1,p-1} \right) \tag{D-6}$$

$$q_{i+1/2}^{j+1,p-1} = -\left(K_{i+1/2}^{j+1,p-1} \frac{h_{i}^{j+1,p-1} - h_{i+1}^{j+1,p-1}}{\Delta z_L} - K_{i+1/2}^{j+1,p-1} \right) \tag{D-7}$$

The terms $q_{v,i-1/2}^{j+1,p-1}$ and $q_{v,i+1/2}^{j+1,p-1}$ are as follows:

$$q_{v,i-1/2}^{j+1,p-1} = -\left(D_{v,i-1/2}^{j+1,p-1} h_{r,i}^{j+1,p-1} s_{v,i}^{j+1,p-1} \frac{T_{i-1}^{j+1,p-1} - T_i^{j+1,p-1}}{\Delta z_u} \right) - D_{v,i-1/2}^{j+1,p-1} \rho_{vs,i}^{j+1,p-1} \varsigma_i^{j+1,p-1} \frac{h_{r,i-1}^{j+1,p-1} - h_{r,i}^{j+1,p-1}}{\Delta z_u} \tag{(D-8)}$$

$$q_{v,i+1/2}^{j+1,p-1} = -\left(D_{v,i+1/2}^{j+1,p-1} h_{r,i}^{j+1,p-1} s_{v,i}^{j+1,p-1} \frac{T_{i}^{j+1,p-1} - T_{i+1}^{j+1,p-1}}{\Delta z_L} \right) - D_{v,i+1/2}^{j+1,p-1} \rho_{vs,i}^{j+1,p-1} \varsigma_i^{j+1,p-1} \frac{h_{r,i}^{j+1,p-1} - h_{r,i+1}^{j+1,p-1}}{\Delta z_L} \tag{D-9}$$

In addition, we have:

$$\frac{\partial\left(q_{v,i-1/2}^{j+1} - q_{v,i+1/2}^{j+1}\right)}{\partial h_{i-1}} = \frac{\partial q_{v,i-1/2}^{j+1}}{\partial h_{i-1}} = -\left(D_{v,i-1/2}^{j+1} \rho_{vs,i}^{j+1} \frac{1}{\Delta z_u} \right) h_{r,i-1}^{j+1} \left(\frac{Mg}{RT_k} \right) \tag{D-10}$$

$$\frac{\partial\left(q_{v,i-1/2}^{j+1,p} - q_{v,i+1/2}^{j+1,p}\right)}{\partial h_{i}} = \left(D_{v,i-1/2}^{j+1} \rho_{vs,i}^{j+1} \frac{1}{\Delta z_u} + D_{v,i+1/2}^{j+1} \rho_{vs,i}^{j+1} \frac{1}{\Delta z_L} \right) h_{r,i}^{j+1} \left(\frac{Mg}{RT_k} \right) \tag{D-11}$$

$$\frac{\partial\left(q^{j+1}_{v,i-1/2}-q^{j+1}_{v,i+1/2}\right)}{\partial h_{i+1}}=-\frac{\partial q^{j+1}_{v,i+1/2}}{\partial h_{i+1}}=-\left(D^{j+1}_{v,i+1/2}\rho^{j+1}_{vs,i}\frac{1}{\Delta z_L}\right)h^{j+1}_{r,i+1}\left(\frac{Mg}{RT_k}\right) \qquad \text{(D-12)}$$

(1) Treatment of top boundary conditions

(a) Flux boundary condition (top node, i=1)

Transforming the right-hand side of Eq. (9.1) yields:

$$\begin{aligned}&\left(\theta_1^{j+1}-\theta_1^{j}\right)+\frac{\rho_i}{\rho_l}\left(\alpha_i^{j+1}-\alpha_i^{j}\right)=\\&\frac{\Delta t}{\Delta z_1}\left(-q_{sur}-K^{j+1}_{i+1/2}\frac{h_1^{j+1}-h_2^{j+1}}{\Delta z_L}-K^{j+1}_{i+1/2}\right)-\frac{\Delta t}{\Delta z_i}\left(0-q^{j+1}_{v,i+1/2}\right)-\Delta t S_1^{j}\end{aligned} \qquad \text{(D-13)}$$

Application of N-RIT to this equation and rearrangement give the coefficients:

$$B_1=\begin{cases}C_1^{j+1,p-1}+\dfrac{\Delta t}{\Delta z_1\Delta z_L}K^{j+1,p-1}_{1+1/2}-\dfrac{\Delta t}{\Delta z_1\rho_l}\dfrac{\partial q^{j+1,p-1}_{v,1+1/2}}{\partial h_1^{j+1}} & \text{otherwise}\\ \dfrac{\rho_i}{\rho_l} & \text{if node } i \text{ is frozen}\end{cases} \qquad \text{(D-14)}$$

$$C_1=\begin{cases}0 & \text{if } i+1 \text{ is frozen}\\ -\dfrac{\Delta t}{\Delta z_i\Delta z_L}K^{j+1,p-1}_{1+1/2}-\dfrac{\Delta t}{\Delta z_i\rho_l}\dfrac{\partial q^{j+1,p-1}_{v,1+1/2}}{\partial h_2} & \text{otherwise}\end{cases} \qquad \text{(D-15)}$$

$$\begin{aligned}D_1=&-\left(\theta_1^{j+1,p-1}-\theta_1^{j}\right)-\frac{\rho_1}{\rho_l}\left(\alpha_1^{j+1,p-1}-\alpha_1^{j}\right)+\frac{\Delta t}{\Delta z_1}\left(-q_{sur}+q^{j+1,p-1}_{1+1/2}\right)\\&-\frac{\Delta t}{\Delta z_1}\left(0-q^{j+1,p-1}_{v,1+1/2}\right)-\Delta t S_1^{j}\end{aligned} \qquad \text{(D-16)}$$

(b) Third-type boundary condition (i=1)

Transforming the right-hand side of Eq. (9.1) yields:

$$\begin{aligned}&\left(\theta_1^{j+1}-\theta_1^{j}\right)+\frac{\rho_i}{\rho_l}\left(\alpha_i^{j+1}-\alpha_i^{j}\right)=\\&\frac{\Delta t}{\Delta z_1}\left(K^{j+1}_{1-1/2}\frac{h_{sur}-h_1^{j+1}}{\Delta z_u}+K^{j+1}_{1-1/2}-K^{j+1}_{1+1/2}\frac{h_1^{j+1}-h_2^{j+1}}{\Delta z_L}-K^{j+1}_{1+1/2}\right)-\frac{\Delta t}{\Delta z_1}\left(0-q^{j+1}_{v,1+1/2}\right)-\Delta t S_1^{j}\end{aligned} \qquad \text{(D-17)}$$

Application of N-RIT to this equation and rearrangement give the coefficients:

$$B_1 = \begin{cases} C_1^{j+1,p-1} + \dfrac{\Delta t}{\Delta z_1 \Delta z_u} K_{1-1/2}^{j+1,p-1} + \dfrac{\Delta t}{\Delta z_1 \Delta z_L} K_{1+1/2}^{j+1,p-1} - \dfrac{\Delta t}{\Delta z_1 \rho_l} \dfrac{\partial q_{v,1+1/2}^{j+1,p-1}}{\partial h_1^{j+1}} & \text{otherwise} \\ \dfrac{\rho_1}{\rho_l} & \text{if node } i \text{ is frozen} \end{cases} \tag{D-18}$$

$$C_1 = \begin{cases} 0 & \text{if node } i\text{+1 is frozen} \\ -\dfrac{\Delta t}{\Delta z_1 \Delta z_L} K_{1+1/2}^{j+1,p-1} - \dfrac{\Delta t}{\Delta z_i \rho_l} \dfrac{\partial q_{v,1+1/2}^{j+1,p-1}}{\partial h_2} & \text{otherwise} \end{cases} \tag{D-19}$$

$$\begin{aligned} D_1 = &-\left(\theta_1^{\,j+1,p-1} - \theta_1^{\,j}\right) - \frac{\rho_i}{\rho_l}\left(\alpha_1^{\,j+1,p-1} - \alpha_1^{\,j}\right) + \frac{\Delta t}{\Delta z_1}\left(-q_{1-1/2}^{j+1,p-1} + q_{1+1/2}^{j+1,p-1}\right) \\ &- \frac{\Delta t}{\Delta z_1}\left(0 - q_{v,1+1/2}^{j+1,p-1}\right) - \Delta t S_1^j \end{aligned} \tag{D-20}$$

where the term $q_{1-1/2}^{j+1,p-1}$ is as follows:

$$q_{1-1/2}^{j+1,p-1} = -\left(K_{1-1/2}^{j+1,p-1} \frac{h_{sur} - h_1^{j+1,p-1}}{\Delta z_u} + K_{1-1/2}^{j+1,p-1}\right) \tag{D-21}$$

(2) Treatment of bottom boundary conditions

(a) Pressure head boundary condition (i=n)

If a first-type (Dirichlet) boundary condition is specified at the bottom of the soil profile, then the term B_N is equal to unity, A_N reduces to zero, and D_N is equal to zero. Thus, the calculated pressure head for node N is always equal to the prescribed pressure head, h_0.

(b) Flux boundary condition (i=n)

Transforming the right-hand side of Eq. (9.1) yields:

$$\begin{aligned} &\left(\theta_N^{\,j+1} - \theta_N^{\,j}\right) + \frac{\rho_i}{\rho_l}\left(\alpha_N^{\,j+1} - \alpha_N^{\,j}\right) = \\ &\frac{\Delta t}{\Delta z_N}\left(K_{N-1/2}^{j+1} \frac{h_{N-1}^{j+1} - h_N^{j+1}}{\Delta z_u} + K_{N-1/2}^{j+1} + q_{bot}\right) - \frac{\Delta t}{\Delta z_N}\left(q_{v,N-1/2}^{j+1} - 0\right) - \Delta t S_N^j \end{aligned} \tag{D-22}$$

Application of N-RIT to to this equation and rearrangement give the coefficients:

$$A_N = \begin{cases} 0 & \text{if node } n\text{-1 is frozen} \\ -\dfrac{\Delta t}{\Delta z_N \Delta z_u} K_{N-1/2}^{j+1,p-1} + \dfrac{\Delta t}{\Delta z_N \rho_l} \dfrac{\partial q_{v,N-1/2}^{j+1,p-1}}{\partial h_{N-1}} & \text{otherwise} \end{cases} \tag{D-23}$$

$$B_N = \begin{cases} C_N^{j+1,p-1} + \dfrac{\Delta t}{\Delta z_N \Delta z_u} K_{N-1/2}^{j+1,p-1} + \dfrac{\Delta t}{\Delta z_N \rho_l}\left(\dfrac{\partial q_{v,N-1/2}^{j+1,p-1}}{\partial h_N^{j+1}}\right) & \text{otherwise} \\ \dfrac{\rho_i}{\rho_l} & \text{if node } N \text{ is frozen} \end{cases} \tag{D-24}$$

$$D_N = -\left(\theta_N{}^{j+1,p-1} - \theta_N{}^{j}\right) - \frac{\rho_i}{\rho_l}\left(\alpha_N{}^{j+1,p} - \alpha_N{}^{j}\right) + \frac{\Delta t}{\Delta z_N}\left(-q_{N-1/2}^{j+1,p-1} + q_{bot}\right) - \frac{\Delta t}{\Delta z_N}\left(q_{v,N-1/2}^{j+1,p-1} - 0\right) - \Delta t S_N^j \tag{D-25}$$

(c) Third-type boundary condition (i=N)

Transforming the right-hand side of Eq. (9.1) yields:

$$\left(\theta_N{}^{j+1} - \theta_N{}^{j}\right) + \frac{\rho_i}{\rho_l}\left(\alpha_N{}^{j+1} - \alpha_N{}^{j}\right) = \frac{\Delta t}{\Delta z_N}\left(K_{N-1/2}^{j+1}\frac{h_{N-1}^{j+1} - h_N^{j+1}}{\Delta z_u} + K_{N-1/2}^{j+1} - K_{N+1/2}^{j+1}\frac{h_N^{j+1} - h_{bot}}{\Delta z_L} - K_{N+1/2}^{j+1}\right) - \frac{\Delta t}{\Delta z_N}\left(q_{v,N-1/2}^{j+1} - 0\right) - \Delta t S_N^j \tag{D-26}$$

Application of N-RIT to to this equation and rearrangement give the coefficients:

$$A_N = \begin{cases} 0 & \text{if node } N\text{-1 is frozen} \\ -\dfrac{\Delta t}{\Delta z_N \Delta z_u} K_{N-1/2}^{j+1,p-1} + \dfrac{\Delta t}{\Delta z_N \rho_l}\dfrac{\partial q_{v,N-1/2}^{j+1,p-1}}{\partial h_{N-1}} & \text{otherwise} \end{cases} \tag{D-27}$$

$$B_N = \begin{cases} \dfrac{\rho_i}{\rho_l} & \text{if node } N \text{ is frozen} \\ C_N^{j+1,p-1} + \dfrac{\Delta t}{\Delta z_N \Delta z_u} K_{N-1/2}^{j+1,p-1} + \dfrac{\Delta t}{\Delta z_N \Delta z_L} K_{N+1/2}^{j+1,p-1} \\ \quad + \dfrac{\Delta t}{\Delta z_N \rho_l}\dfrac{\partial q_{v,N-1/2}^{j+1,p-1}}{\partial h_N^{j+1}} & \text{otherwise} \end{cases} \tag{D-28}$$

$$D_N = -\left(\theta_N{}^{j+1,p-1} - \theta_N{}^{j}\right) - \frac{\rho_i}{\rho_l}\left(\alpha_N{}^{j+1,p} - \alpha_N{}^{j}\right) + \frac{\Delta t}{\Delta z_N}\left(-q_{N-1/2}^{j+1,p-1} + q_{N+1/2}^{j+1,p-1}\right) - \frac{\Delta t}{\Delta z_N}\left(q_{v,N-1/2}^{j+1,p-1} - 0\right) - \Delta t S_N^j \tag{D-29}$$

where the term $q_{n+1/2}^{j+1,p-1}$ is as follows:

$$q_{N+1/2}^{j+1,p-1} = -\left(K_{N+1/2}^{j+1,p-1}\frac{h_N^{j+1,p-1} - h_{bot}}{\Delta z_L} + K_{N+1/2}^{j+1,p-1}\right) \tag{D-30}$$

Annex E: Calculation of solute concentration (C_{im}) in immobile regions using the Newton iteration method

The discretization form of the solute change equation in the immobile region (Eq. 4.21) is rewritten as follows:

$$\left(\theta_{\mathrm{im}}^{j+1} c_{\mathrm{im}}^{j+1} + \rho_b \frac{k_s \left(c_{\mathrm{im}}^{j+1}\right)^{\beta}}{1+\eta\left(c_{\mathrm{im}}^{j+1}\right)^{\beta}}\right) = \frac{X_{\mathrm{im}}{}^{j} - \left(1-f_{\mathrm{m}}^{j+1}\right)\left(\mu_w \theta_{im} c_{\mathrm{im}}^{j} + \mu_s \rho_b Q_{\mathrm{im}}^{j}\right)\Delta t + G_c \Delta t}{\left(1-f_{\mathrm{m}}^{j+1}\right)} \tag{E-1}$$

We can set a constant X_2 as:

$$X_2 = \frac{X_{\mathrm{im}}{}^{j} - \left(1-f_{\mathrm{m}}^{j+1}\right)\left(\mu_w \theta_{im} c_{\mathrm{im}}^{j} + \mu_s \rho_b Q_{\mathrm{im}}^{j}\right)\Delta t + G_c \Delta t}{\left(1-f_{\mathrm{m}}^{j+1}\right)} \tag{E-2}$$

It yields a concise expression of Eq. (E-1):

$$\theta_{\mathrm{im}} c_{\mathrm{im}} + \frac{\rho_b k_s c_{\mathrm{im}}^{\beta}}{1+\eta c_{\mathrm{im}}^{\beta}} = X_2 \tag{E-3}$$

We could define a function $f(c_{im})$ as below:

$$f(c_{\mathrm{im}}) = \theta_{\mathrm{im}} c_{\mathrm{im}} + \frac{\rho_b k_s c_{\mathrm{im}}^{\beta}}{1+\eta c_{\mathrm{im}}^{\beta}} - X_2 \tag{E-4}$$

The expression for the derivative of $f(c_{im})$ can be obtained as:

$$f'(c_{\mathrm{im}}) = \theta_{\mathrm{im}} + \rho_b \left[k_s \beta \cdot c_{\mathrm{im}}^{\beta-1} \cdot \left(1+\eta c_{\mathrm{im}}^{\beta}\right)^{-1} - k_s \eta \beta \cdot c_{\mathrm{im}}^{2\beta-1} \cdot \left(1+\eta c_{\mathrm{im}}^{\beta}\right)^{-2}\right] \tag{E-5}$$

Then the c_{im} can be calculated iteratively as follows:

$$c_{\mathrm{im}}^{j+1} = c_{\mathrm{im}}^{j} - \frac{f(c_{\mathrm{im}})}{f'(c_{\mathrm{im}})} \tag{E-6}$$

Annex F: Development process and history of the AHC model

Since 2013, we developed a new EPIC crop growth module (with moderate need of crop parameters) for SWAP version 2.0 (Xu et al., 2013); and we also incorporated a nonlinear osmotic head-dependent function to better describe the salinity stress of root water uptake, and introduced a soil moisture-based, mechanism approach to better calculate soil evaporation under shallow groundwater environments (Xu et al., 2013; Xu et al., 2015). At that stage, we referred to the model as SWAP-EPIC to distinguish it from the original SWAP. Next, we incorporated the efficient global methods (i.e., the combination of Latin Hypercube One-factor-At-a-Time (LH-OAT) method and the modified Micro Genetic Algorithm (modified-MGA)) to realize the identification of model parameters in GSA Process and GPE Process; and meanwhile, we developed a new solute transport module (based on fully implicit or Crank-Nicholson, central finite-difference scheme) to replace the original one (based on explicit difference scheme), to increase model stability and calculation efficiency to meet the need of GSA and GPE Processes (Xu et al., 2016). Then, to realize the simulation of nitrogen, we further developed several new modules, mainly involving the soil nitrogen-carbon module, and nitrogen transport and transformation module; meantime, we modified the heat conduction module into the heat convection-dispersion module, and rewritten the sources code of soil water flow module (Xu et al., 2018). At this stage, we further renamed the model as AHC (version 1.0) since model functions have been largely improved and expanded compared to SWAP-EPIC, and the source codes were largely rewritten and recompiled. The relevant paper of AHC v1.0 was published in the academic journal "*Ecological Modeling*" (Xu et al., 2018).

In the following years, we further proposed a method for coupled simulation of soil water-heat, salt and nitrogen to improve the capability of AHC in salt-affected regions. This function has been tested and applied in salt-nitrogen manmangent at spring maize fields (Li et al., 2023a; Qi et al., 2024). During this time, we also improved the model abilities with the description of

CO_2 effect on crop ET and growth and that of CO_2 emission from agro-ecosystems, which has been tested and applied by some field case studies (e.g., Liu et al., 2020; Li et al., 2023a; Li et al., 2023b). Meanwhile, a module for irrigation scheduling is modified from SWAP v2.0 (van Dam et al., 1997) and incorporated into AHC, which has been tested and applied in Qi et al. (2024). At this stage, we updated the AHC to a new version AHC v2.01, since the GUI was improved as well. Later, we made very significant improvements in function for simulating soil freeze and thaw processes, in which the soil water flow module and heat transport module were rewritten again. The vapor transfer and its effects on the soil system were newly included for this version. The two differential equations for soil water and heat flow are now solved using the Newton-Raphson iterative technique, which can particularly make the source code more concise and efficient for soil frozen simulation in AHC. Hence, the updated version can simulate soil freezing and thawing processes using heat-water coupling iteration calculation (considering phase changes of ice and water). Therefore, so far, the AHC has been updated and noted as version 2.1R (Liu et al., 2025). Besides, we have also modified the solute transport for the mobile/immobile regions with a reasonable transfer function, and meantime updated the calculation of immobile solute change using the Newton iteration method. In 2026, we further added the function for auto-fertiliztion for nitrogen management. Therefore, the AHC v2.1R has become a one-dimensional (1-D) physically-based model that, in brief, simulates water flow, solute transport and transformation, soil nitrogen/carbon, heat transport, and crop growth and yield in various/complex agricultural fields, and can be used to assist in the management of field irrigation, drainage and nitrogen fertilizer application.

Annex A: Derivation of various partial derivatives （程序中的单位, 备份用的）

(1) Derivation of C_s

The solutes may dissolve in the soil water and be absorbed by organic matter or clay minerals. Hence, the total solutes can be expressed as:

$$X = \theta c + \rho_b Q \tag{A-1}$$

where X is the total solutes present per volume of soil (g L^{-3}), Q is solute amount absorbed per volume of soil (g kg^{-1}), c is solute concentration in soil water (g L^{-1}), ρ_b is the soil bulk density (g cm^{-3}), and ρ_l is the liquid water bulk density (g cm^{-3}).

We assume a linear adsorption relationship between Q and c such that

$$Q = k_d \cdot c \tag{A-2}$$

where k_d is the empirical coefficient relating the solution and adsorbed concentrations (cm^3 g^{-1}). Hence, Eq. (A-1) becomes:

$$X = \theta c + \rho_b k_d c \tag{A-3}$$

$$c = \frac{X}{\theta + \rho_b k_d} \tag{A-4}$$

$$c = \frac{\dfrac{X}{\rho_b}}{\dfrac{\theta}{\rho_b} + k_d} \tag{A-5}$$

Setting $S = X/\rho_b$, we get

$$c = \frac{S}{k_d + \dfrac{\theta}{\rho_b}} \tag{A-6}$$

with S being the total solutes present per mass of soil (g kg^{-1}).

We differentiate c with respect to θ, to get

$$\frac{\partial c}{\partial \theta} = -\left(k_d + \frac{\theta}{\rho_b}\right)^{-2} \frac{1}{\rho_b} S \tag{A-7}$$

In dilute solutions, the osmotic pressure is proportional to the concentration of the solution and to its temperature according to the following equation (Hillel, 1998):

$$h_s = -100\frac{i_v c_M R T_k}{\rho_l g} = -100\frac{i_v R T_k}{\rho_l g}\frac{c}{M} \tag{A-8}$$

where h_s is the osmotic potential of soil solution (cm), i_v is the van't hoff factor, c_M is the molar concentration of the solute in the solution (mol L^{-1}), R is the universal gas constant (8.3143 J mol^{-1} K^{-1}), T_k is the temperature of the soil in Kelvin (K) with T_k=T+273.16, g is gravitational acceleration (9.81 m s^{-2}), and M_s is the molecular weight of specific solute (g mol^{-1}). When the simulated solute consists of multiple salt ions, the user may need to specify an equivalent representative value for M_s.

We further differentiate h_s with respect to θ, to get:

$$\frac{\partial h_s}{\partial \theta} = \frac{\partial\left(\dfrac{-100 i_v c R T_k}{M\rho_l g}\right)}{\partial \theta} = \frac{-100 i_v R T_k}{M\rho_l g}\frac{\partial c}{\partial \theta} = \frac{100 i_v R T_k}{M\rho_l g}\left(k_d + \frac{\theta}{\rho_b}\right)^{-2}\frac{1}{\rho_b}S \tag{A-9}$$

Here we note a new variable C_s (cm^{-1}) equaling to $\frac{\partial \theta}{\partial h_s}$, with its reciprocal being $\frac{1}{C_s} = \frac{\partial h_s}{\partial \theta}$. C_s will be used in calculating the allowed maximum liquid water content at the freezing point (see Eq. B-6 in Annex B)

(2) Derivation of C_T

According to the freezing point depression equation (Eq. 5.13), the soil matric potential could be expressed as:

$$\varphi = h + h_s = \frac{100 L_f}{g}\left(\frac{T}{T_k}\right) \qquad \varphi = h + h_s = \frac{100 L_f}{g T_f}T \tag{A-10}$$

$$h = \frac{100 L_f}{g}\left(\frac{T}{T_k}\right) - h_s \qquad h = \frac{100 L_f}{g T_f}T - h_s \tag{A-11}$$

where φ is the total potential of the soil water with ice present (cm), and T is the actual soil temperature (°C).

We differentiate h with respect to T, to get:

$$\frac{\partial h}{\partial T}=100\left(\frac{L_f}{g}\frac{T_k-T}{T_k^2}+\frac{c_M R}{\rho_l g}+\frac{RT_k}{\rho_l g}\frac{\partial c}{\partial T}\right)=100\left(\frac{L_f}{g}\frac{T_f}{T_k^2}+\frac{c_M R}{\rho_l g}+\frac{RT_k}{\rho_l g}\frac{\partial c}{\partial \theta}\frac{\partial \theta}{\partial T}\right)$$

$$\frac{\partial h}{\partial T}=100\left(\frac{L_f}{gT_f}+\frac{c_M R}{\rho_l g}+\frac{RT_k}{\rho_l g}\frac{\partial c}{\partial T}\right)=100\left(\frac{L_f}{gT_f}+\frac{c_M R}{\rho_l g}+\frac{RT_k}{\rho_l g}\frac{\partial c}{\partial \theta_l}\frac{\partial \theta_l}{\partial T}\right) \tag{A-12}$$

We define C_T as the derivate of θ_l with respect to T, and then obtain:

$$C_T=\frac{\partial \theta_l}{\partial T}=\frac{\partial \theta_l}{\partial h}\frac{\partial h}{\partial T}=C_w\frac{\partial h}{\partial T}=C_w\left(\frac{L_f T_f}{gT_k^2}+\frac{c_M R}{\rho_l g}+\frac{RT_k}{\rho_l g}\frac{\partial c_M}{\partial \theta}\frac{\partial \theta_l}{\partial T}\right)100$$

$$C_T=\frac{\partial \theta_l}{\partial T}=\frac{\partial \theta_l}{\partial h}\frac{\partial h}{\partial T}=C_w\frac{\partial h}{\partial T}=C_w\left(\frac{L_f}{gT_f}+\frac{c_M R}{\rho_l g}+\frac{RT_k}{\rho_l g}\frac{\partial c_M}{\partial \theta}\frac{\partial \theta_l}{\partial T}\right)100 \tag{A-13}$$

where C_T is the change of liquid soil water content when soil temperature rises or falls by unit degree (K^{-1}), and C_w is soil water capacity in unit of cm^{-1}. Rewriting Eq. (A-13), we obtain C_T as:

$$\frac{\partial \theta_l}{\partial T}=C_w\left(\frac{L_f T_f}{gT_k^2}+\frac{c_M R}{\rho_l g}+\frac{RT_k}{\rho_l g}\frac{\partial c_M}{\partial \theta}\frac{\partial \theta_l}{\partial T}\right)100 \qquad \frac{\partial \theta_l}{\partial T}=C_w\left(\frac{L_f}{gT_f}+\frac{c_M R}{\rho_l g}+\frac{RT_k}{\rho_l g}\frac{\partial c_M}{\partial \theta}\frac{\partial \theta_l}{\partial T}\right)100 \tag{A-14}$$

$$\frac{\partial \theta_l}{\partial T}\left(\frac{1}{100C_w}-\frac{RT_k}{\rho_l g}\frac{\partial c_M}{\partial \theta}\right)=\frac{L_f T_f}{gT_k^2}+\frac{c_M R}{\rho g} \qquad \frac{\partial \theta_l}{\partial T}\left(\frac{1}{100C_w}-\frac{RT_k}{\rho_l g}\frac{\partial c_M}{\partial \theta}\right)=\frac{L_f}{gT_f}+\frac{c_M R}{\rho g} \tag{A-15}$$

$$\frac{\partial \theta_l}{\partial T}=\left(\frac{L_f T_f}{gT_k^2}+\frac{c_M R}{\rho_l g}\right)\left(\frac{1}{100C_w}-\frac{RT_k}{\rho_l g}\frac{\partial c_M}{\partial \theta}\right)^{-1} \qquad \frac{\partial \theta_l}{\partial T}=\left(\frac{L_f}{gT_f}+\frac{c_M R}{\rho_l g}\right)\left(\frac{1}{100C_w}-\frac{RT_k}{\rho_l g}\frac{\partial c_M}{\partial \theta}\right)^{-1} \tag{A-16}$$

$$\text{i.e. }\frac{\partial \theta_l}{\partial T}=\left(\frac{L_f T_f}{gT_k^2}+\frac{cR}{M\rho_l g}\right)\left(\frac{1}{100C_w}-\frac{RT_k}{M\rho_l g}\frac{\partial c}{\partial \theta}\right)^{-1} \qquad \frac{\partial \theta_l}{\partial T}=\left(\frac{L_f}{gT_f}+\frac{cR}{M\rho_l g}\right)\left(\frac{1}{100C_w}-\frac{RT_k}{M\rho_l g}\frac{\partial c}{\partial \theta}\right)^{-1} \tag{A-17}$$

When using L_{fw}=335 kJ kg^{-1} instead of above L_f=335000 J kg^{-1}, we can obtain the C_T expression used in AHC, as follows:

$$C_T=\frac{\partial \theta_l}{\partial T}=\left(\frac{10^3 L_{fw}T_f}{gT_k^2}+\frac{cR}{M\rho_l g}\right)\left(\frac{10^{-2}}{C_w}-\frac{RT_k}{M\rho_l g}\frac{\partial c}{\partial \theta}\right)^{-1} \text{ or}$$

$$C_T = \frac{\partial \theta_l}{\partial T} = \left(\frac{10^5 L_{fw} T_f}{g T_k^2} + \frac{10^2 cR}{M \rho_l g} \right) \left(\frac{1}{C_w} - \frac{10^2 R T_k}{M \rho_l g} \frac{\partial c}{\partial \theta} \right)^{-1}$$

$$C_T = \frac{\partial \theta_l}{\partial T} = \left(\frac{10^3 L_{fw}}{g T_f} + \frac{cR}{M \rho_l g} \right) \left(\frac{10^{-2}}{C_w} - \frac{R T_k}{M \rho_l g} \frac{\partial c}{\partial \theta} \right)^{-1} \text{ or}$$

$$C_T = \frac{\partial \theta_l}{\partial T} = \left(\frac{10^5 L_{fw}}{g T_f} + \frac{10^2 cR}{M \rho_l g} \right) \left(\frac{1}{C_w} - \frac{10^2 R T_k}{M \rho_l g} \frac{\partial c}{\partial \theta} \right)^{-1} \qquad \text{(A-18)}$$

Note that C_T will be used in solving heat transport equation (see Eqs. 9.72-9.73).

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